OPTN/SRTR 2023 Annual Data Report: Liver

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Abstract

The number of liver transplants performed in 2023 in the United States reached another record high, totaling 10,659 overall, of which 10,125 (95.0%) were in adult recipients and 534 (5.0%) were in pediatric recipients. This growth was driven by increased recovery of livers from older donors and donation after circulatory death (DCD) donors—likely related to the wider availability of machine perfusion technologies. The overall nonuse rate, or percent of livers recovered for transplant and not transplanted, was 9.7%, a decrease from the preceding years, and 16.7% of transplant recipients accepted DCD livers. There was also growth in living donation, representing 5.7% of adult transplants and 14.6% of pediatric transplants. In July 2023, the model for end-stage liver disease (MELD) 3.0 and pediatric end-stage liver disease (PELD)–creatinine scoring systems were updated from MELD-sodium and PELD, respectively, and criteria for status 1B qualification for pediatric candidates were updated. A major goal of MELD 3.0 was to address the sex

disparity in deceased donor transplant rates. In 2023, the gap in deceased donor liver transplant rates between sexes narrowed, although the rate remained higher for adult male candidates compared with female candidates, and pretransplant mortality rates were higher among adult female candidates compared with male candidates. Alcohol-associated liver disease and metabolic dysfunction–associated steatohepatitis remained the leading indications for liver transplant.
Keywords: Liver transplantation, liver donation, outcomes, waiting list

1 Introduction

As we survey the totality of the 2023 liver transplant data, we observe macro trends that may reflect the impact of recent developments in liver transplantation on pretransplant and posttransplant outcomes. Before we delve into detailed data in the body of the report, we would like to highlight three areas that may be noteworthy.

First, since the first approval by the US Food and Drug Administration in 2021, machine perfusion devices have had substantial impact on the practice and outcome of liver transplantation. Compared with the traditional cold storage method, the technology enhances organ viability, reduces the risk of ischemic injury, and allows real-time functional assessment of the graft. Its largest impact has been seen in donation after circulatory death (DCD) graft, leading to a substantial expansion of donor pools. In this year's report, we note that removals from the waiting list for transplant among adult liver transplant candidates in the United States outpaced the number of new registrations, leaving fewer patients remaining on the list at the end of 2023. Waiting times are shorter, and there are more recipients with model for end-stage liver disease (MELD) scores of 14 or lower at the time of transplant and fewer in the higher MELD categories. The data indicate that this is not because of reduced waitlist registration, as the number of patients newly added to the list actually increased.

Second, the MELD score has been used to prioritize patients on the liver transplant waiting list since 2002. The latest version of the score, MELD 3.0, incorporates new variables including female sex and serum albumin and updates the coefficients for serum creatinine, international normalized ratio, bilirubin, and sodium. Although the pretransplant mortality rate for liver transplant candidates decreased with MELD-based organ allocation, women had experienced lower transplant rates and higher waitlist mortality. MELD 3.0 was adopted for policy implementation in July 2023. In this report, concurrent with implementation of MELD 3.0, we begin to see that the disparity in transplant rates between sexes has decreased, foretelling an encouraging impact of the new system.

The third trend, seen on a longer time scale compared with the other two, is simultaneous liver-kidney (SLK) transplant. As recognized by the inclusion of serum creatinine in the MELD score, renal dysfunction represents a vitally important extrahepatic manifestation of end-stage liver disease. In addition to having a significant impact on pretransplant mortality, renal impairment at the time of transplant also affects posttransplant outcomes, which justifies SLK in some of the liver transplant recipients. To achieve equitable use of renal allografts between kidney and liver transplant candidates, standardized medical criteria for SLK transplant were established in 2017. At the same time, a “safety net” mechanism was put in place, so that a kidney graft would be preferentially allocated in the event the renal function does not recover soon after liver transplant. In this 2023 data report, the proportion of adult SLK transplants of total liver transplants has decreased, and the number of new kidney registrations and the number of kidney transplants in adults with a previous history of liver transplant have increased since implementation of the safety net policy in 2017. The net effect may be the more judicious use of kidney allografts in patients with end-stage liver disease and improved organ utility.

2 Adult Liver Transplant

2.1 Waiting List

In 2023, there were a total of 24,492 adult liver candidates ever waiting in the United States, with a record 13,954 candidates newly added to the waiting list, an 8.5% increase over the previous year (Figure LI 1 and Figure LI 2). By the end of the year, 14,747 were removed from the list—64.5% for deceased donor liver transplant, 3.9% for living donor liver transplant, 6.4% for death, 6.6% for being too sick, 7.5% for improved condition, and 11.1% for other reasons—and 9,745 were left waiting (Table LI 4 and Table LI 5), a 7.5% decrease from the end of 2022. Compared with previous years, a greater proportion of patients were removed for transplant, and a smaller proportion were removed for death or being too sick for transplant.

Overall, adult candidates in 2023 were younger than in previous years: 6.3% were aged 18-34 years, 20.1% aged 35-49 years, 46.7% aged 50-64 years, and 26.9% aged 65 years or older (Figure LI 3). In terms of sex distribution, female candidates made up an increasing proportion of the waiting list, although still there were more male than female candidates being listed (60.6% and 39.4%, respectively) (Figure LI 4). The racial and ethnic composition was relatively unchanged: 67.8% White, 19.0% Hispanic, 6.5% Black, 4.5% Asian, 1.2% Native American, 0.6%

Multiracial, and 0.3% unreported (Figure LI 5).

The most common primary diagnosis among adult waitlist candidates was alcohol-associated liver disease, with 34.9% categorized as alcohol-associated cirrhosis and 4.1% as alcohol-associated hepatitis—increases from 22.8% and 0.1% in 2013, respectively (Figure LI 6). In August 2022, the diagnosis for "alcoholic cirrhosis" was updated to "alcohol-associated cirrhosis without acute alcohol-associated hepatitis," and the diagnosis of "alcoholic hepatitis" was updated to "acute alcohol-associated hepatitis with or without cirrhosis." Accordingly, the prevalence of waitlisting for alcohol-associated hepatitis increased from 1.7% in 2021, to 2.1% in 2022, to the 4.1% in 2023, likely related to not only practice changes but also the updated nomenclature. In 2023, 20.4% of registrations had metabolic dysfunction–associated steatohepatitis (MASH) as the primary indication for waitlisting, an increase from 9.6% in 2013. Other specified diagnoses were less common, including hepatocellular carcinoma (HCC) (10.3%), cholestatic liver disease (7.2%), hepatitis C (6.2%), and acute liver failure (1.6%).

Most adult liver transplant candidates in 2023 were overweight by body mass index (BMI); there were 1.6% with BMI of <18.5 kg/m^2, 23.0% with BMI of 18.5-<25 kg/m^2, 33.8% with BMI of 25-<30 kg/m^2, 23.7% with BMI of 30-<35 kg/m^2, and 17.3% with BMI of 35 kg/m^2 or greater (Figure LI 8). The most common blood type was O (48.4%), followed by type A (38.0%), type B (10.6%), and type AB (3.1%) (Figure LI 9), mirroring the distribution of the general US population. Similar to previous years, 3.2% of candidates had a history of prior transplant (Figure LI 10).

2.2 Waitlist Outcomes

The overall deceased donor liver transplant rate for adults increased to 94.7 transplants per 100 patient-years in 2023, from 36.1 transplants per 100 patient-years in 2013 (Figure LI 11). This increase occurred across all age and racial and ethnic groups (Figure LI 12 and Figure LI 13). Candidates aged 18-34 years and 35-49 years continued to have higher deceased donor transplant rates than older candidates. White and Black candidates had higher deceased donor transplant rates than Asian or Hispanic candidates. The deceased donor liver transplant rate for patients with blood type AB (254.6 transplants per 100 patient-years) was nearly twice more than that for patients with blood type B and three times more than that for patients with blood types A or O (Figure LI 14).

Deceased donor transplant rates for adult candidates with and without HCC exception points have been similar since 2020, after many years of greater transplant access for patients with HCC and a series of policy changes lengthening their waiting time and decreasing their waitlist priority (Figure LI 15).

In July 2023, MELD 3.0 replaced MELD-sodium (MELD-Na) to represent medical urgency in the liver transplant waiting list, which was intended to address, in part, the sex disparity in deceased donor transplant rates. In 2023, with the second half of the year reflecting the policy change, the gap in adult deceased donor liver transplant rates between sexes had already started to narrow, although the rate was still higher for male candidates compared with female candidates (96.9 versus 91.4 transplants per 100 patient-years) (Figure LI 16).

Among adult candidates newly listed in 2018-2020, only 7.0% remained on the waiting list after 3 years, 58.3% received a deceased donor liver transplant, 3.3% received a living donor liver transplant, 8.6% died, and 22.8% were removed from the list for other reasons (Figure LI 17). Of candidates listed in 2022, 41.7% underwent deceased donor liver transplant within 3 months; 48.4%, within 6 months; and 58.2%, within 1 year—all increases compared with previous years (Figure LI 18). The 1-year probability of deceased donor liver transplant increased to 58.2% from 40.6% 10 years prior.

The pretransplant mortality rate among adult waitlist candidates has decreased from 16.9 to 12.9 deaths per 100 patient-years over the past decade and has been overall stable in the past several years (Figure LI 19). In 2023, female candidates still had higher pretransplant mortality rates compared with male candidates (13.9 versus 12.2 deaths per 100 patient-years); MELD 3.0 was in effect for only the latter half of the year, and height- or body size–based disparities may still have contributed (Figure LI 22). By etiology, higher pretransplant mortality rates were observed among those with alcohol-associated hepatitis (29.4 deaths per 100 patient-years), acute liver failure (21.5 deaths per 100 patient-years), and MASH (16.5 deaths per 100 patient-years) (Figure LI 23). Despite the noted allocation policy changes that have lengthened waiting time and lowered waitlist priority for patients with HCC exception points, pretransplant mortality rates remained lower for patients with an HCC exception compared to those without (11.2 versus 13.1 deaths per 100 patient-years) (Figure LI 25). Pretransplant mortality rates still varied widely across the 51 active donation service areas, with a median of 13.1 deaths per 100 patient-years (interquartile range, 10.2-17.2) and ranging from 3.9 to 39.7 deaths per 100 patient-years (Figure LI 26).

Among adults removed from the liver transplant waiting list for reasons other than transplant or death, 14.3% died within 6 months after removal (Figure LI 27). The risk of death within 6 months after removal was higher among those with MELD scores greater than 25 and patients aged 65 years or older (Figure LI 28 and Figure LI 29).

2.3 Transplants

In 2023, there were 10,125 adult liver transplant recipients, another all-time high, representing a 71% increase in the past decade (Figure LI 33). Both deceased and living donation increased, with 9,545 deceased donor and 580 living donor liver transplants performed (Figure LI 34).

Alcohol-associated liver disease continued to be the most common indication for adult liver transplant, with 41.1% of recipients having a primary diagnosis of alcohol-associated liver disease (34.6% alcohol-associated cirrhosis and 6.5% alcohol-associated hepatitis), followed by MASH at 20.3%. Other specified etiologies included HCC (10.4%), hepatitis C virus (HCV) (4.2%), cholestatic liver disease (7.4%), and acute liver failure (2.0%) (Figure LI 38 and Table LI 7). Only 14.6% of patients underwent transplant with HCC MELD exception points, a decrease from 26.4% in 2013. By urgency status, 2.1% underwent transplant as status 1A, 9.7% with an allocation MELD score of 40 or greater, 10.8% with a MELD of 35-39, 26.8% with a MELD of 25-34, 30.9% with a MELD of 15-24, and 19.5% with a MELD of 14 or lower. Compared with 2013, there were more recipients in the lowest urgency category (MELD 14 or lower) and fewer in the higher urgency categories (MELD 25 or higher). Waiting times were also shorter: 63.8% of transplant recipients waited less than 90 days, 10.8% waited 3-<6 months, 13.6% waited 6-<12 months, 7.2% waited 1-<2 years, and 4.5% waited 2 years or longer (Table LI 8).

The proportion of female adult liver transplant recipients has increased: 38.8% in 2023 compared with 34.1% in 2013 (Figure LI 36). In 2023, 69.8% of liver transplant recipients were White, 17.5% Hispanic, 6.6% Black, 4.1% Asian, 0.6% Multiracial, and 0.4% unreported (Figure LI 37). With the decline in hepatitis C and the rise in alcohol-associated liver disease and MASH, the age composition of the waiting list has also shifted, with a higher proportion of recipients aged 35-49 years (23.3% in 2023 versus 16.5% in 2013) and 65 years or older (22.4% versus 16.3%), and fewer aged 50-64 years (46.8% versus 61.6%) (Table LI 6). There were more recipients in the highest BMI categories: BMI of 35 kg/m^2 or greater, 15.5% in 2023 versus 13.0% in 2013; BMI of 30-<35 kg/m^2, 22.4% in 2023 versus 21.5% in 2013. Only 3.4% of recipients had a history of previous transplant in 2023, compared with 5.0% in 2013 (Figure LI 40 and Table LI 8).

Most adult liver transplants were covered by private insurance (50.9%), followed by Medicare (26.0%) and Medicaid (19.0%). Based on the rural-urban commuting area designation of their home zip code, 82.8% of recipients lived in metropolitan areas; 56.7% lived less than 50 miles from the transplant center, 17.6% within 50-<100 miles, 9.2% within 100-<150 miles, 7.8% within 150-<250 miles, and 7.9% 250 miles or farther (Table LI 6).

Recipients of livers procured from DCD made up 16.7% of adult liver transplants in 2023, a marked increase from preceding years (11.3% in 2022 and 5.2% in 2013), likely related to the wider availability of machine perfusion technologies (Figure LI 39). Only 91 adults received split deceased donor livers (0.9% of adult transplants in 2023 versus 1.2% in 2013) (Table LI 8).

In 2023, there were 804 adult recipients of SLK transplants, representing 7.9% of total liver transplants, a decrease from 9.6% in 2017 when standardized medical eligibility criteria were introduced (Figure LI 41). By far, patients who received SLK transplant were listed via the chronic kidney disease criteria (89.7%) rather than acute kidney injury (9.5%) or metabolic disease (0.2%) criteria (Figure LI 42).

Regarding immunosuppression, induction therapy was used in 30.3% of adult liver transplants, most often with interleukin-2 receptor antibody alone (24.6%) rather than T-cell–depleting agent alone (5.6%) (Figure LI 46 and Figure LI 47). Most liver transplant recipients were maintained on a combination of tacrolimus, a mycophenolate agent, and steroids (66.6%); the remainder used tacrolimus and mycophenolate (18.7%), tacrolimus and steroids (4.4%), or another regimen (10.3%) (Figure LI 48).

2.4 Outcomes

Graft failure was reported in 6.3% at 6 months and 7.9% at 1 year for adult deceased donor liver transplant recipients in 2022, 15.8% at 3 years for those who received transplant in 2020, 20.7% at 5 years for those who received transplant in 2018, and 36.7% at 10 years for those who received transplant in 2013 (Figure LI 49). Overall patient outcomes followed a similar pattern, with mortality of 5.0% at 6 months, 6.5% at 1 year, 14.0% at 3 years, 19.0% at 5 years, and 34.6% at 10 years (Figure LI 51). Over the past decade, 1-year graft failure and patient mortality have decreased; these values were 7.9% and 6.5%, respectively, for patients who underwent transplant in 2022, compared with 11.6% and 10.0%, respectively, for patients who underwent transplant in 2012.

Among adult recipients of deceased donor liver transplant in 2016-2018, 5-year graft survival was 75.0% among those aged 65 years or older, 79.3% for those aged 50-64 years, 82.2% for those aged 35-49 years, and 80.8% for those aged 18-34 years. Patient survival followed a similar pattern (Figure LI 52 and Figure LI 68). Lower graft and patient survival were observed in Black or Other race compared with White, Hispanic, or Asian recipients (Figure LI 53 and Figure LI 69); in male recipients compared with female recipients (Figure LI 54 and Figure LI 70); in those with HCC and MASH diagnoses compared with other etiologies (Figure LI 55 and Figure LI 71); and in those with MELD scores of 40 or greater compared with other MELD categories (39 or lower) (Figure LI 56 and Figure LI 72). In all of these groups, however, 5-year graft survival still exceeded 74.1% and patient survival exceeded 75.1%. The 5-year graft survival rate for DCD livers was 75.6%, compared with 79.2% for recipients of donation after brain death (DBD) livers (Figure LI 57). Liver transplant recipients with HCC MELD exception had a 5-year graft survival rate of 77.6% compared with 79.3% in those without (Figure LI 58).

Graft outcomes among adult living donor liver transplant recipients were: 5.8% graft failure at 6 months, 6.6% at 1 year, 13.6% at 3 years, 21.2% at 5 years, and 40.8% at 10 years (Figure LI 50). The 5-year graft survival among living donor liver transplant recipients was highest for cholestatic liver disease (85.8%) and MASH (83.9%) compared with other etiologies (Figure LI 63). Older recipients (aged 65 years or older) had worse outcomes, with 5-year patient survival of 76.6%, compared with 82.5%, 88.4%, and 94.2% for ages 50-64 years, 35-49 years, and 18-34 years, respectively (Figure LI 73).

Among liver recipients who underwent transplant in 2022, the incidence of reported acute rejection by 1-year posttransplant was higher among younger adult patients (20.2% for age 18-34 years) and lower among older patients (7.9% for age 65 years or older) (Figure LI 65). Among adult liver transplant recipients in 2012-2018, the 5-year incidence of reported posttransplant lymphoproliferative disorder was 1.0% in those with positive Epstein-Barr virus (EBV) status at transplant and 2.0% in those with negative EBV status at transplant (Figure LI 67).

In 2023, there were 734 new adult registrations for kidney transplant with a previous history of liver transplant (Figure LI 43). The most common primary diagnosis for kidney transplant waitlisting among liver transplant recipients was hepatorenal syndrome (31.1%), followed by Other/unknown (20.2%), diabetes (17.7%), and calcineurin nephrotoxicity (16.6%) (Figure LI 44). Hepatorenal syndrome as the primary diagnosis for kidney transplant increased notably after implementation of the SLK eligibility criteria, as did safety net listings (i.e., waitlisting 60-<365 days after liver transplant), which accounted for 327 (44.6%) new registrations (Figure LI 43). There were 418 kidney transplants in recipients with a previous history of liver transplant, which increased from 274 since implementation of the safety net policy in 2017 (Figure LI 45).

3 Donation

In 2023, there were 10,967 deceased donors whose livers were recovered for transplant in the United States, another record high (Figure LI 77). This was driven by an increase in older donors (40 years or older), whereas the number of younger donors (pediatric [younger than 18 years] and adults aged 18-29 years) remained relatively stable (Figure LI 78). Only 5.8% of donors were younger than 18 years, 17.1% were aged 18-29 years, 20.3% were aged 30-39 years, 29.5% were aged 40-54 years, and 27.3% were aged 55 years or older (Figure LI 80). The sex distribution of donors was similar to that in previous years at 61.3% male and 38.7% female (Figure LI 81), as were racial and ethnic categories: 62.3% White, 18.3% Black, 14.9% Hispanic, 3.0% Asian, and 1.5% Other (Figure LI 82). Although the absolute number of HCV-positive donors by antibody testing increased, the overall proportion decreased, representing 8.8% of deceased liver donors in 2023 (Figure LI 79); by both antibody and nucleic acid test (NAT), 91.1% of donor livers were HCV negative (Ab-/NAT-), 4.7% were HCV positive with Ab+ and NAT-, and 4.1% were HCV positive with NAT+ (Figure LI 83). The most common cause of death among liver donors remained anoxia (49.0%), followed by cerebrovascular accident/stroke (24.4%), head trauma (23.8%), and Other/unknown (2.7%) (Figure LI 86).

In recent years, the proportion of DCD livers recovered for transplant has increased markedly, representing 20.1% of deceased donor livers in 2023, an increase from 14.1% in 2022, 9.0% in 2018, and 6.3% in 2013 (Figure LI 84); this increase reflects the promise of machine perfusion technologies to reduce the risk of ischemic injury and improve organ viability. There are also more donor livers undergoing liver biopsy (46.9%): 37.1% reported <11% macrovesicular steatosis, 7.0% had 11%-<31%, and 2.8% had 31% or greater (Figure LI 85).

The overall nonuse rate, or percent of livers recovered for transplant and not transplanted, was 9.7% in 2023, a slight decrease from the preceding years (Figure LI 87). Livers from older donors were less likely to be used (12.4% for donors aged 55 years or older versus 3.2% for donors younger than 18 years) (Figure LI 88). Livers from HCV NAT+ donors were more likely to be recovered and not transplanted (14.6%), while livers from HCV Ab+/NAT- donors were used at similar rates as those from HCV Ab-/NAT- donors (9.2% and 9.5%, respectively) (Figure LI 92). Livers with risk factors present by US Public Health Service criteria were not less likely to be used than those without this designation (Figure LI 93). DCD livers remained over three times more likely than DBD livers to be unused in 2023, although the nonuse rate for both DCD and DBD livers has decreased compared with years prior (22.8% in 2023 versus 30.0% in 2019 for DCD; 6.5% in 2023 versus 7.1% in 2019 for DBD) (Figure LI 94). Livers with macrovesicular steatosis were also more likely to be unused (48.7% for livers with 31% fat or greater, 18.2% for livers with 11%-<31% fat, and 10.1% for livers with <11% fat versus 6.4% for livers without liver biopsy values reported) (Figure LI 95).

There were 659 living donors in 2023, an all-time high—driven by an increase in directed (26.3%) or Other donor-relation donors (15.2%), while the number of donor livers from related persons or spouses/partners remained stable (Figure LI 96). A small but growing number of living donor liver transplants (49 [7.5%]) were performed as a paired donation. Living donors were somewhat older than in previous years, with 7.3% aged 55 years or older, 35.8% aged 40-54 years, 36.7% aged 30-39 years, 20.0% aged 18-29 years, and 0.2% younger than 18 years (Figure LI 97). Living donors were more likely to be female than male (59.2% versus 40.8%) and more likely to be White than all other races and ethnicities combined (76.8% versus 23.2%) (Figure LI 98 and Figure LI 99).

Living donors most often donated the right lobe (80.7%), an increasing trend; 1.1% of living liver donations were recorded as whole domino (Figure LI 100).

4 Pediatric Liver Transplant

4.1 Waiting List

In 2023, there were 1,142 pediatric liver candidates listed at any time during the year in the United States, with 438 on the list at the start of the year and 704 candidates newly added (Figure LI 102, Figure LI 103, and Table LI 12). Over the course of the year, 738 patients were removed (63.7% for deceased donor liver transplant, 10.7% for living donor liver transplant, 3.7% for death, 1.8% for being too sick, 13.0% for improved condition, and 7.2% for a reason of Other), leaving 404 candidates still waiting, an 8% decrease from 2022 (Table LI 12 and Table LI 13).

Among pediatric candidates, the largest age category was 1-5 years (35.3%), followed by 12-17 years (21.3%) and younger than 1 year (18.4%) (Figure LI 104). Compared with 10 years ago, there were more liver transplant candidates in the younger age groups and fewer adolescents (12 years or older). In terms of race and ethnicity, most candidates were White (45.9%), followed by Hispanic (24.8%), Black (17.5%), Asian (6.0%), Multiracial (4.1%), Native American (1.1%), and unreported (0.7%) (Figure LI 105).

Among pediatric candidates who had been listed in 2018-2020, 66.9% received deceased donor liver transplant, 9.8% received living donor liver transplant, 3.2% died, 16.0% were removed from the list, and 4.1% were still waiting (Figure LI 107).

Overall deceased donor liver transplant rates for pediatric candidates remained stable in 2023 at 110.9 transplants per 100 patient-years, which had increased slightly after the allocation policy change in 2020 that prioritized pediatric donors for pediatric recipients (Figure LI 108). Deceased donor liver transplant rates were higher for age younger than 1 year compared with other pediatric age categories (Figure LI 109) and for the White, Black, and Other racial and ethnic categories compared with the Hispanic and Asian categories (Figure LI 110).

The overall pretransplant mortality rate for pediatric candidates in 2023 was 6.7 deaths per 100 patient-years, which has increased from a nadir of 4.9 deaths per 100 patients-years in 2020 (Figure LI 111). Despite the higher deceased donor liver transplant rate, the highest pretransplant mortality was still among the age category of younger than 1 year (18.6 deaths per 100 patient-years; Figure LI 112) and among the Black and Other racial and ethnic categories compared with Hispanic, White, and Asian (Figure LI 113).

4.2 Transplants

In 2023, there were 534 pediatric liver transplants, including retransplant and multiorgan recipients, a relatively stable number compared with previous years (Figure LI 114). The proportion of living donation has increased, to 14.6% in 2023 from 7.7% in 2013 (Figure LI 115).

Among all pediatric liver transplant recipients, 24.9% were aged younger than 1 year, 37.6% were aged 1-5 years, 16.9% were aged 6-11 years, and 20.6% were aged 12-17 years (Figure LI 116 and Table LI 14). The most common diagnosis among these liver transplant recipients was biliary atresia (37.5%), followed by Other/unknown etiologies (24.3%), metabolic disease (14.8%), hepatoblastoma (8.1%), acute liver failure (8.1%), and other cholestatic disease (7.3%) (Table LI 15). Similar to adult liver transplant, the proportion of pediatric recipients with a history of prior transplant decreased: 4.5% in 2023 compared with 8.8% in 2013 (Table LI 16).

In terms of medical urgency, 9.7% of pediatric liver transplant recipients underwent transplant as status 1A; 22.3%, as status 1B; 1.7%, with MELD or PELD score of 40 or greater; 1.1%, with MELD/PELD of 35-39; 6.7%, with MELD/PELD of 25-34; 16.9%, with MELD/PELD of 15-24; and 41.2%, with MELD/PELD of 14 or lower (Table LI 15). In July 2023, criteria for status 1B qualification were updated, along with PELD updating to PELD-creatinine, but this has not yet resulted in any detectable changes in the medical urgency status. Among pediatric liver transplant recipients in 2023, 65.4% had waited fewer than 90 days, 16.7% waited for 3-<6 months, 10.5% waited for 6-<12 months, 5.1% waited for 1-<2 years, and 2.4% waited for 2 or more years (Table LI 16). Overall, waiting times were shorter than in 2013. The proportion of ABO-incompatible pediatric transplants increased: 4.3% in 2023 compared with 3.6% in 2013. No pediatric patients received a liver from a DCD donor in 2023. Most received whole liver (58.4%), followed by partial liver (21.7%) and split liver (19.9%), which represents a modest increase in split liver acceptance since 2021 (Figure LI 117).

The most common insurance among pediatric liver transplant recipients in 2023 was Medicaid (50.7%), followed by private insurance (37.8%). Based on the recipient's reported zip code, 82.5% lived in metropolitan areas; 45.1% lived less than 50 miles from the transplant center, 16.7% within 50-<100 miles, 12.2% within 100-<150 miles, 11.4% within 150-<250 miles, and 12.9% were 250 miles or farther (Table LI 14).

Induction agent use was reported in 40.4% of pediatric liver transplant recipients in 2023 (Figure LI 118), more commonly interleukin-2 receptor antibody than T-cell depleting agent (Figure LI 119). The most common immunosuppression regimens were combination tacrolimus, mycophenolate, and steroids (39.0%) or combination tacrolimus and steroids (36.9%) (Figure LI 120).

4.3 Outcomes

Short-term graft failure rates among pediatric deceased donor liver transplant recipients increased in 2022 (9.1% at 6 months and 10.7% at 1 year) compared with previous years (Figure LI 121). Longer-term graft failure rates were similar: 10.6% at 3 years for transplants in 2020, 12.2% at 5 years for transplants in 2018, and 23.1% at 10 years for transplants in 2013. Mirroring the graft outcomes, short-term patient mortality worsened (7.6% at 6 months, 8.9% at 1 year, and 9.8% at 3 years) compared with previous years (Figure LI 129). Longer-term outcomes were stable or improved, with a patient mortality rate of 7.6% at 5 years for transplants in 2018 and 13.9% at 10 years for transplants in 2013.

Among pediatric deceased donor liver transplant recipients in 2016-2018, overall 5-year graft survival was 86.3% and patient survival was 90.3% (Figure LI 126, Figure LI 130, and Figure LI 133). Five-year graft survival was highest among those with metabolic disease (95.5%), biliary atresia (90.9%), or other cholestatic disease (87.6%) and was lowest among those with a diagnosis of hepatoblastoma (78.8%) or Other/unknown etiology (75.0%) (Figure LI 124).

Outcomes among pediatric living donor liver transplant recipients were superior to those among deceased donor recipients, with graft failure rates of 5.7% at 6 months, 5.7% at 1 year, 7.6% at 3 years, 9.7% at 5 years, and 17.1% at 10 years (Figure LI 122); overall 5-year patient survival was 94.9% (Figure LI 126 and Figure LI 133).

The incidence of acute rejection by 1-year posttransplant for pediatric liver transplant recipients was 19.3% for those younger than 1 year, 25.9% for ages 1-5 years, 22.9% for ages 6-11 years, and

15.6% for ages 12-17 years (Figure LI 127), rates that are higher than those observed in adult recipients. The overall 5-year incidence of posttransplant lymphoproliferative disorder for pediatric recipients who underwent transplant in 2012-2018 was 4.4%: 5.2% among those with EBV-negative status and 3.1% among those with EBV-positive status at the time of transplant (Figure LI 128).

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Publications based on data in this report or supplied on request must include a citation and the following statement: The data and analyses reported in the *OPTN/SRTR 2023 Annual Data Report* have been supplied by the United Network for Organ Sharing and Hennepin Healthcare Research Institute under contract with HHS/HRSA. The authors alone are responsible for reporting and interpreting these data; the views expressed herein are those of the authors and not necessarily those of the U.S. government.

This report is available at https://srtr.transplant.hrsa.gov/annualdatareports. Individual chapters may be downloaded.

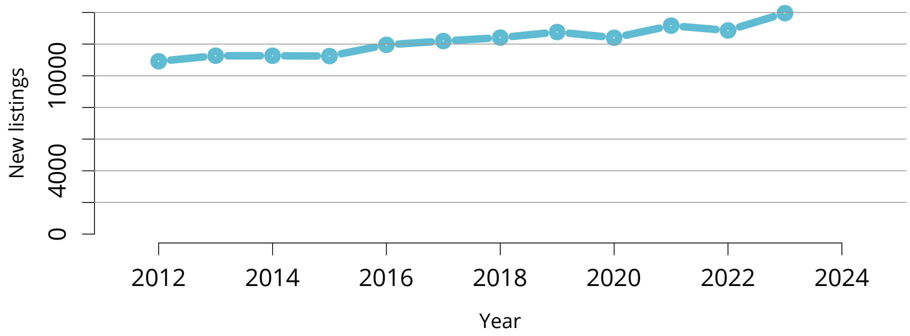

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Figure LI 1: New adult candidates added to the liver transplant waiting list. A new candidate is one who first joined the list during the given year, without having been listed in a previous year. Previously listed candidates who underwent transplant and subsequently relisted are considered new. Active and inactive patients are included.

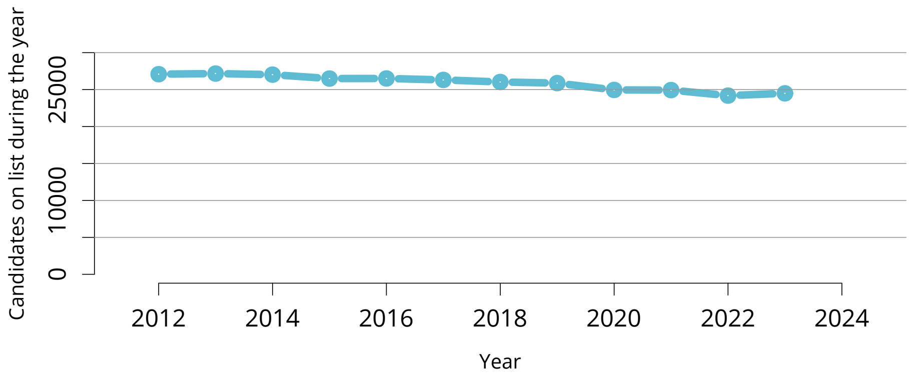

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Figure LI 2: All adult candidates on the liver transplant waiting list. Adult candidates on the list at any time during the year. Candidates listed at more than one center are counted once per listing.

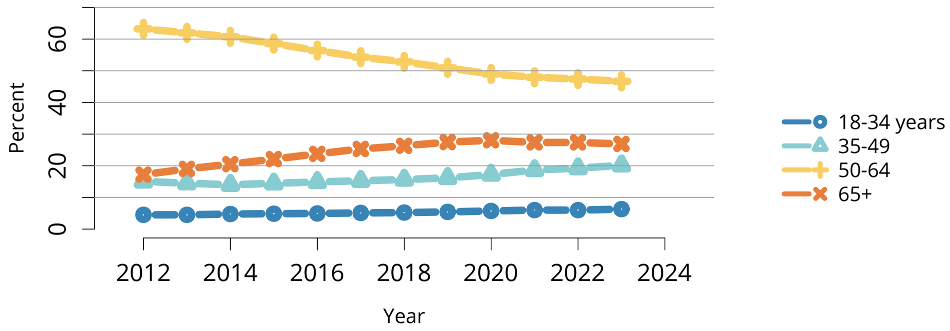

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Figure LI 3: Distribution of adults waiting for liver transplant by age. Candidates waiting for transplant at any time in the given year. Candidates listed at more than one center are counted once per listing. Active and inactive candidates are included. Age is determined at the earliest of transplant, death, removal, or December 31 of the year.

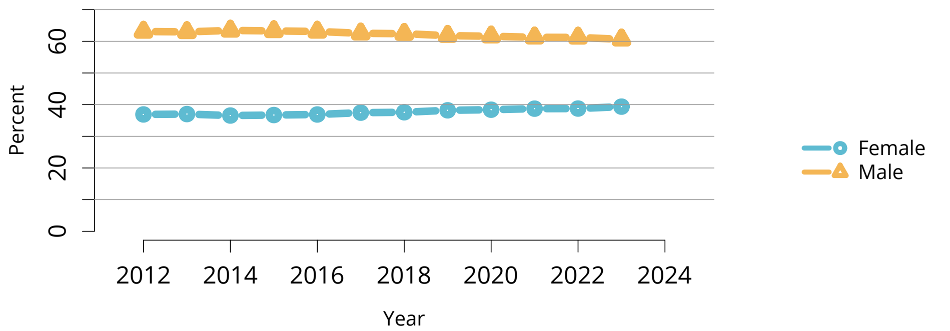

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Figure LI 4: Distribution of adults waiting for liver transplant by sex. Candidates waiting for transplant at any time in the given year. Candidates listed at more than one center are counted once per listing. Active and inactive patients are included.

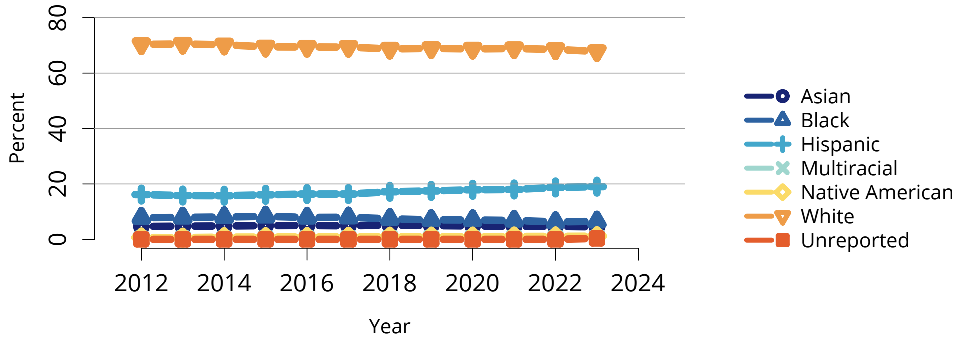

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Figure LI 5: Distribution of adults waiting for liver transplant by race and ethnicity. Candidates waiting for transplant at any time in the given year. Candidates listed at more than one center are counted once per listing. Active and inactive patients are included.

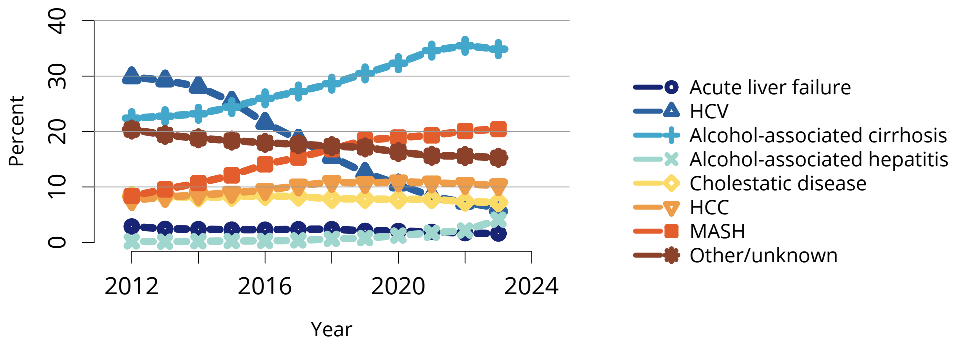

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Figure LI 6: Distribution of adults waiting for liver transplant by diagnosis. Candidates waiting for transplant at any time in the given year. Candidates listed at more than one center are counted once per listing. Active and inactive patients are included. HCC, hepatocellular carcinoma; HCV, hepatitis C virus; MASH, metabolic dysfunction–associated steatohepatitis.

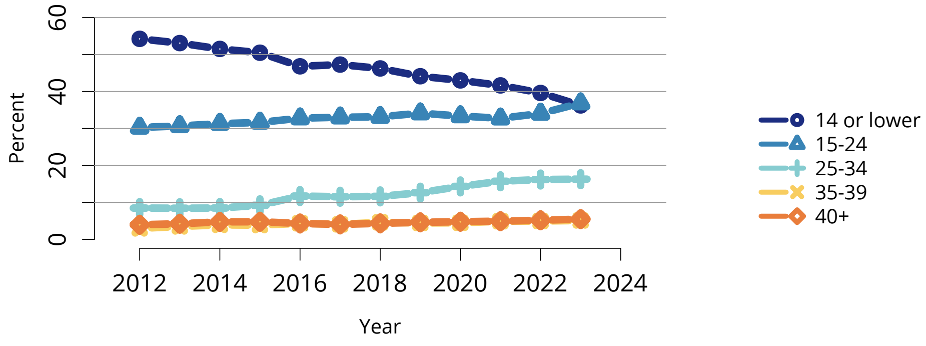

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Figure LI 7: Distribution of adults waiting for liver transplant by last laboratory MELD score in the year. Candidates waiting for transplant at any time in the given year. Candidates listed at more than one center are counted once per listing. Active and inactive patients are included. MELD, model for end-stage liver disease.

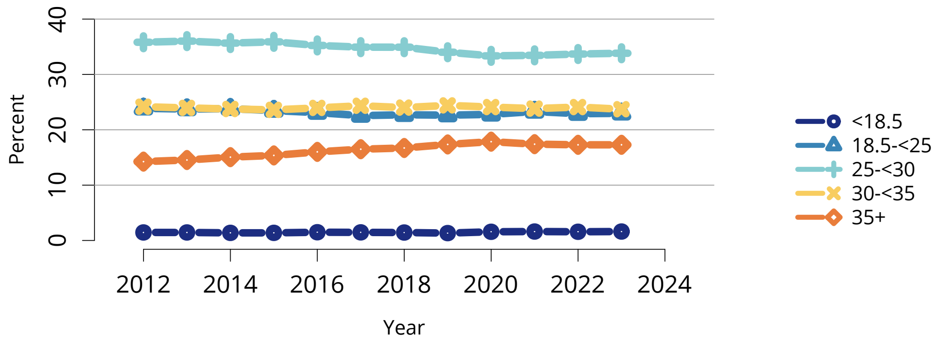

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Figure LI 8: Distribution of adults waiting for liver transplant by BMI. Candidates waiting for transplant at any time in the given year. Candidates listed at more than one center are counted once per listing. Active and inactive patients are included. BMI, body mass index.

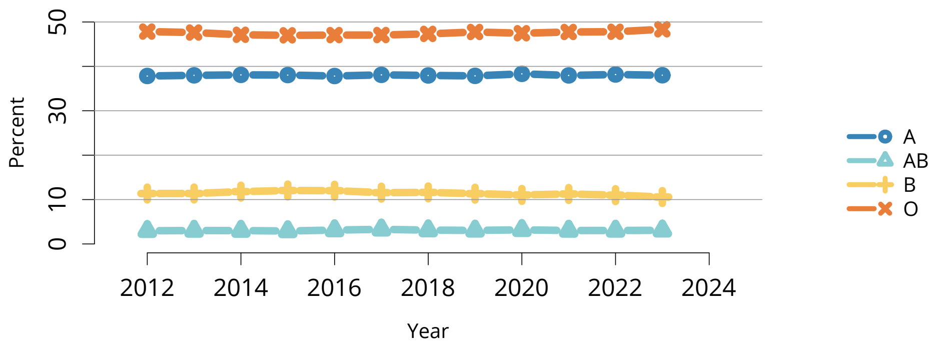

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Figure LI 9: Distribution of adults waiting for liver transplant by blood type. Candidates waiting for transplant at any time in the given year. Candidates listed at more than one center are counted once per listing. Active and inactive patients are included.

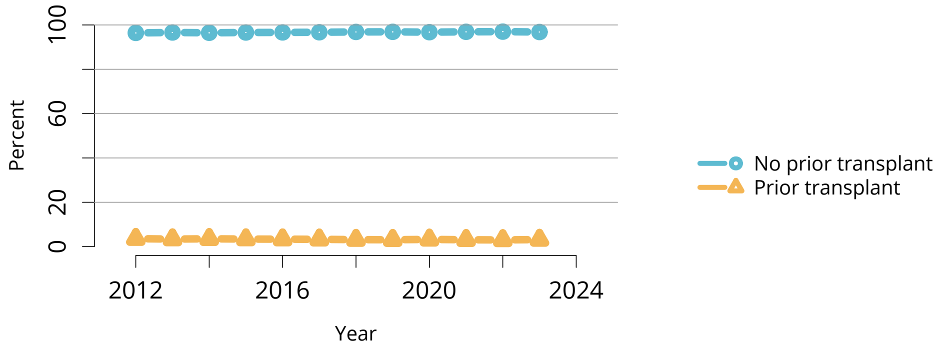

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Figure LI 10: Distribution of adults waiting for liver transplant by prior transplant status. Candidates waiting for transplant at any time in the given year. Candidates listed at more than one center are counted once per listing. Active and inactive patients are included.

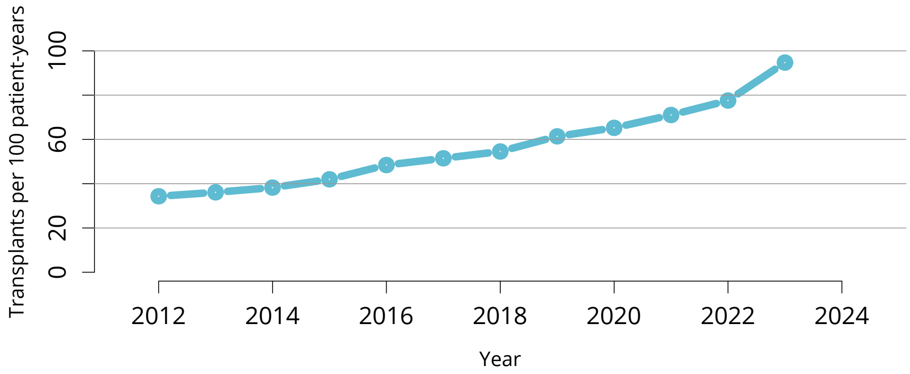

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Figure LI 11: Overall deceased donor liver transplant rates among adult waitlist candidates. Transplant rates are computed as the number of deceased donor transplants per 100 patient-years of waiting time in a given year. Individual listings are counted separately.

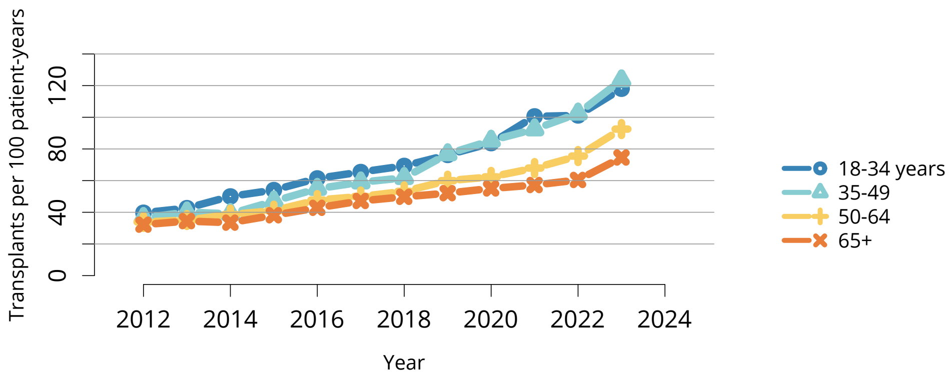

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Figure LI 12: Deceased donor liver transplant rates among adult waitlist candidates by age. Transplant rates are computed as the number of deceased donor transplants per 100 patient-years of waiting time in a given year. Individual listings are counted separately. Age is determined at the later of listing date or January 1 of the given year.

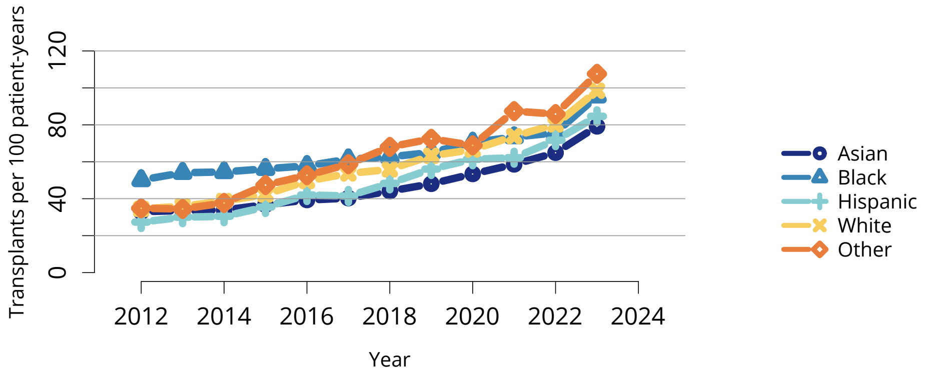

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Figure LI 13: Deceased donor liver transplant rates among adult waitlist candidates by race and ethnicity. Transplant rates are computed as the number of deceased donor transplants per 100 patient-years of waiting time in a given year. Individual listings are counted separately. The Other race category is composed of Native American and Multiracial categories.

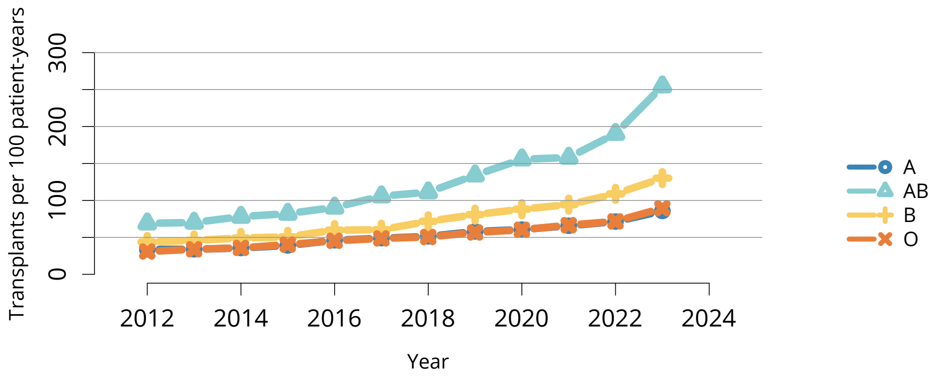

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Figure LI 14: Deceased donor liver transplant rates among adult waitlist candidates by blood type. Transplant rates are computed as the number of deceased donor transplants per 100 patient-years of waiting time in a given year. Individual listings are counted separately.

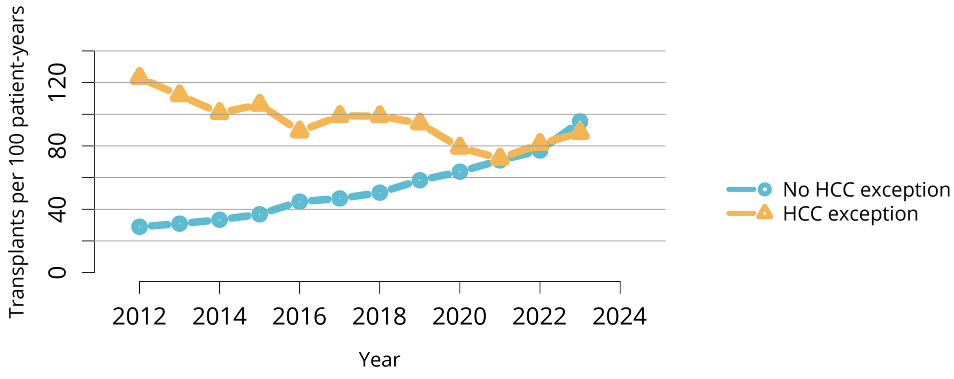

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Figure LI 15: Deceased donor liver transplant rates among adult waitlist candidates by HCC exception status. Transplant rates are computed as the number of deceased donor transplants per 100 patient-years of waiting time in a given year. Individual listings are counted separately. HCC is determined at the later of listing date or January 1 of the year. HCC, hepatocellular carcinoma.

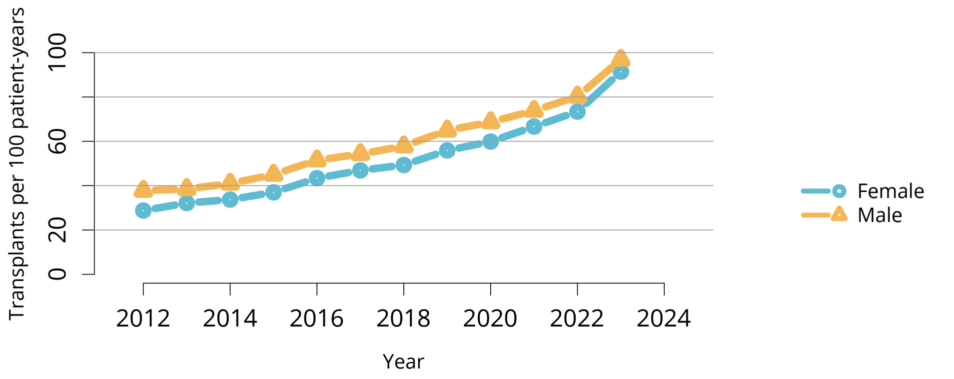

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Figure LI 16: Deceased donor liver transplant rates among adult waitlist candidates by sex. Transplant rates are computed as the number of deceased donor transplants per 100 patient-years of waiting time in a given year. Individual listings are counted separately.

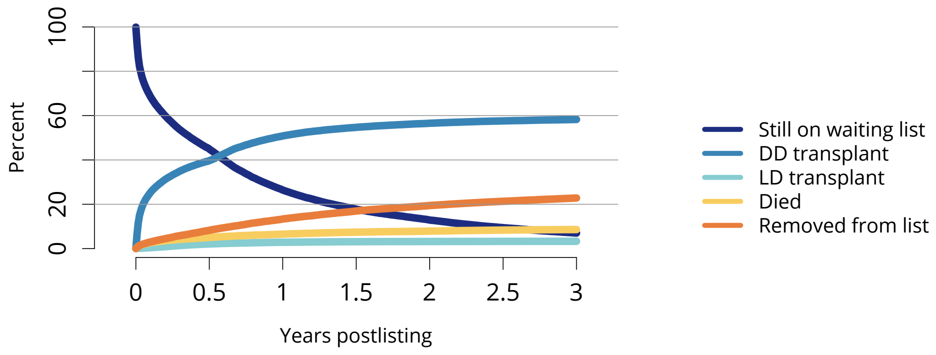

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Figure LI 17: Three-year outcomes for adults waiting for liver transplant, new listings in 2018-2020. Candidates listed at more than one center are counted once per listing. Removed from list includes all reasons except transplant and death. DD, deceased donor; LD, living donor.

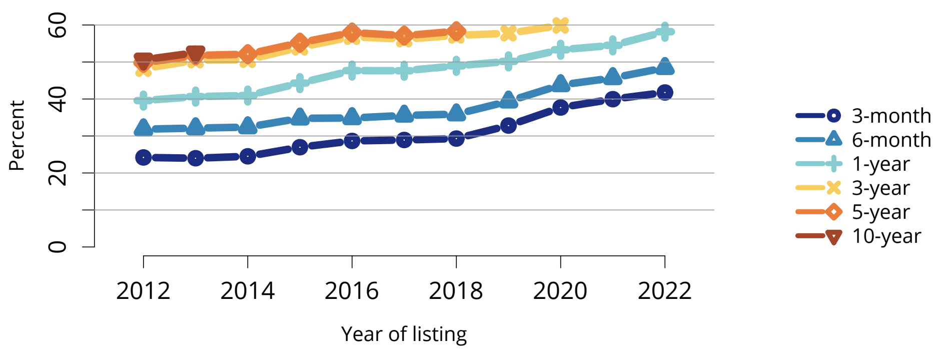

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Figure LI 18: Percentage of adults who underwent deceased donor liver transplant within a given period of listing. Candidates listed at more than one center are counted once per listing.

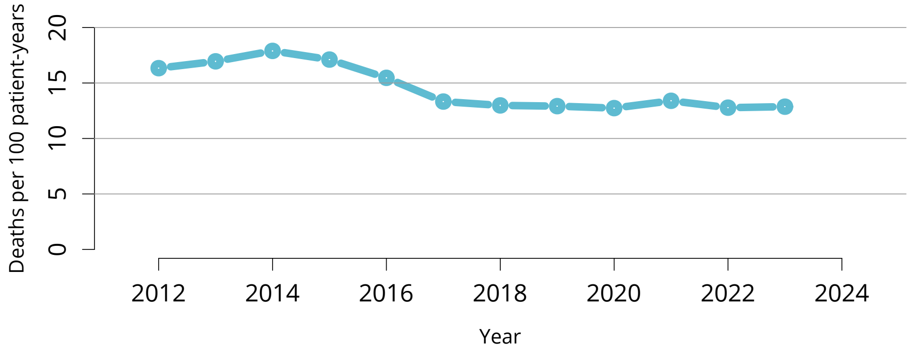

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Figure LI 19: Overall pretransplant mortality rates among adults waitlisted for liver transplant. Mortality rates are computed as the number of deaths per 100 patient-years of waiting time in the given year. Waiting time is censored at transplant, death, transfer to another program, removal because of improved condition, or end of cohort. Individual listings are counted separately.

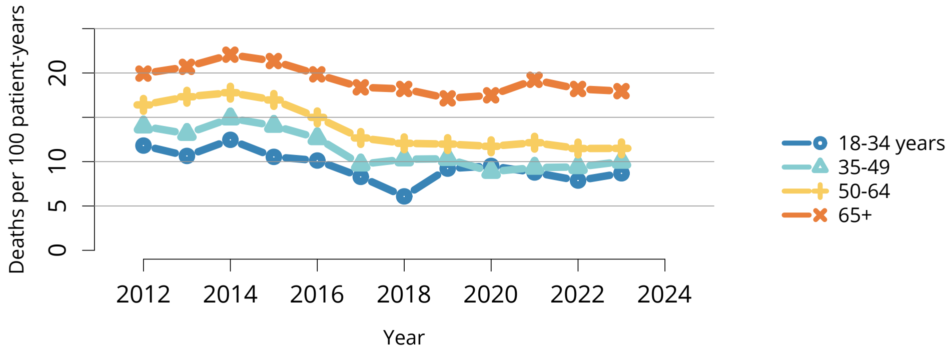

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Figure LI 20: Pretransplant mortality rates among adults waitlisted for liver transplant by age. Mortality rates are computed as the number of deaths per 100 patient-years of waiting time in the given year. Waiting time is censored at transplant, death, transfer to another program, removal because of improved condition, or end of cohort. Individual listings are counted separately. Age is determined at the later of listing date or January 1 of the given year.

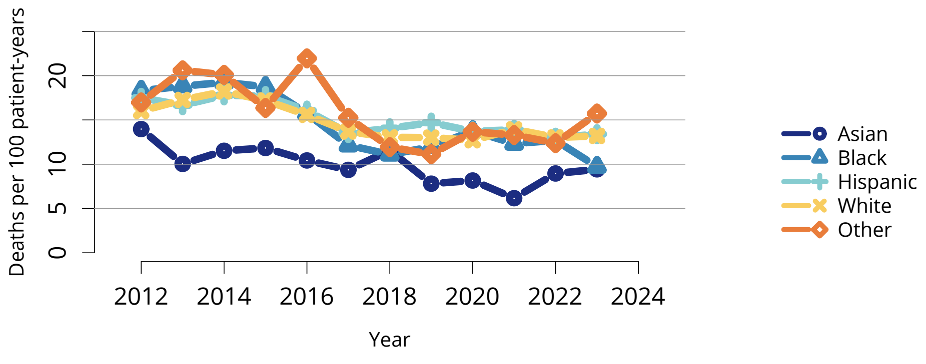

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Figure LI 21: Pretransplant mortality rates among adults waitlisted for liver transplant by race and ethnicity. Mortality rates are computed as the number of deaths per 100 patient-years of waiting time in the given year. Waiting time is censored at transplant, death, transfer to another program, removal because of improved condition, or end of cohort. Individual listings are counted separately. The Other race category is composed of Native American and Multiracial categories.

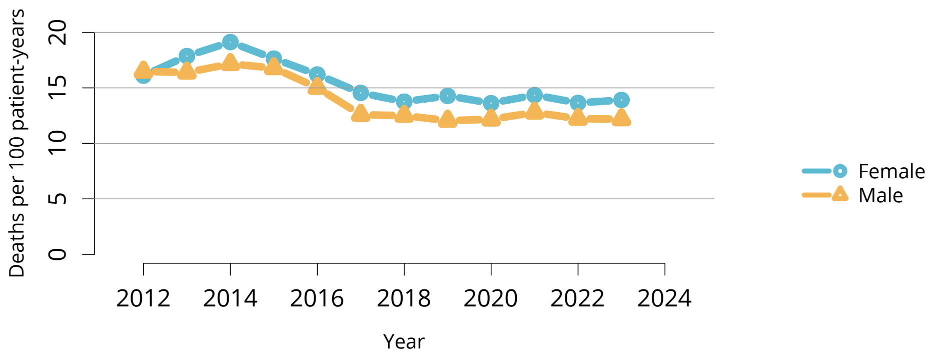

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Figure LI 22: Pretransplant mortality rates among adults waitlisted for liver transplant by sex. Mortality rates are computed as the number of deaths per 100 patient-years of waiting time in the given year. Waiting time is censored at transplant, death, transfer to another program, removal because of improved condition, or end of cohort. Individual listings are counted separately.

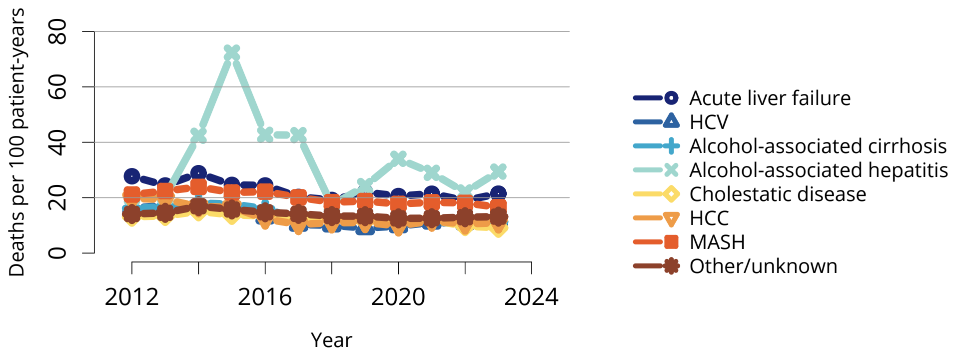

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Figure LI 23: Pretransplant mortality rates among adults waitlisted for liver transplant by diagnosis. Mortality rates are computed as the number of deaths per 100 patient-years of waiting time in the given year. Waiting time is censored at transplant, death, transfer to another program, removal because of improved condition, or end of cohort. Individual listings are counted separately. HCC, hepatocellular carcinoma; HCV, hepatitis C virus; MASH, metabolic dysfunction–associated steatohepatitis.

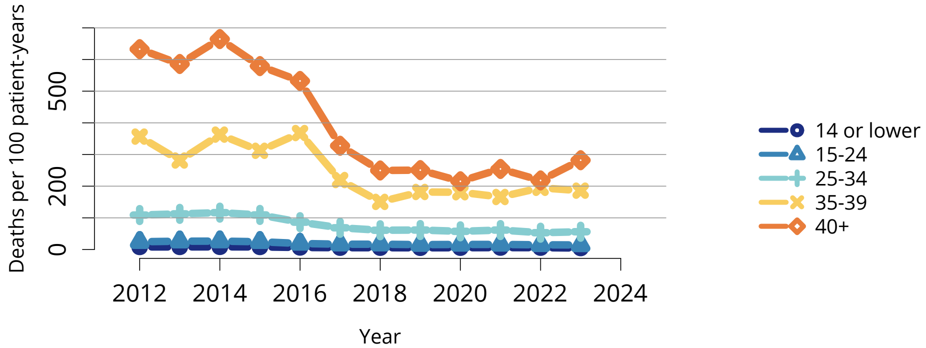

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Figure LI 24: Pretransplant mortality rates among adults waitlisted for liver transplant by first laboratory MELD score in the year. Mortality rates are computed as the number of deaths per 100 patient-years of waiting time in the given year. Waiting time is censored at transplant, death, transfer to another program, removal because of improved condition, or end of cohort. Individual listings are counted separately. Medical urgency is determined at the later of listing date or January 1 of the year. MELD, model for end-stage liver disease.

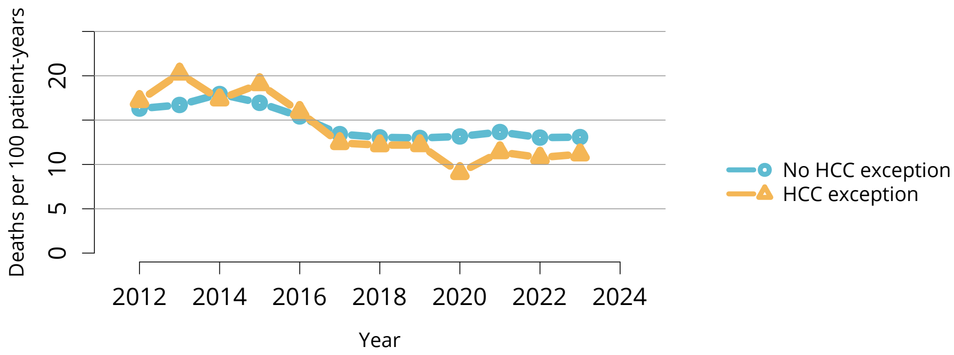

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Figure LI 25: Pretransplant mortality rates among adults waitlisted for liver by HCC exception status. Mortality rates are computed as the number of deaths per 100 patient-years of waiting time in the given year. Waiting time is censored at transplant, death, transfer to another program, removal because of improved condition, or end of cohort. Individual listings are counted separately. HCC is determined at the later of listing date or January 1 of the year. HCC, hepatocellular carcinoma.

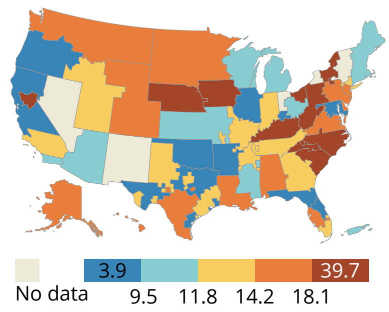

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Figure LI 26: Pretransplant mortality rates among adults waitlisted for liver transplant in 2023 by DSA. Mortality rates are computed as the number of deaths per 100 patient-years of waiting time in the given year. Waiting time is censored at transplant, death, transfer to another program, removal because of improved condition, or end of cohort. Individual listings are counted separately. DSA, donation service area.

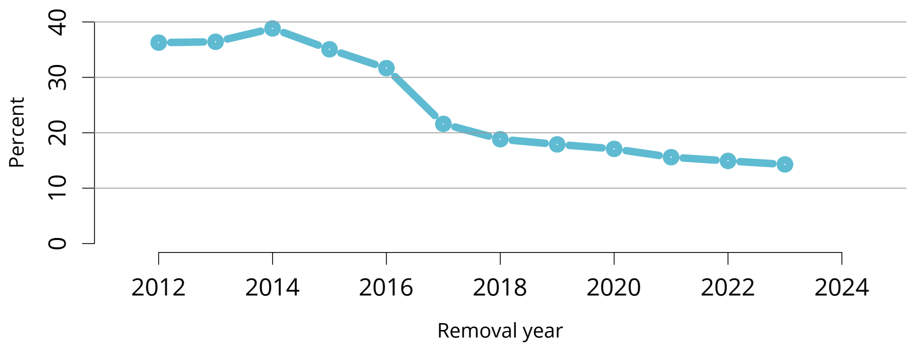

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Figure LI 27: Deaths within 6 months after removal among adult liver waitlist candidates, overall. Denominator includes only candidates removed from the waiting list for reasons other than transplant or death while on the list.

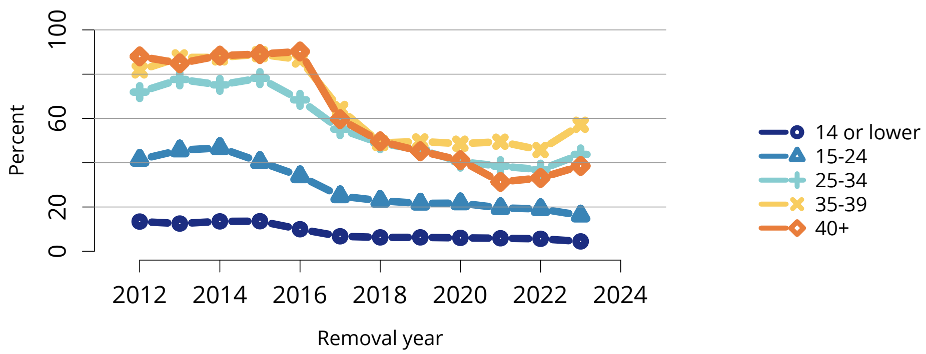

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Figure LI 28: Deaths within 6 months after removal among adult liver waitlist candidates, by laboratory MELD score at removal. Denominator includes only candidates removed from the waiting list for reasons other than transplant or death while on the list. MELD, model for end-stage liver disease.

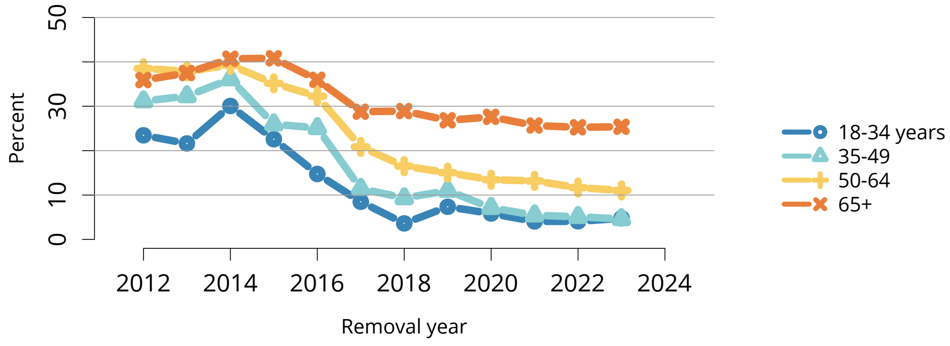

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Figure LI 29: Deaths within 6 months after removal among adult liver waitlist candidates, by age. Denominator includes only candidates removed from the waiting list for reasons other than transplant or death while on the list. Age is determined at removal.

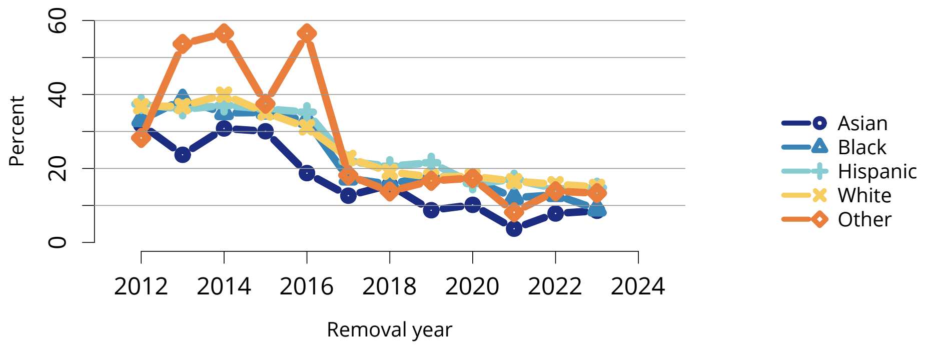

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Figure LI 30: Deaths within 6 months after removal among adult liver waitlist candidates, by race and ethnicity. Denominator includes only candidates removed from the waiting list for reasons other than transplant or death while on the list. The Other race category is composed of Native American and Multiracial categories.

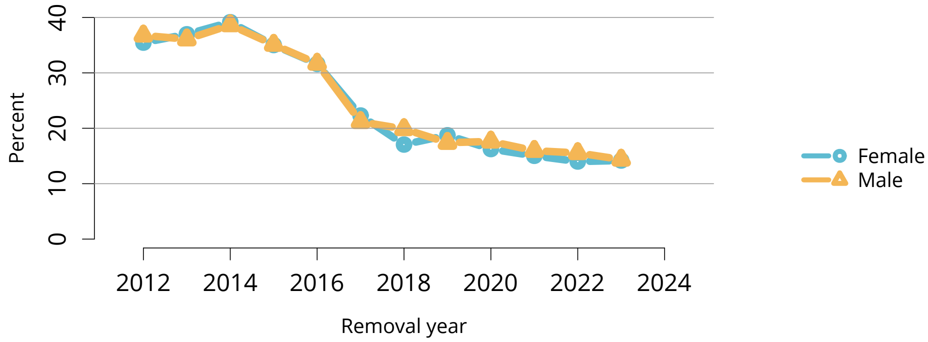

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Figure LI 31: Deaths within 6 months after removal among adult liver waitlist candidates, by sex. Denominator includes only candidates removed from the waiting list for reasons other than transplant or death while on the list.

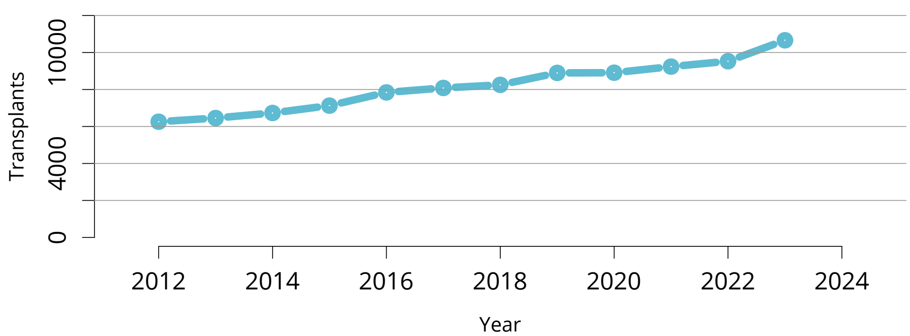

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Figure LI 32: Overall liver transplants. All liver transplant recipients, including adult and pediatric, re-transplant, and multiorgan recipients.

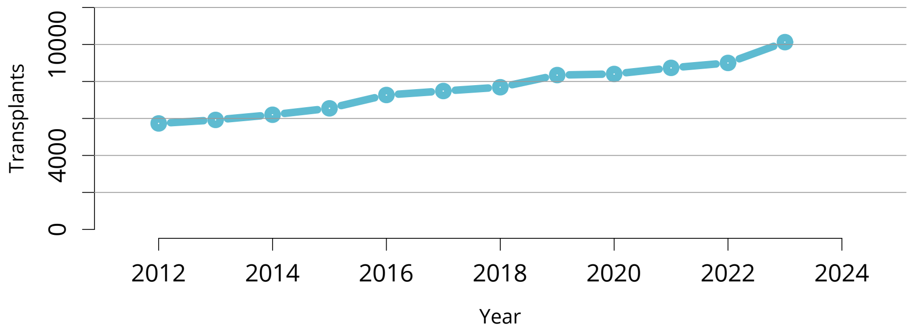

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Figure LI 33: Overall adult liver transplants. All adult liver transplant recipients, including retransplant and multiorgan recipients.

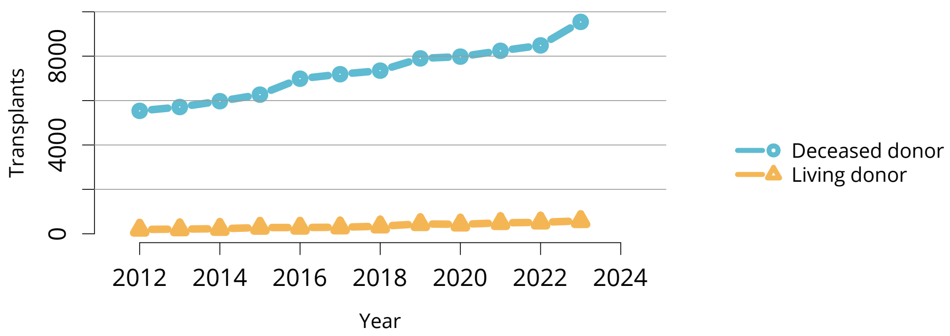

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Figure LI 34: Adult liver transplants by donor type. Adult liver transplant recipients, including retransplant and multiorgan recipients.

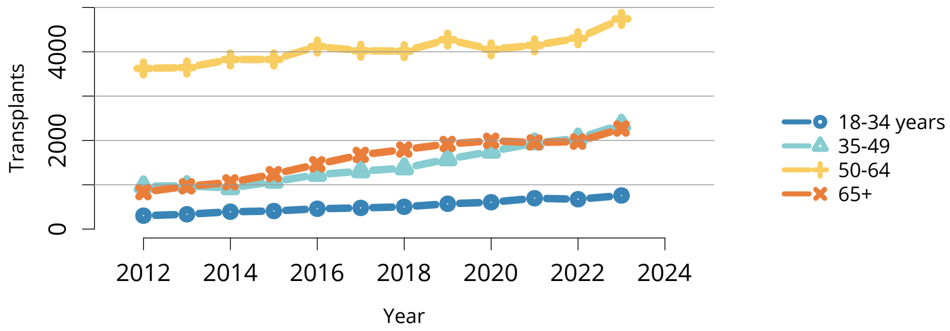

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Figure LI 35: Adult liver transplants by age. Adult liver transplant recipients, including retransplant and multiorgan recipients.

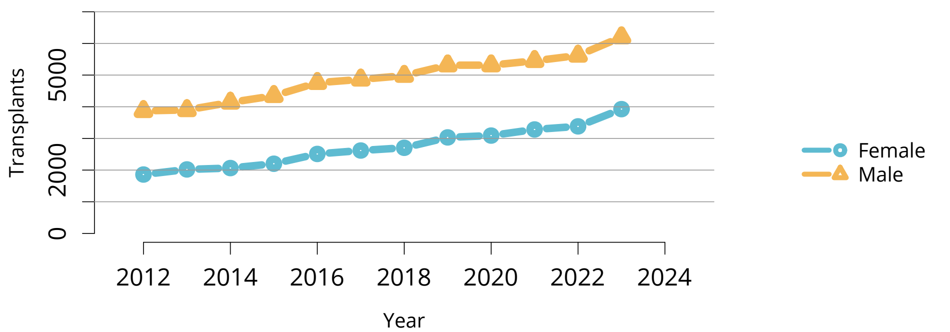

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Figure LI 36: Adult liver transplants by sex. Adult liver transplant recipients, including retransplant and multiorgan recipients.

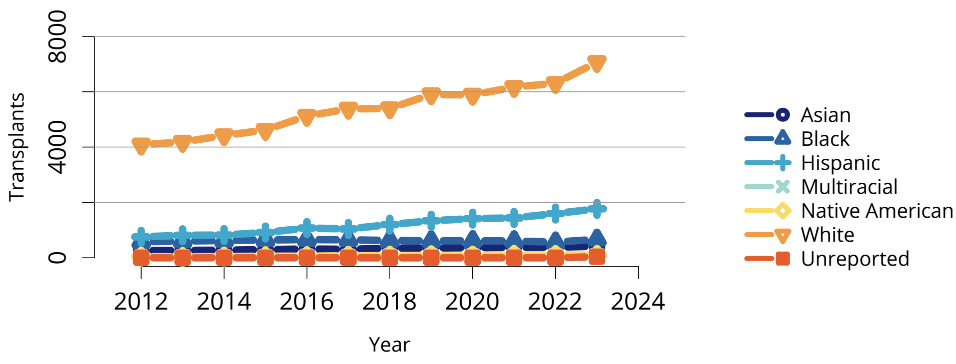

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Figure LI 37: Adult liver transplants by race and ethnicity. Adult liver transplant recipients, including retransplant and multiorgan recipients.

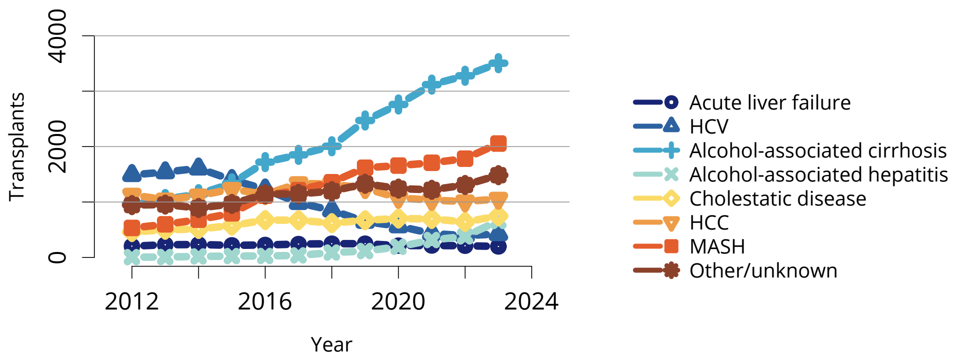

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Figure LI 38: Adult liver transplants by diagnosis. Adult liver transplant recipients, including retransplant and multiorgan recipients. HCC, hepatocellular carcinoma; HCV, hepatitis C virus; MASH, metabolic dysfunction–associated steatohepatitis.

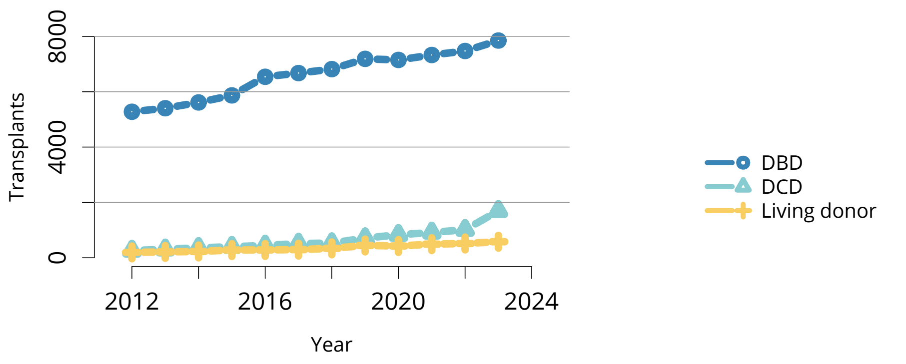

Figure LI 39: Adult liver transplants by DCD status. Adult liver transplant recipients, including retransplant and multiorgan recipients. DBD, donation after brain death; DCD, donation after circulatory death.

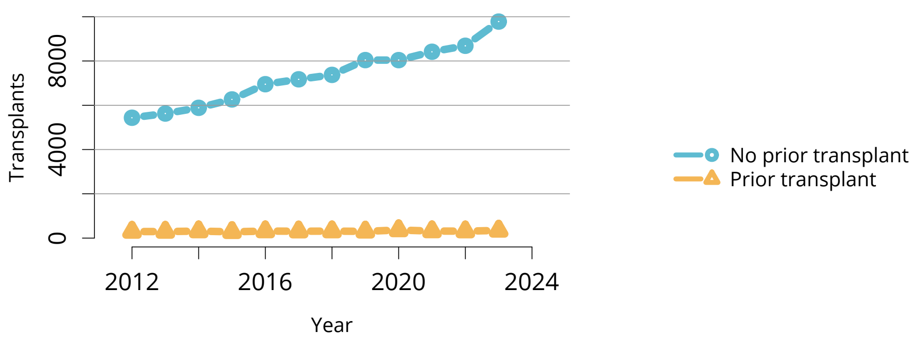

Figure LI 40: Adult liver transplants by prior transplant status. Adult liver transplant recipients, including retransplant and multiorgan recipients.

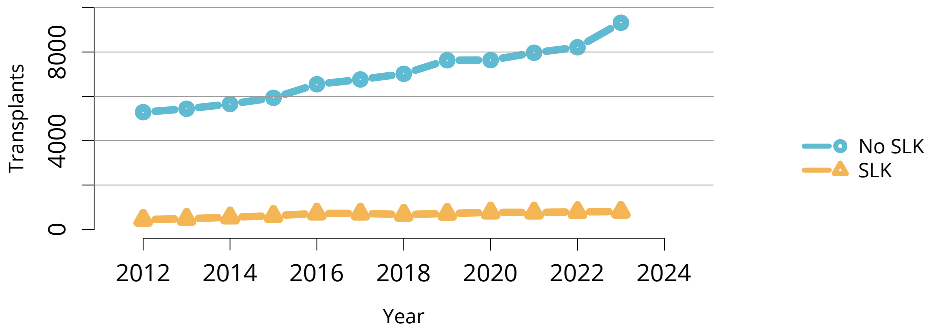

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Figure LI 41: Adult liver transplants by SLK. Adult liver transplant recipients, including retransplant and multiorgan recipients. SLK transplants are in recipients with a liver and kidney transplant from the same donor. SLK, simultaneous liver-kidney.

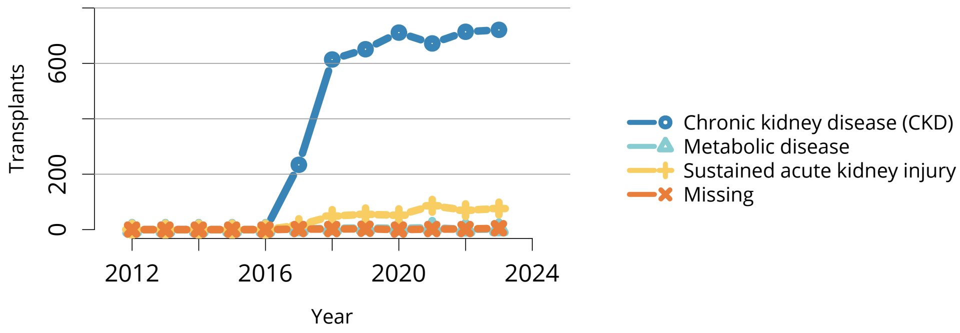

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Figure LI 42: Adult SLK transplants by SLK diagnosis. Adult SLK transplant recipients, including retransplant and multiorgan recipients. SLK transplants are in recipients with a liver and kidney transplant from the same donor. SLK, simultaneous liver-kidney.

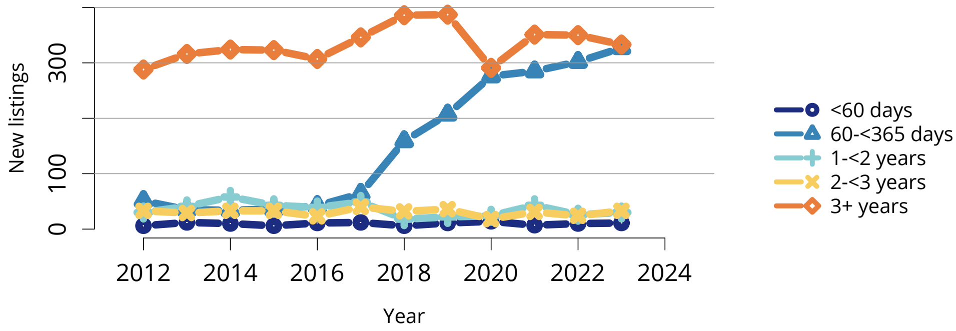

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Figure LI 43: New adult candidates added to the kidney transplant waiting list after liver transplant by time to kidney listing from liver transplant. A new candidate is one who first joined the list during the given year, without having been listed in a previous year. Previously listed candidates who underwent transplant and subsequently relisted are considered new. Active and inactive patients are included.

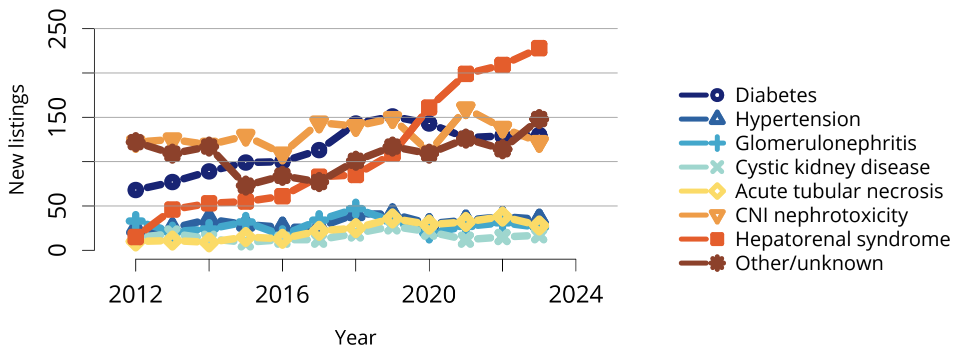

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Figure LI 44: New adult candidates added to the kidney transplant waiting list after liver transplant by diagnosis. A new candidate is one who first joined the list during the given year, without having been listed in a previous year. Previously listed candidates who underwent transplant and subsequently relisted are considered new. Active and inactive patients are included. CNI, calcineurin inhibitor.

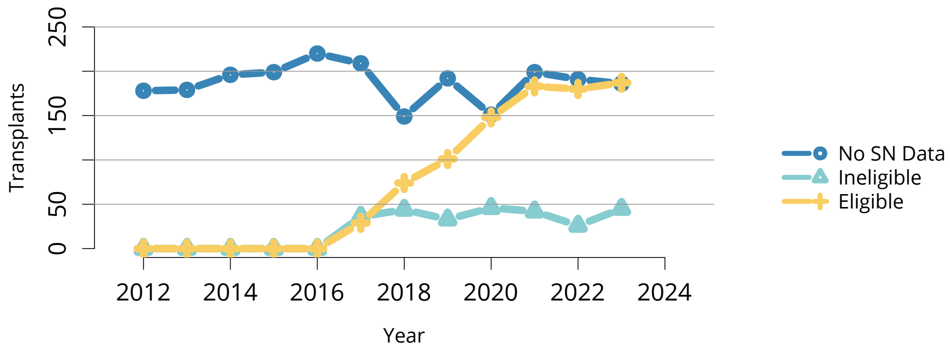

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Figure LI 45: Adult kidney transplants by SLK safety net eligibility. Adult kidney transplant recipients, including retransplant and multiorgan recipients. SLK transplants are in recipients with a liver and kidney transplant from the same donor. SLK, simultaneous liver-kidney; SN, safety net.

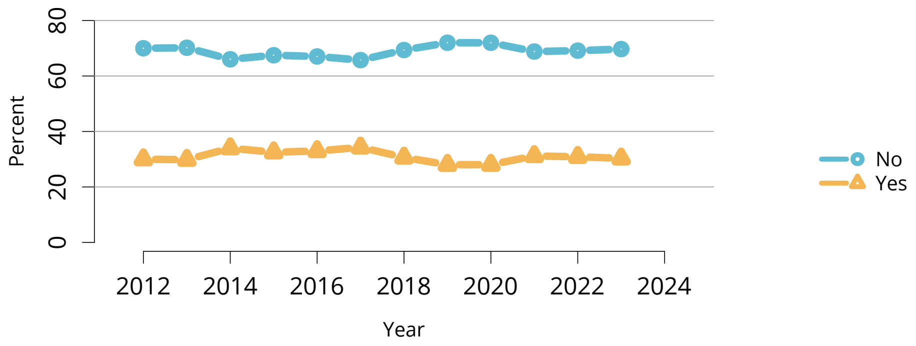

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Figure LI 46: Induction agent use in adult liver transplant recipients. Immunosuppression at transplant reported to the OPTN.

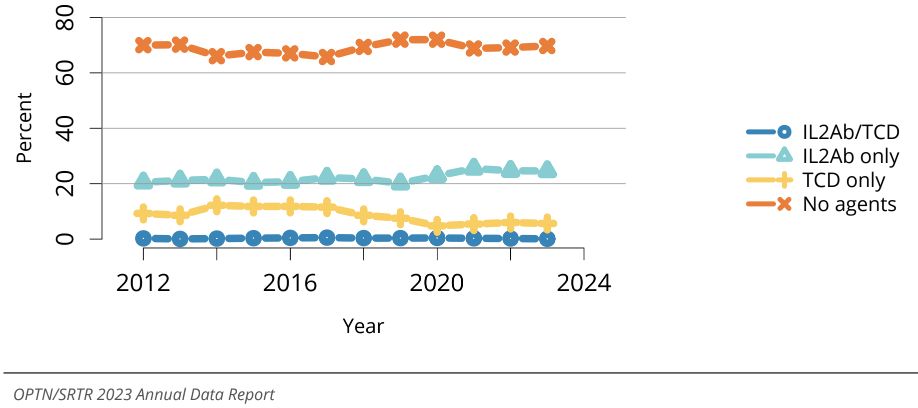

Figure LI 47: Type of induction agent use in adult liver transplant recipients. Immunosuppression at transplant reported to the OPTN. IL2Ab, interleukin-2 receptor antibody; TCD, T-cell depleting.

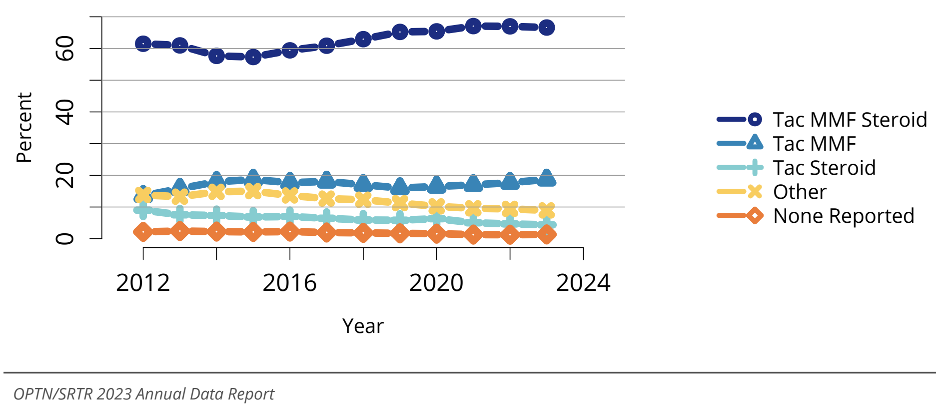

Figure LI 48: Immunosuppression regimen use in adult liver transplant recipients. Immunosuppression regimen at transplant reported to the OPTN. MMF, all mycophenolate agents; Tac, tacrolimus.

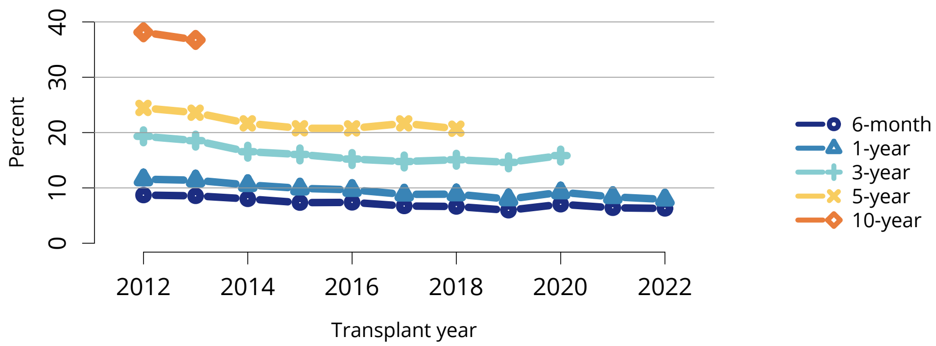

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Figure LI 49: Graft failure among adult deceased donor liver transplant recipients. All adult recipients of deceased donor livers, including multiorgan transplant recipients.

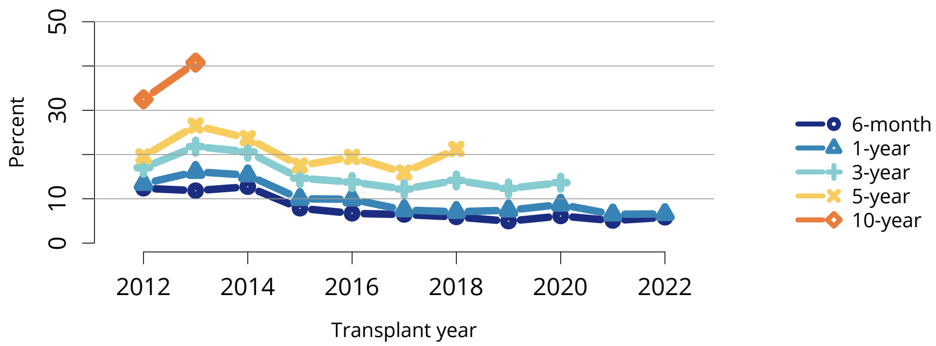

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Figure LI 50: Graft failure among adult living donor liver transplant recipients. All adult recipients of living donor livers, including multiorgan transplant recipients.

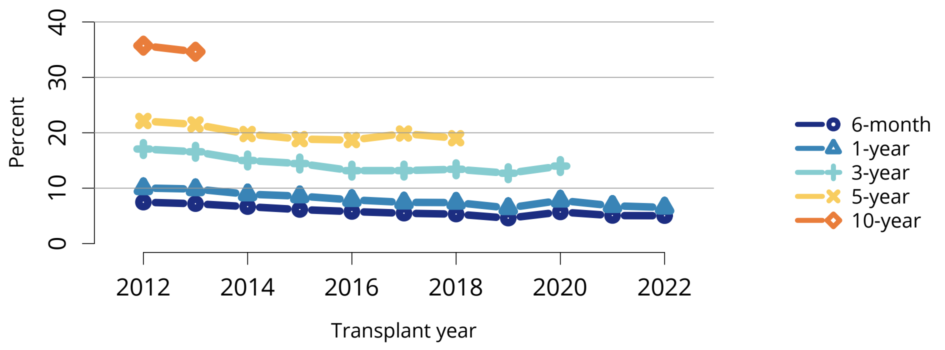

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Figure LI 51: Patient death among adult liver transplant recipients. All adult recipients of deceased donor livers, including multiorgan transplant recipients.

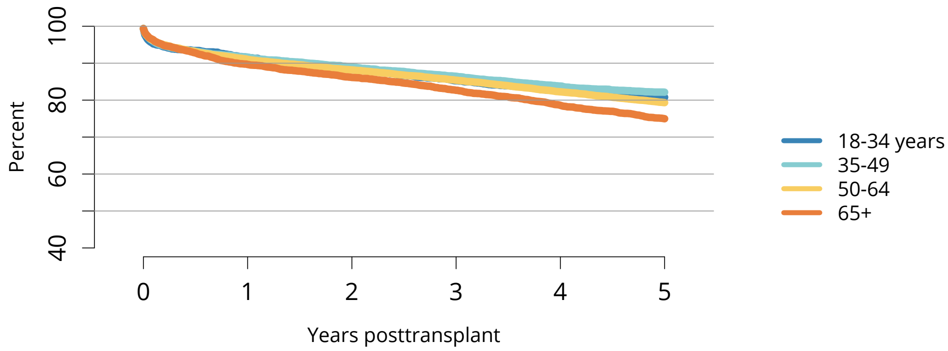

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Figure LI 52: Graft survival among adult deceased donor liver transplant recipients, 2016-2018, by age. Graft survival estimated using unadjusted Kaplan-Meier methods.

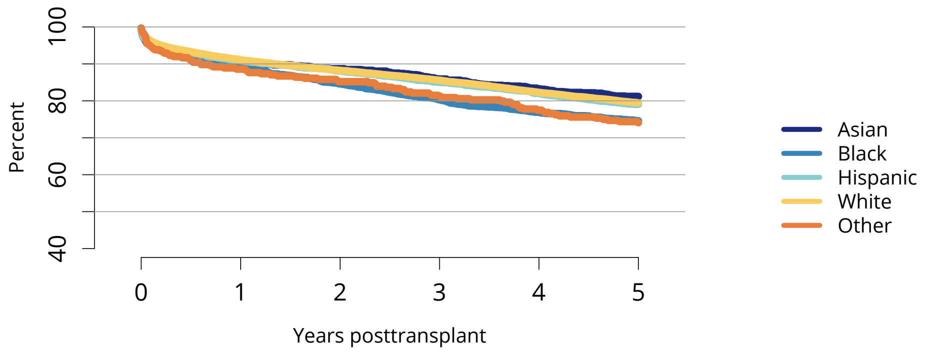

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Figure LI 53: Graft survival among adult deceased donor liver transplant recipients, 2016-2018, by race and ethnicity. Graft survival estimated using unadjusted Kaplan-Meier methods. The Other race category is composed of Native American and Multiracial categories.

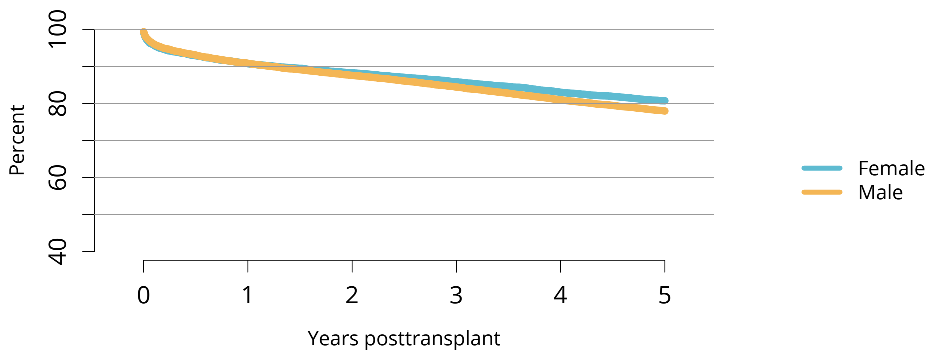

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Figure LI 54: Graft survival among adult deceased donor liver transplant recipients, 2016-2018, by sex. Graft survival estimated using unadjusted Kaplan-Meier methods.

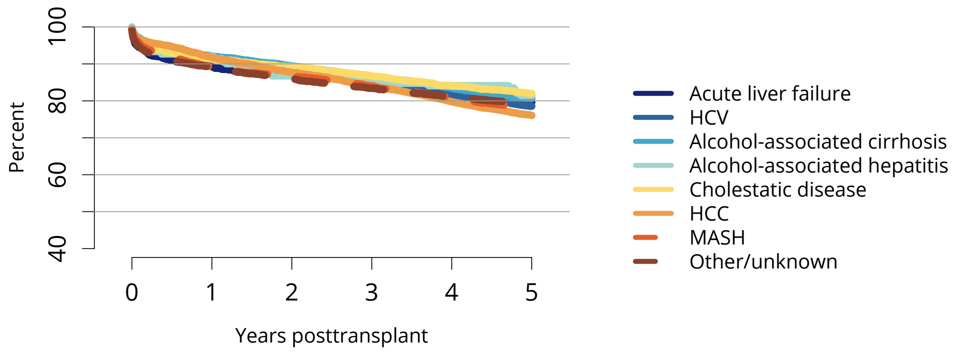

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Figure LI 55: Graft survival among adult deceased donor liver transplant recipients, 2016-2018, by diagnosis. Graft survival estimated using unadjusted Kaplan-Meier methods. HCC, hepatocellular carcinoma; HCV, hepatitis C virus; MASH, metabolic dysfunction–associated steatohepatitis.

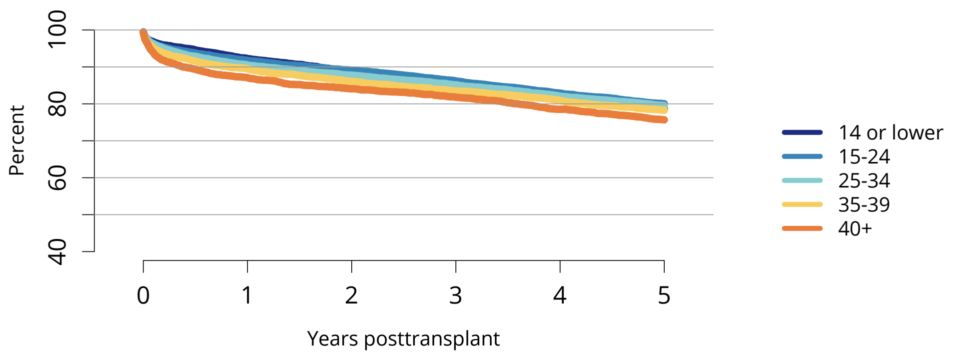

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Figure LI 56: Graft survival among adult deceased donor liver transplant recipients, 2016-2018, by laboratory MELD score. Graft survival estimated using unadjusted Kaplan-Meier methods. MELD, model for end-stage liver disease.

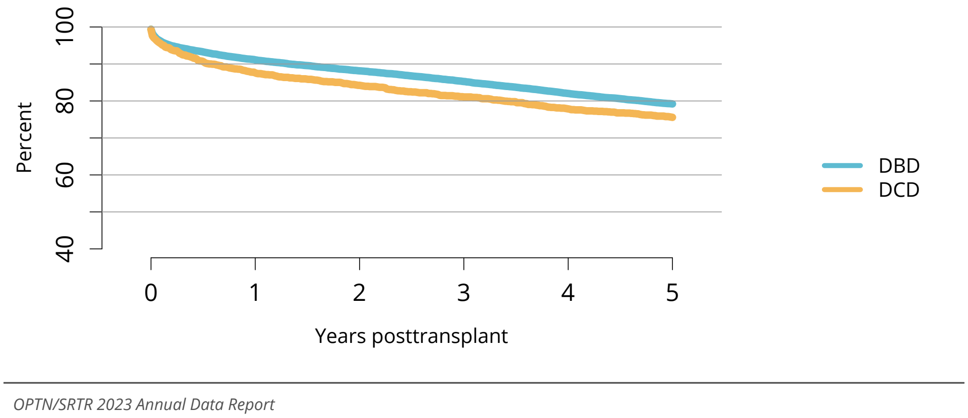

Figure LI 57: Graft survival among adult deceased donor liver transplant recipients, 2016-2018, by DCD status. Graft survival estimated using unadjusted Kaplan-Meier methods. DBD, donation after brain death; DCD, donation after circulatory death.

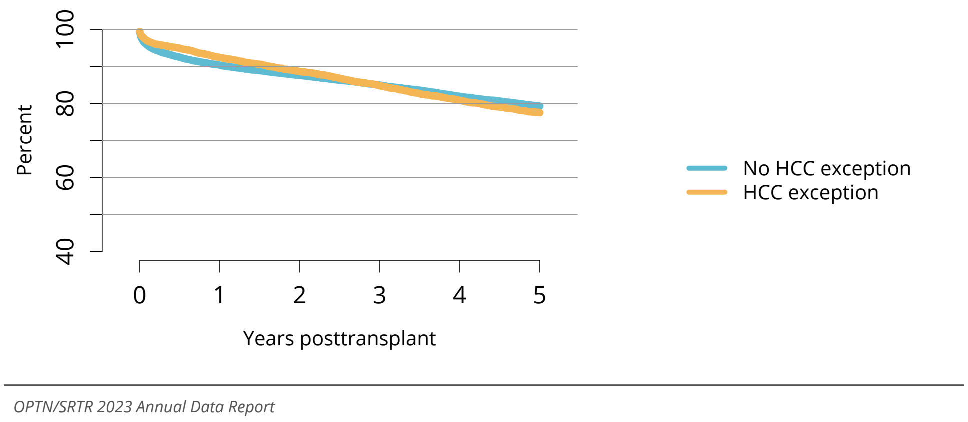

Figure LI 58: Graft survival among adult deceased donor liver transplant recipients, 2016-2018, by HCC status. Graft survival estimated using unadjusted Kaplan-Meier methods. HCC, hepatocellular carcinoma.

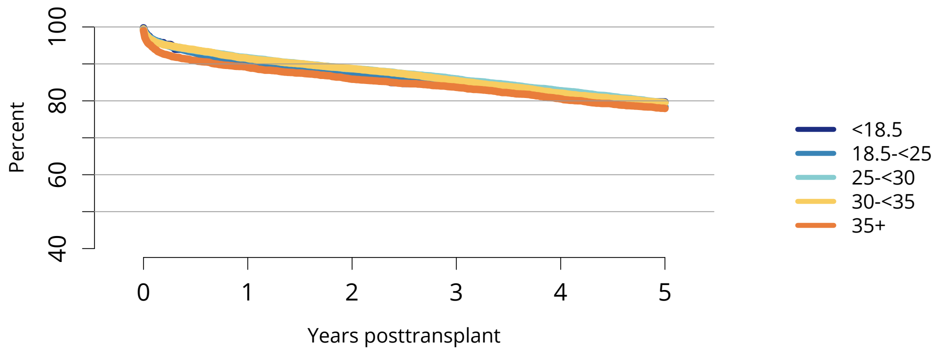

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Figure LI 59: Graft survival among adult deceased donor liver transplant recipients, 2016-2018, by BMI. Graft survival estimated using unadjusted Kaplan-Meier methods. BMI, body mass index.

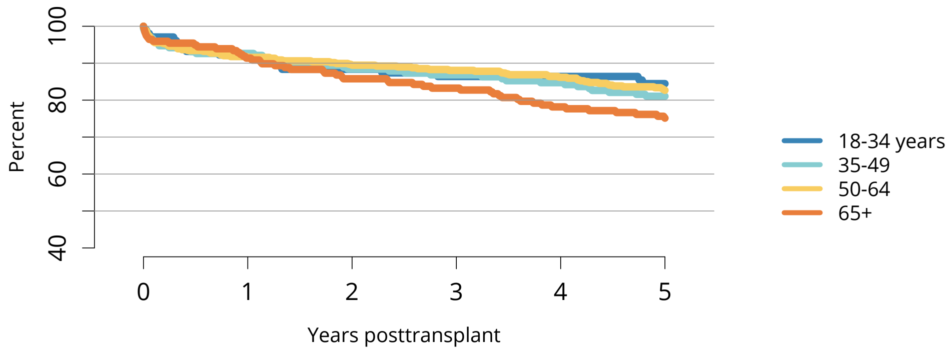

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Figure LI 60: Graft survival among adult living donor liver transplant recipients, 2016-2018, by age. Graft survival estimated using unadjusted Kaplan-Meier methods.

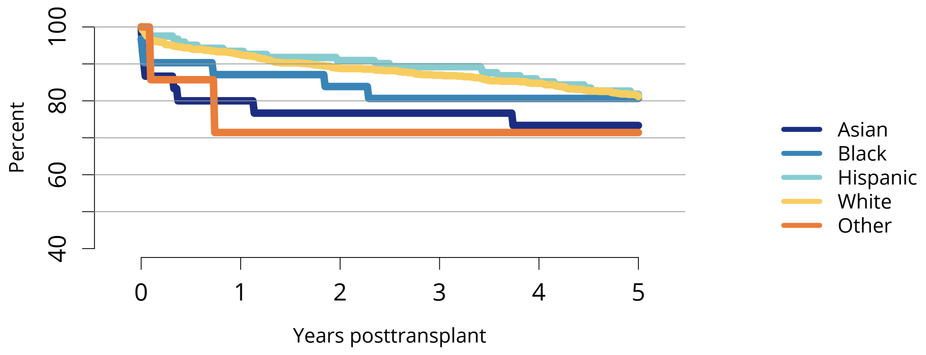

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Figure LI 61: Graft survival among adult living donor liver transplant recipients, 2016-2018, by race and ethnicity. Graft survival estimated using unadjusted Kaplan-Meier methods. The Other race category is composed of Native American and Multiracial categories.

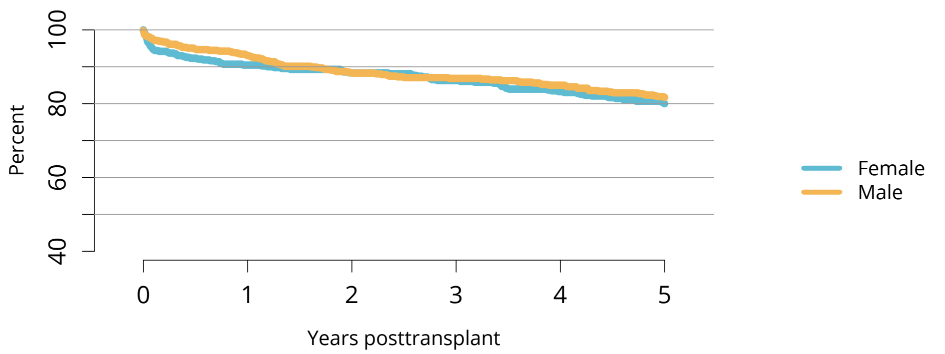

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Figure LI 62: Graft survival among adult living donor liver transplant recipients, 2016-2018, by sex. Graft survival estimated using unadjusted Kaplan-Meier methods.

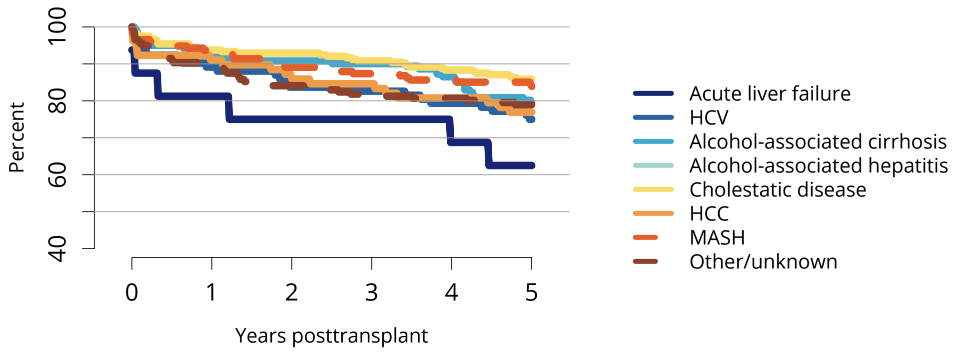

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Figure LI 63: Graft survival among adult living donor liver transplant recipients, 2016-2018, by diagnosis. Graft survival estimated using unadjusted Kaplan-Meier methods. HCC, hepatocellular carcinoma; HCV, hepatitis C virus; MASH, metabolic dysfunction–associated steatohepatitis.

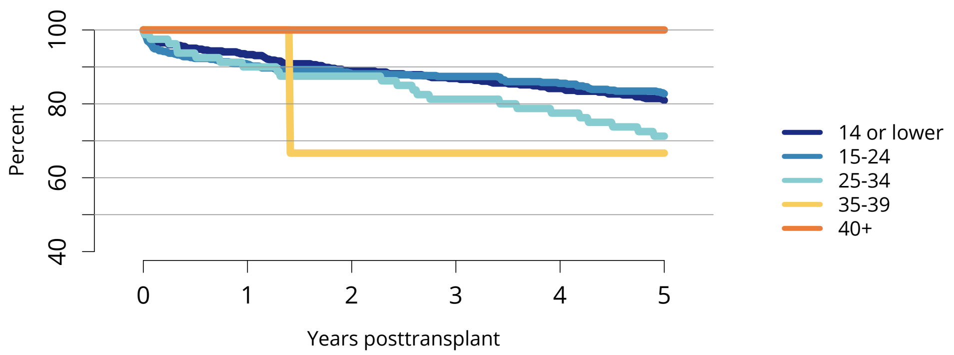

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Figure LI 64: Graft survival among adult living donor liver transplant recipients, 2016-2018, by laboratory MELD score. Graft survival estimated using unadjusted Kaplan-Meier methods. MELD, model for end-stage liver disease.

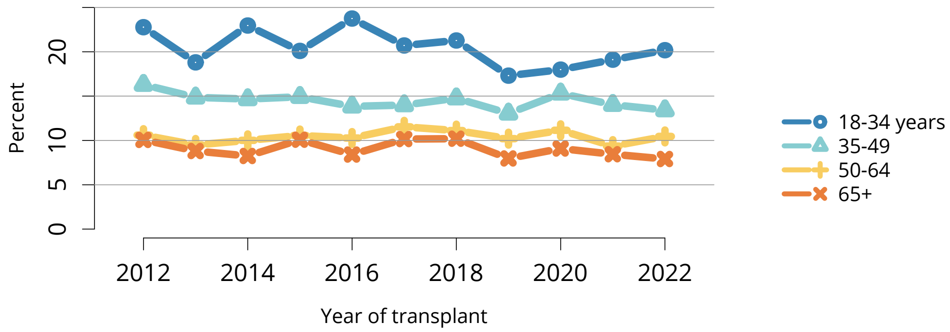

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Figure LI 65: Incidence of acute rejection by 1 year posttransplant among adult liver transplant recipients by age. Only the first reported rejection event is counted. Cumulative incidence is estimated using the Kaplan-Meier method.

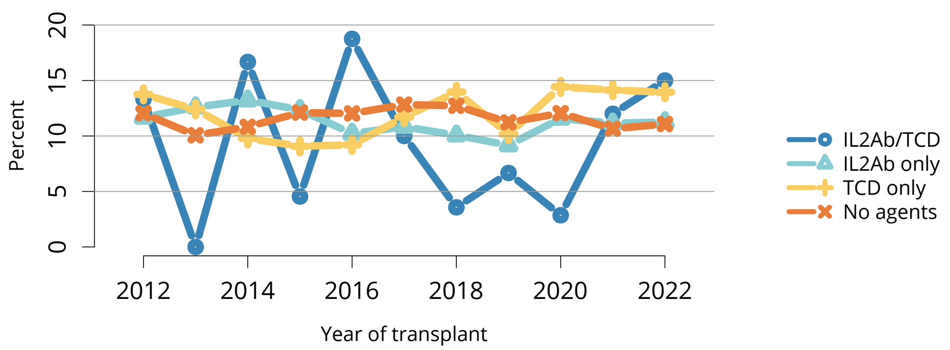

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Figure LI 66: Incidence of acute rejection by 1 year posttransplant among adult liver transplant recipients by induction agent. Only the first reported rejection event is counted. Cumulative incidence is estimated using the Kaplan-Meier method. IL2Ab, interleukin-2 receptor antibody; TCD, T-cell depleting.

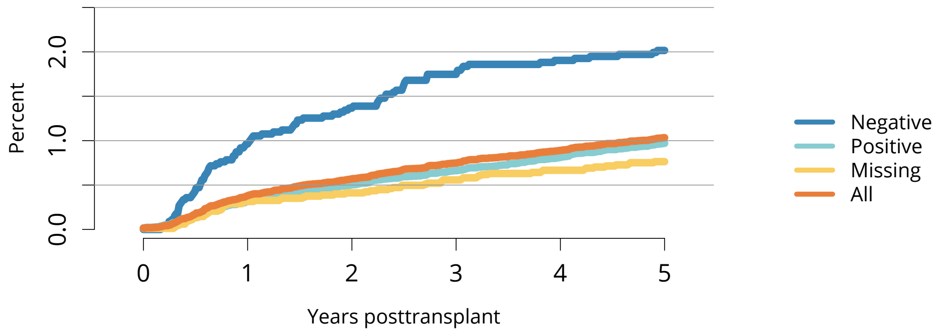

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Figure LI 67: Incidence of PTLD among adult liver transplant recipients by recipient EBV status at transplant, 2012-2018. Cumulative incidence is estimated using the Kaplan-Meier method. PTLD is identified as a reported complication or cause of death on the OPTN Transplant Recipient Follow-up Form or the Posttransplant Malignancy Form as polymorphic PTLD, monomorphic PTLD, or Hodgkin's disease. Only the earliest date of PTLD diagnosis is considered. EBV, Epstein-Barr virus; PTLD, posttransplant lymphoproliferative disorder.

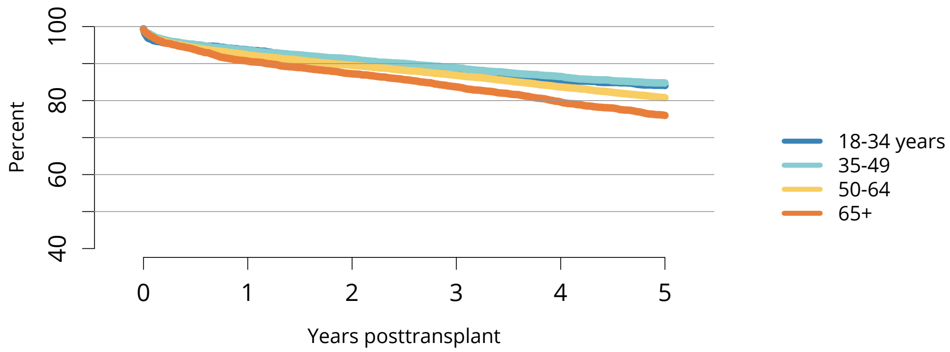

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Figure LI 68: Patient survival among adult deceased donor liver transplant recipients, 2016-2018, by age. Patient survival estimated using unadjusted Kaplan-Meier methods.

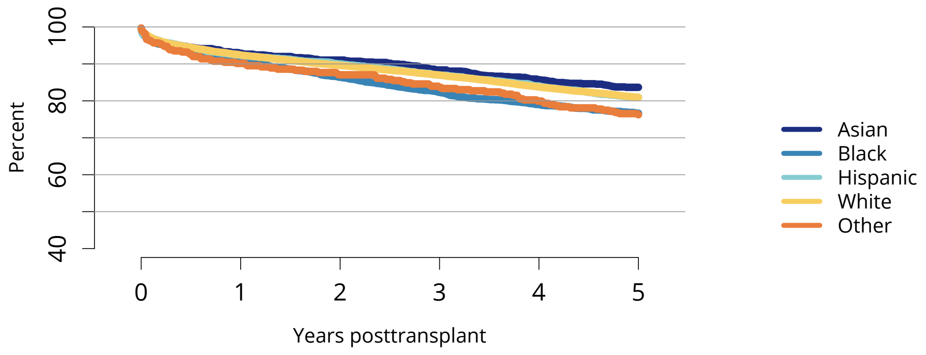

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Figure LI 69: Patient survival among adult deceased donor liver transplant recipients, 2016-2018, by race and ethnicity. Patient survival estimated using unadjusted Kaplan-Meier methods. The Other race category is composed of Native American and Multiracial categories.

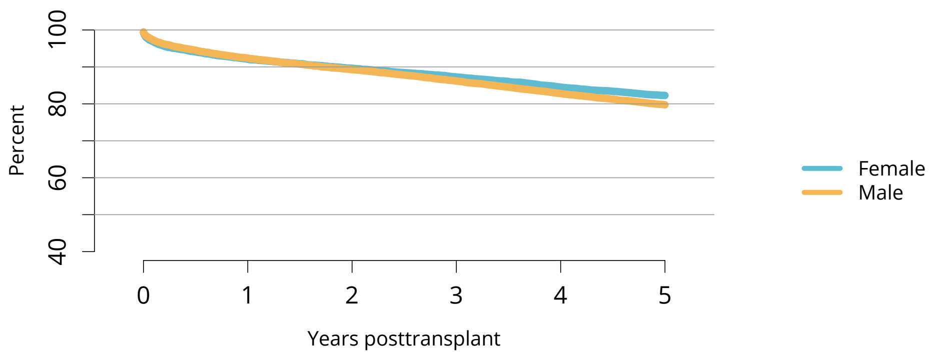

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Figure LI 70: Patient survival among adult deceased donor liver transplant recipients, 2016-2018, by sex. Patient survival estimated using unadjusted Kaplan-Meier methods.

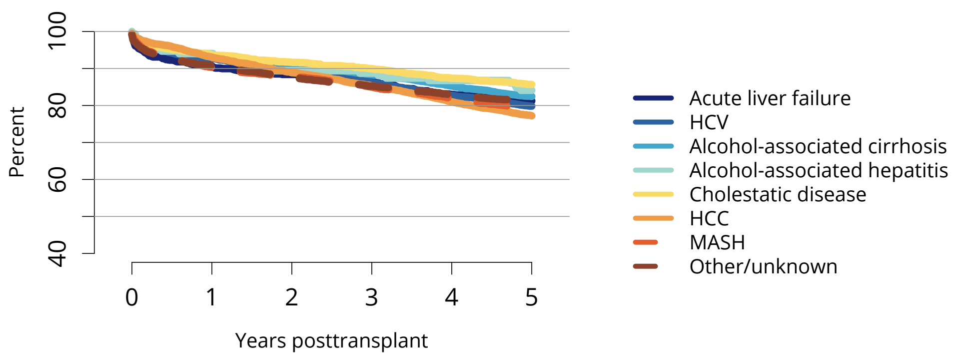

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Figure LI 71: Patient survival among adult deceased donor liver transplant recipients, 2016-2018, by diagnosis. Patient survival estimated using unadjusted Kaplan-Meier methods. HCC, hepatocellular carcinoma; HCV, hepatitis C virus; MASH, metabolic dysfunction–associated steatohepatitis.

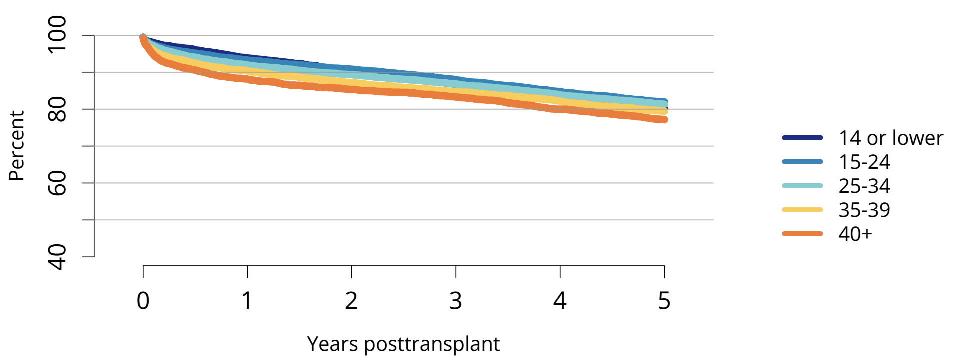

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Figure LI 72: Patient survival among adult deceased donor liver transplant recipients, 2016-2018, by laboratory MELD score. Patient survival estimated using unadjusted Kaplan-Meier methods. MELD, model for end-stage liver disease.

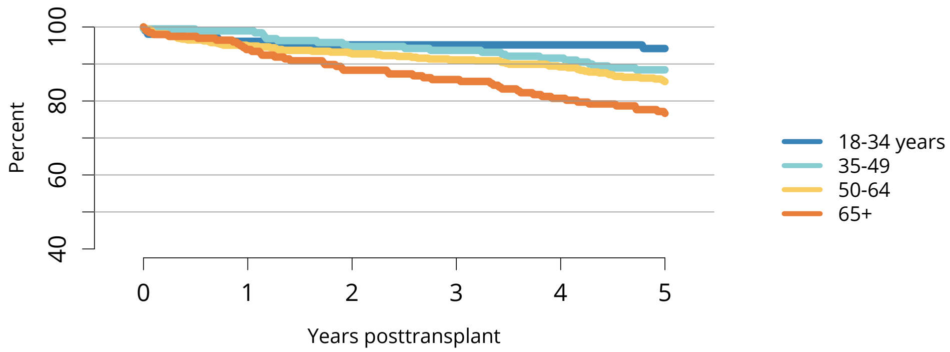

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Figure LI 73: Patient survival among adult living donor liver transplant recipients, 2016-2018, by age. Patient survival estimated using unadjusted Kaplan-Meier methods.

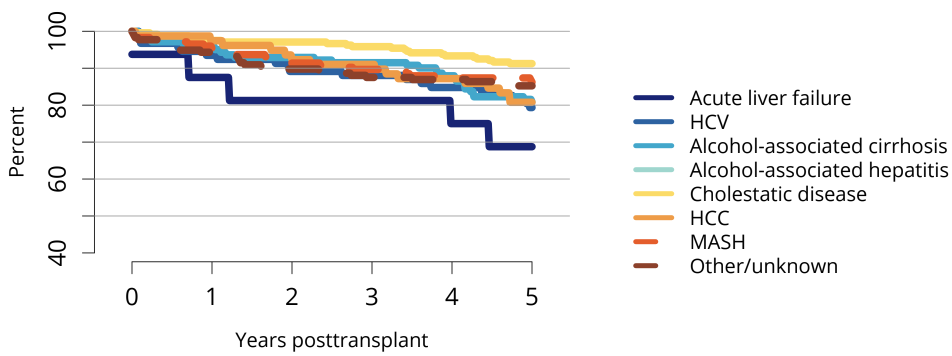

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Figure LI 74: Patient survival among adult living donor liver transplant recipients, 2016-2018, by diagnosis. Patient survival estimated using unadjusted Kaplan-Meier methods. HCC, hepatocellular carcinoma; HCV, hepatitis C virus; MASH, metabolic dysfunction–associated steatohepatitis.

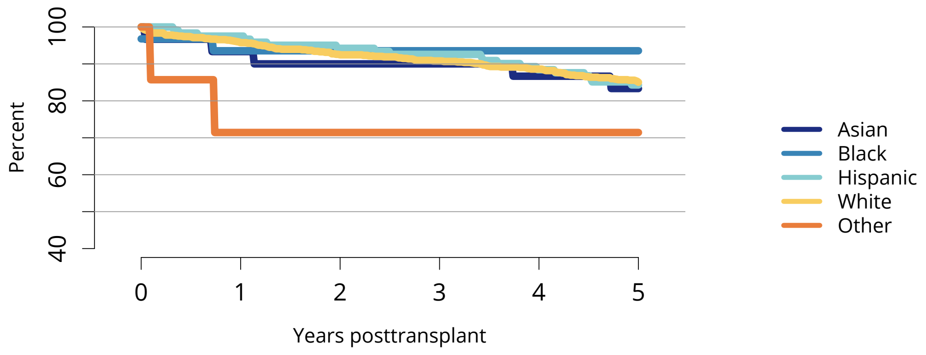

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Figure LI 75: Patient survival among adult living donor liver transplant recipients, 2016-2018, by race and ethnicity. Patient survival estimated using unadjusted Kaplan-Meier methods. The Other race category is composed of Native American and Multiracial categories.

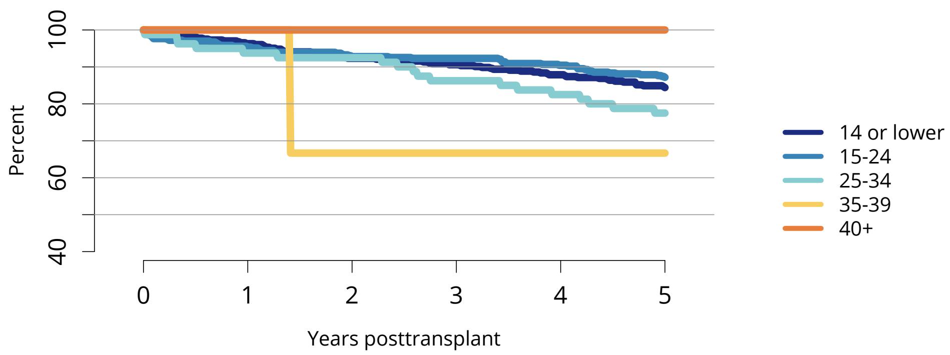

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Figure LI 76: Patient survival among adult living donor liver transplant recipients, 2016-2018, by laboratory MELD score. Patient survival estimated using unadjusted Kaplan-Meier methods. MELD, model for end-stage liver disease.

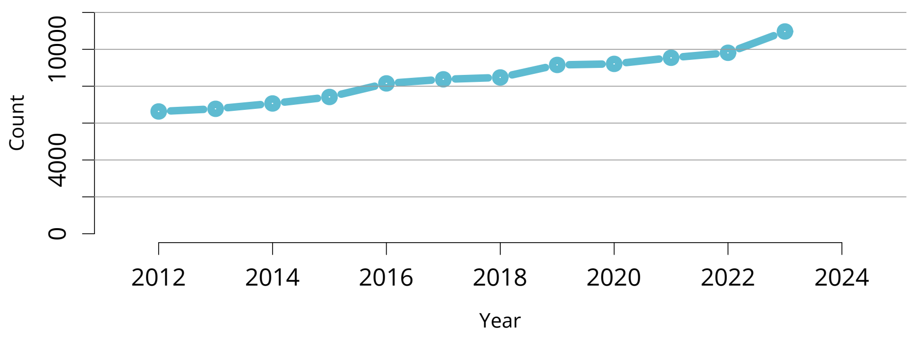

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Figure LI 77: Overall deceased liver donor count. Count of deceased donors whose livers were recovered for transplant.

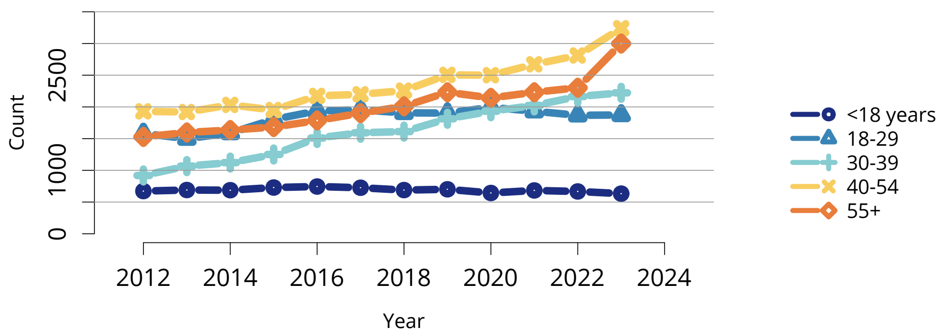

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Figure LI 78: Deceased liver donor count by age. Count of deceased donors whose livers were recovered for transplant.

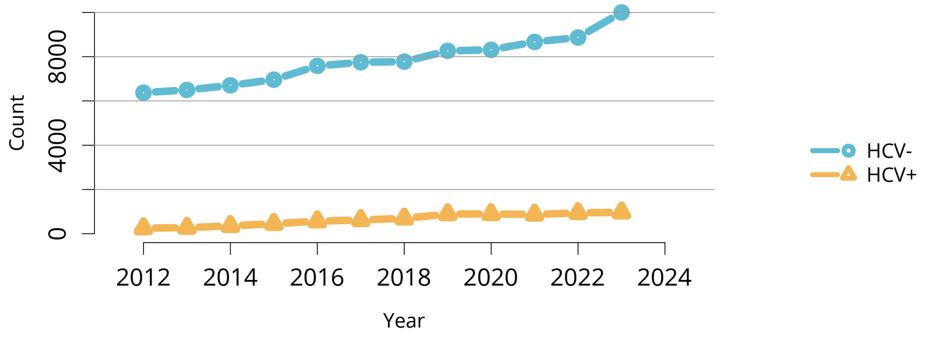

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Figure LI 79: Deceased liver donor count by HCV status. Count of deceased donors whose livers were recovered for transplant. Donor HCV status was based on an antibody test. HCV, hepatitis C virus.

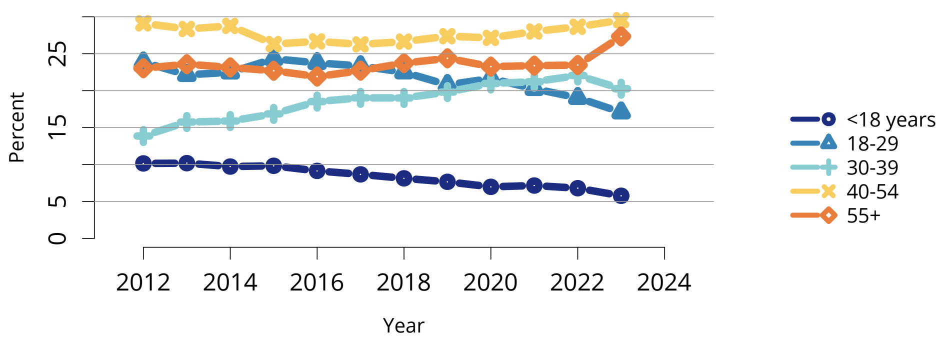

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Figure LI 80: Distribution of deceased liver donors by age. Deceased donors whose livers were recovered for transplant.

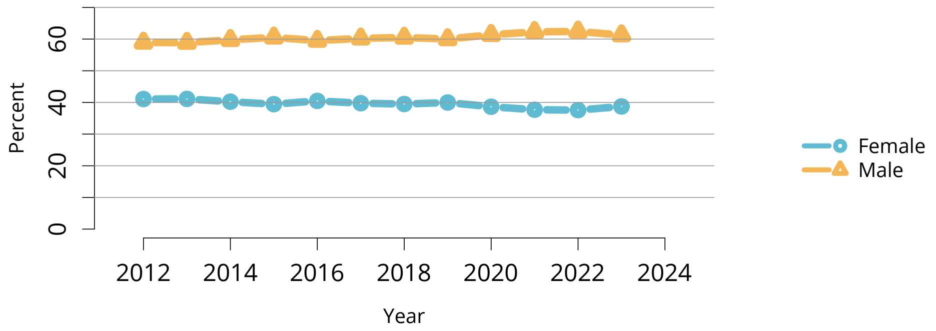

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Figure LI 81: Distribution of deceased liver donors by sex. Deceased donors whose livers were recovered for transplant.

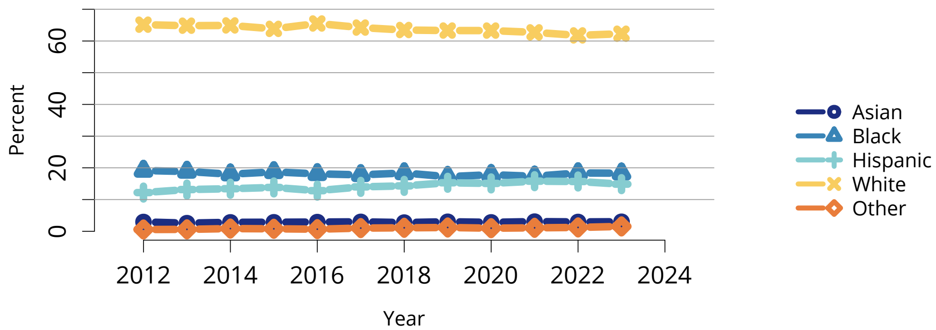

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Figure LI 82: Distribution of deceased liver donors by race and ethnicity. Deceased donors whose livers were recovered for transplant. The Other race category is composed of Native American and Multiracial categories.

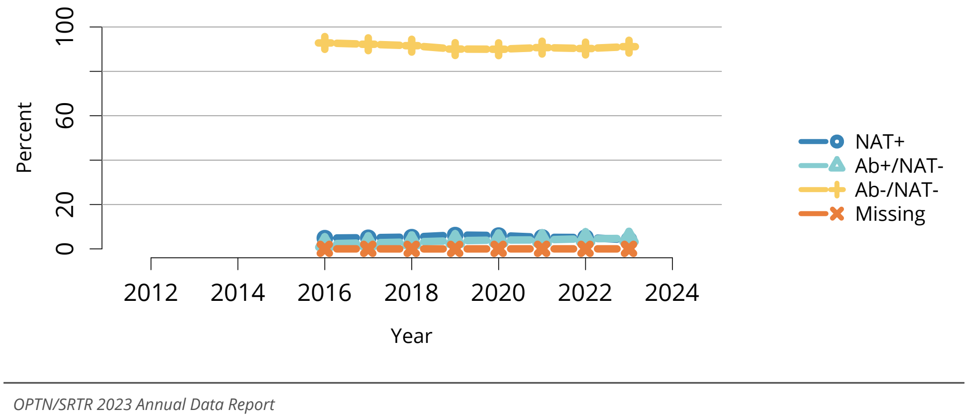

Figure LI 83: Distribution of deceased liver donors by donor HCV status. Deceased donors whose livers were recovered for transplant. Donor HCV status was based on NAT and antibody tests. Ab, antibody; HCV, hepatitis C virus; NAT, nucleic acid test.

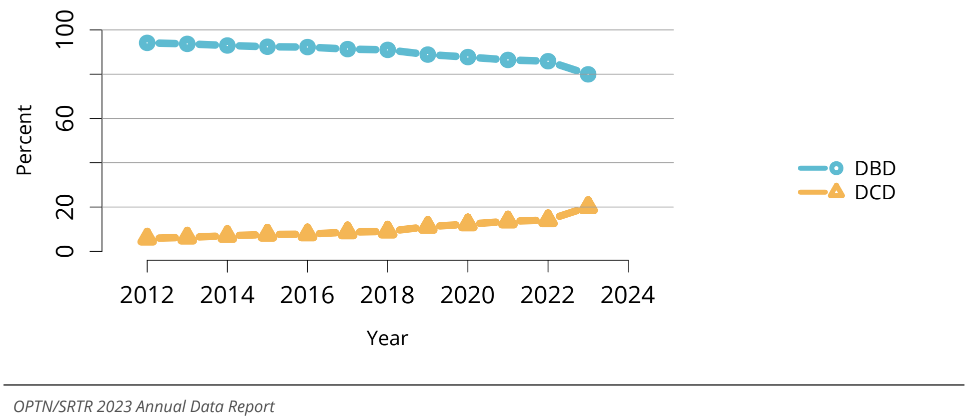

Figure LI 84: Distribution of deceased liver donors by DCD status. Deceased donors whose livers were recovered for transplant. DBD, donation after brain death; DCD, donation after circulatory death.

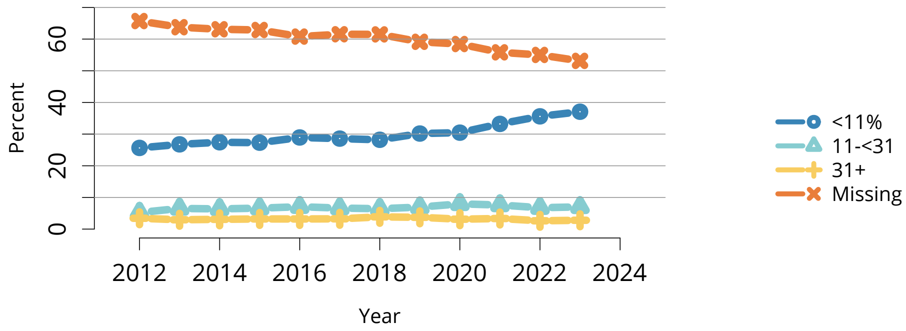

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Figure LI 85: Distribution of deceased liver donors by macrovesicular fat. Deceased donors whose livers were recovered for transplant.

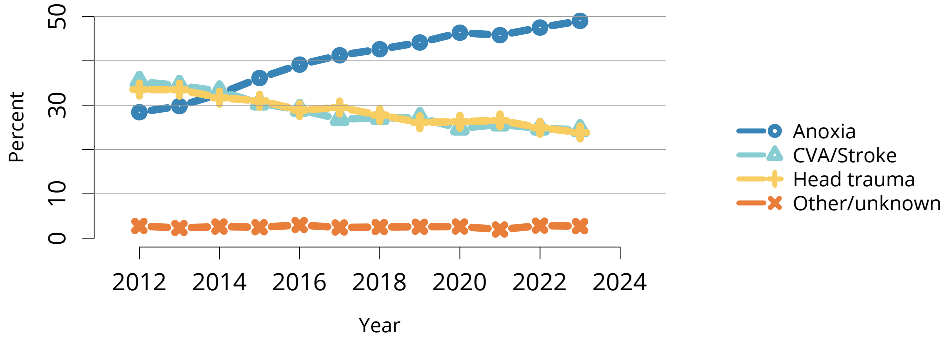

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Figure LI 86: Cause of death among deceased liver donors. Deceased donors with a liver recovered for the purposes of transplant. CVA, cerebrovascular accident.

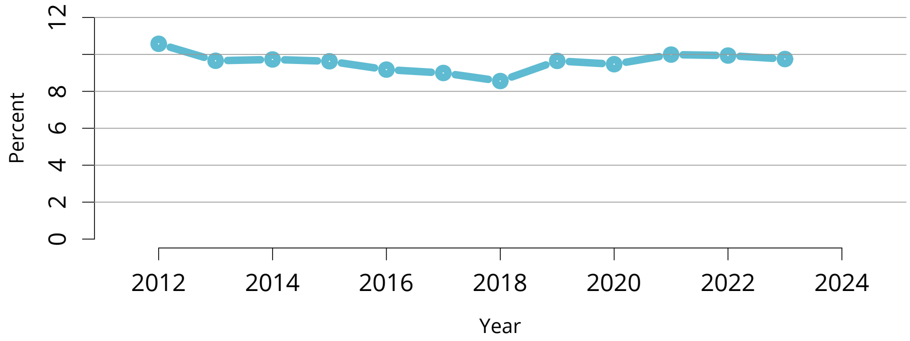

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Figure LI 87: Overall percent of livers recovered for transplant and not transplanted. Percentages of livers not transplanted out of all livers recovered for transplant.

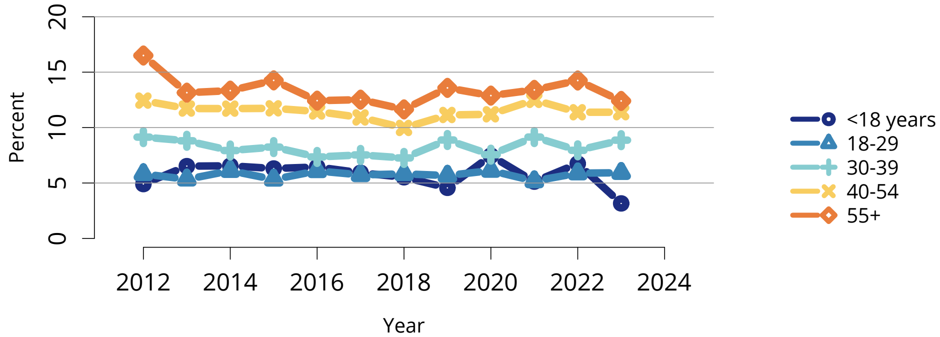

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Figure LI 88: Percent of livers recovered for transplant and not transplanted by donor age. Percentages of livers not transplanted out of all livers recovered for transplant.

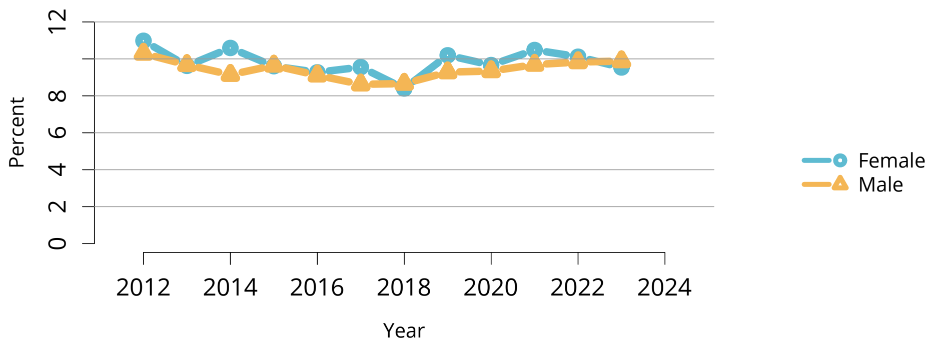

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Figure LI 89: Percent of livers recovered for transplant and not transplanted by donor sex. Percentages of livers not transplanted out of all livers recovered for transplant.

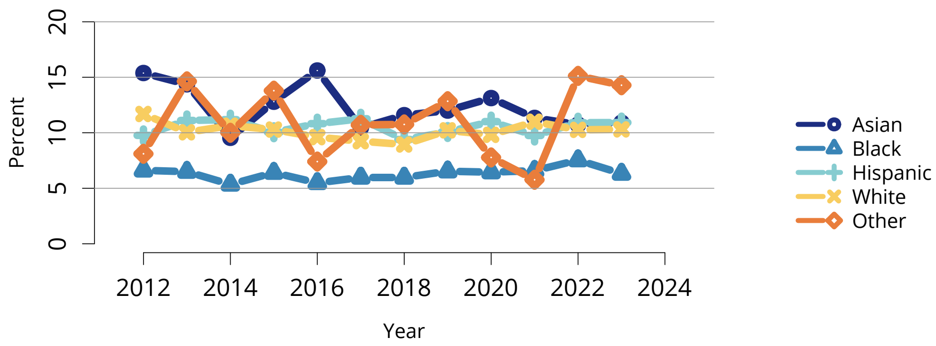

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Figure LI 90: Percent of livers recovered for transplant and not transplanted by donor race and ethnicity. Percentages of livers not transplanted out of all livers recovered for transplant. The Other race category is composed of Native American and Multiracial categories.

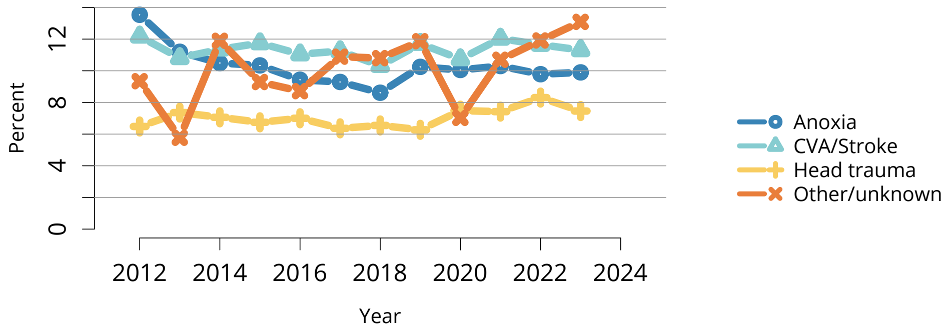

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Figure LI 91: Percent of livers recovered for transplant and not transplanted by donor cause of death. Percentages of livers not transplanted out of all livers recovered for transplant. CVA, cerebrovascular accident.

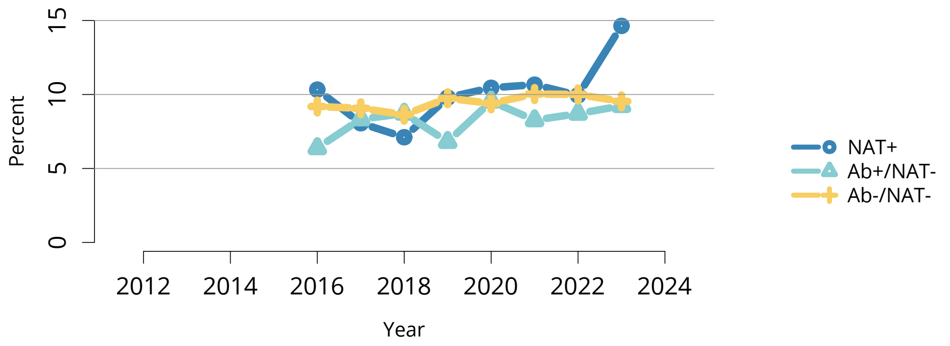

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Figure LI 92: Percent of livers recovered for transplant and not transplanted by donor HCV status. Percentages of livers not transplanted out of all livers recovered for transplant. Donor HCV status was based on NAT and antibody tests. Ab, antibody; HCV, hepatitis C virus; NAT, nucleic acid test.

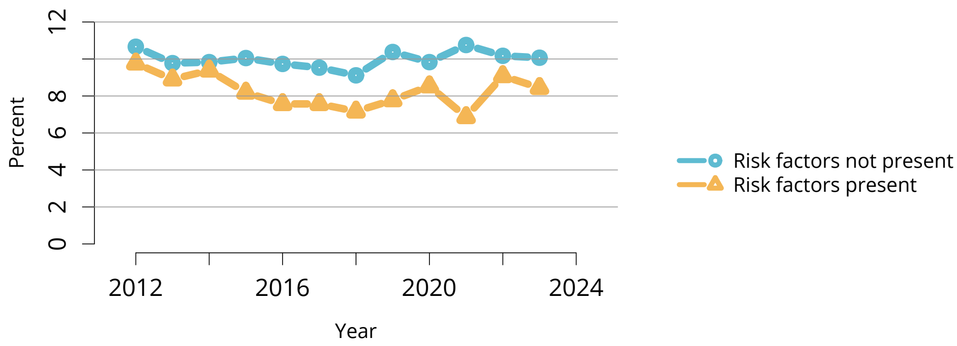

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Figure LI 93: Percent of livers recovered for transplant and not transplanted, by donor risk of disease transmission. Percentages of livers not transplanted out of all livers recovered for transplant. "Risk factors" refers to risk criteria for acute transmission of human immunodeficiency virus, hepatitis B virus, or hepatitis C virus from the US Public Health Service Guideline.

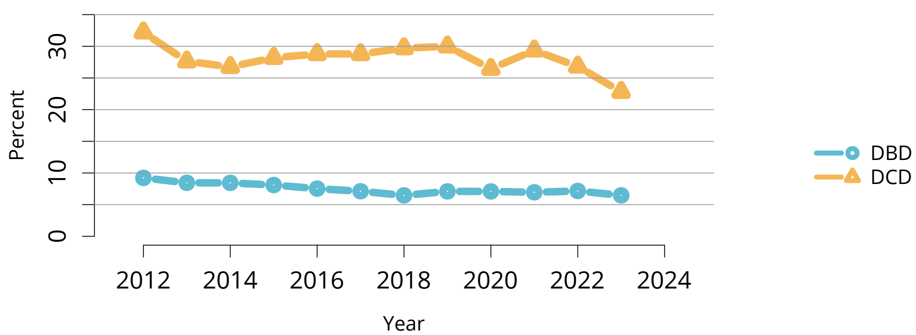

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Figure LI 94: Percent of livers recovered for transplant and not transplanted by DCD status. Percentages of livers not transplanted out of all livers recovered for transplant. DBD, donation after brain death; DCD, donation after circulatory death.

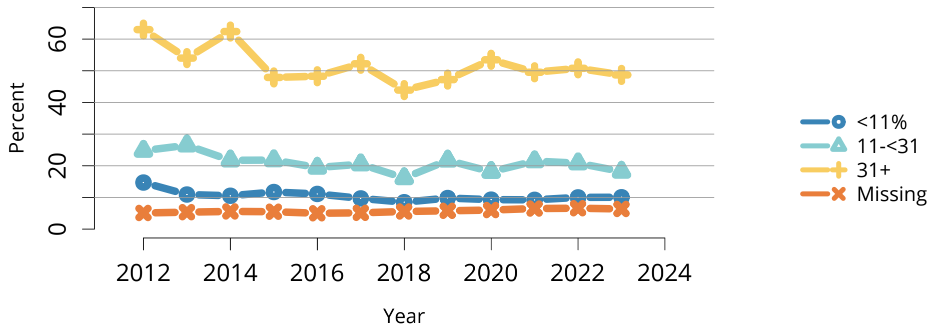

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Figure LI 95: Percent of livers recovered for transplant and not transplanted by macrovesicular fat. Percentages of livers not transplanted out of all livers recovered for transplant.

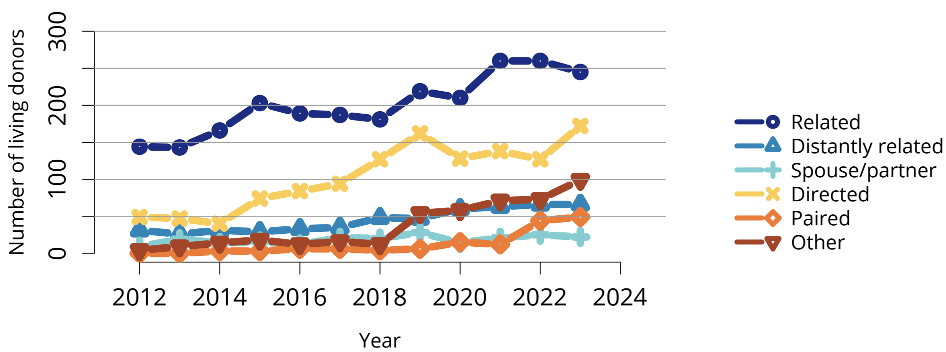

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Figure LI 96: Number of living liver donors by donor relation. Numbers of living donor donations, excluding domino livers, as reported on the OPTN Living Donor Registration Form.

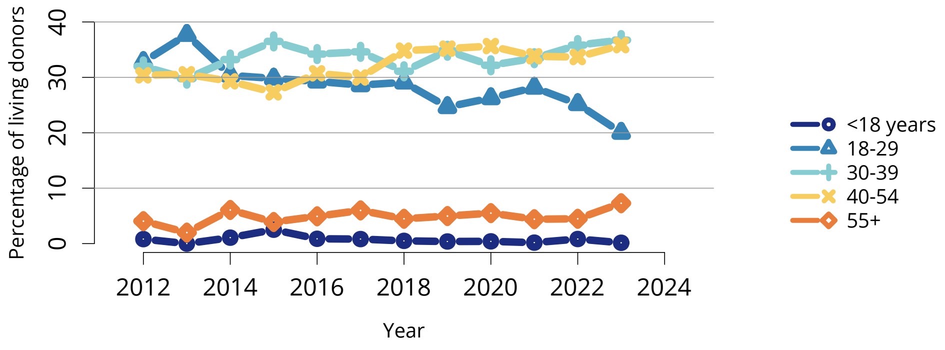

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Figure LI 97: Living liver donors by age. As reported on the OPTN Living Donor Registration Form. Domino liver donors excluded.

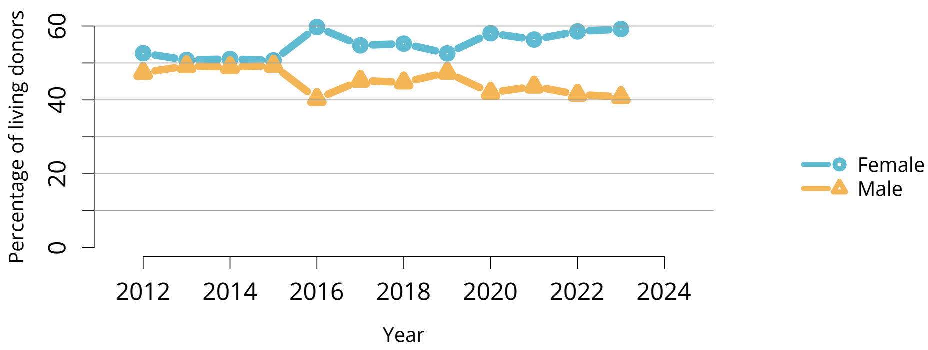

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Figure LI 98: Living liver donors by sex. As reported on the OPTN Living Donor Registration Form. Domino liver donors excluded.

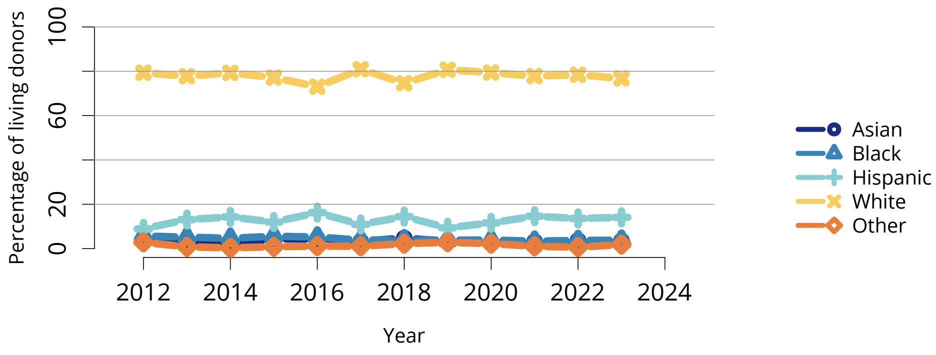

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Figure LI 99: Living liver donors by race and ethnicity. As reported on the OPTN Living Donor Registration Form. Domino liver donors excluded. The Other race category is composed of Native American and Multiracial categories.

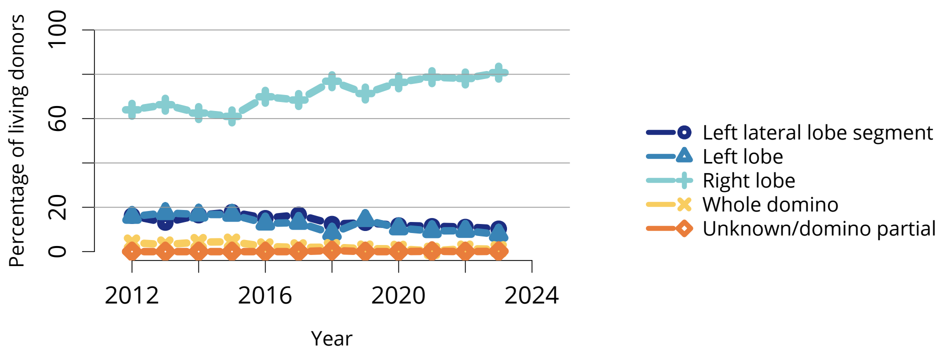

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Figure LI 100: Living donor liver transplant graft type. As reported on the OPTN Living Donor Registration Form.

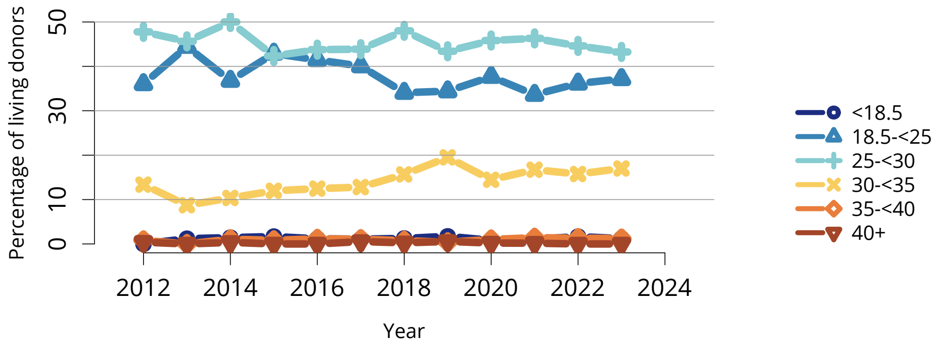

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Figure LI 101: BMI among living liver donors. Donor height and weight reported on the OPTN Living Donor Registration Form. Domino liver donors excluded. BMI, body mass index.

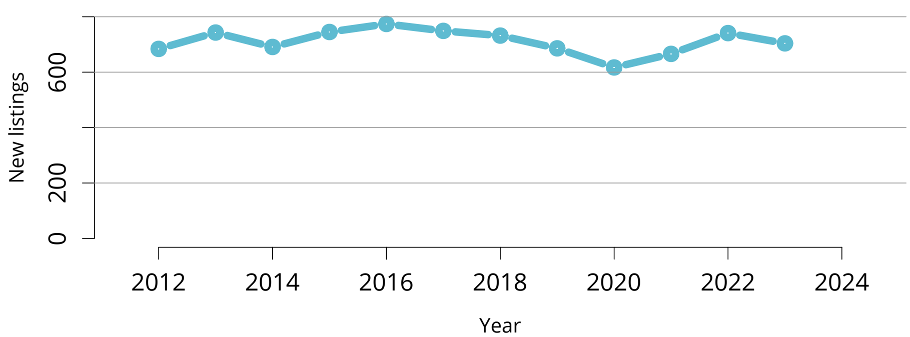

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Figure LI 102: New pediatric candidates added to the liver transplant waiting list. A new candidate is one who first joined the list during the given year, without having been listed in a previous year. Previously listed candidates who underwent transplant and were subsequently relisted are considered new. Candidates listed at more than one center are counted once per listing. Active and inactive patients are included.

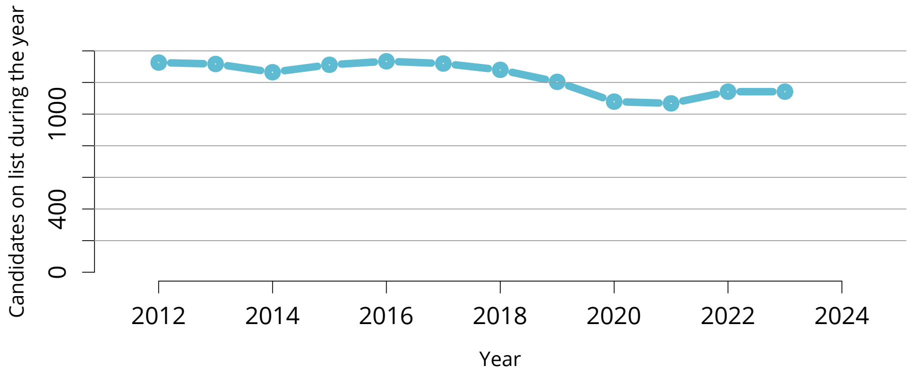

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Figure LI 103: All pediatric candidates on the liver transplant waiting list. Pediatric candidates listed at any time during the year. Candidates listed at more than one center are counted once per listing.

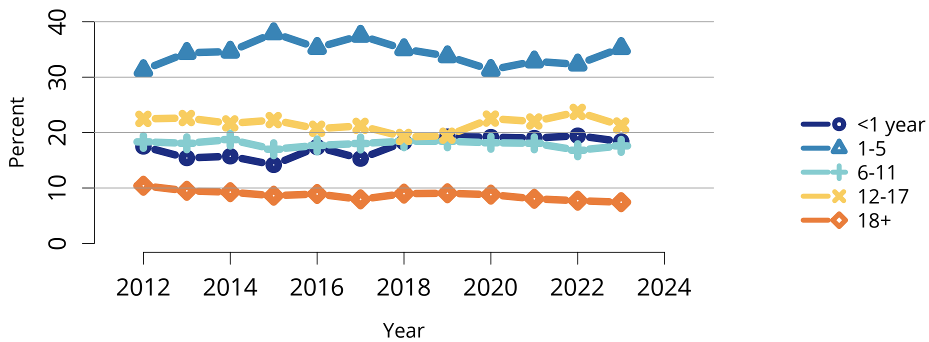

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Figure LI 104: Distribution of pediatric candidates waiting for liver transplant by age. Candidates waiting for transplant at any time in the given year. Candidates listed at more than one center are counted once per listing. Active and inactive candidates are included. Age is determined at the earliest of transplant, death, removal, or December 31 of the year. The 18+ category is for candidates who turned age 18 while waiting.

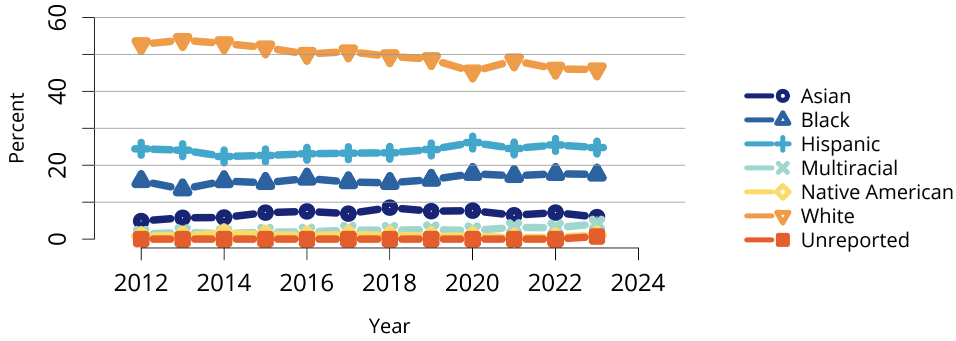

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Figure LI 105: Distribution of pediatric candidates waiting for liver transplant by race and ethnicity. Candidates waiting for transplant any time in the given year. Candidates listed at more than one center are counted once per listing. Active and inactive candidates are included.

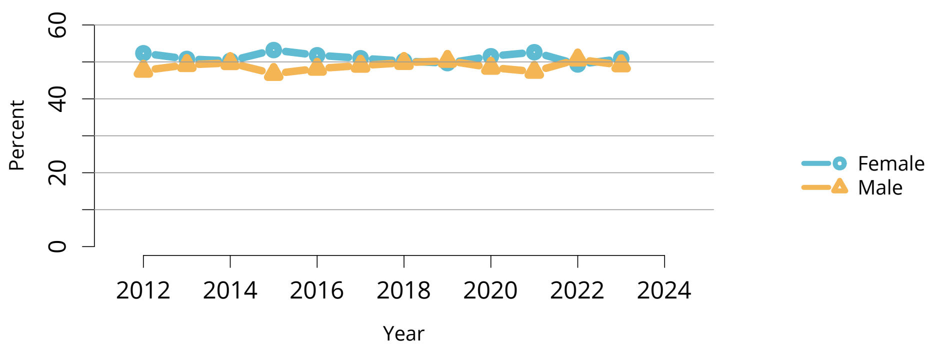

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Figure LI 106: Distribution of pediatric candidates waiting for liver transplant by sex. Candidates waiting for transplant any time in the given year. Candidates listed at more than one center are counted once per listing. Active and inactive patients are included.

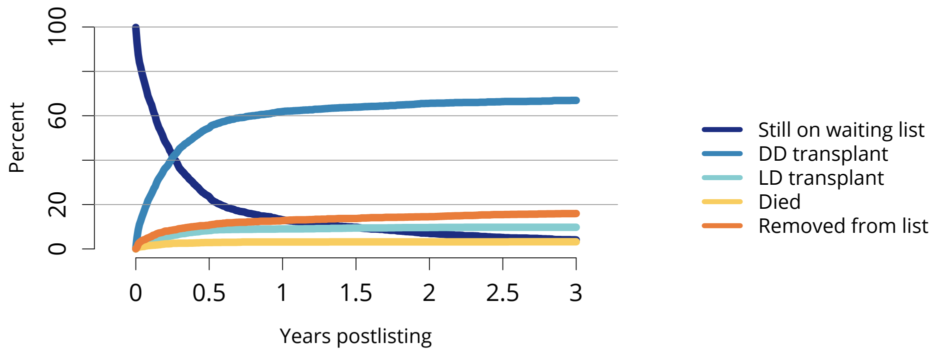

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Figure LI 107: Three-year outcomes for newly listed pediatric candidates waiting for liver transplant, 2018-2020. Pediatric candidates who joined the waiting list in 2018-2020. Candidates listed at more than one center are counted once per listing. DD, deceased donor; LD, living donor.

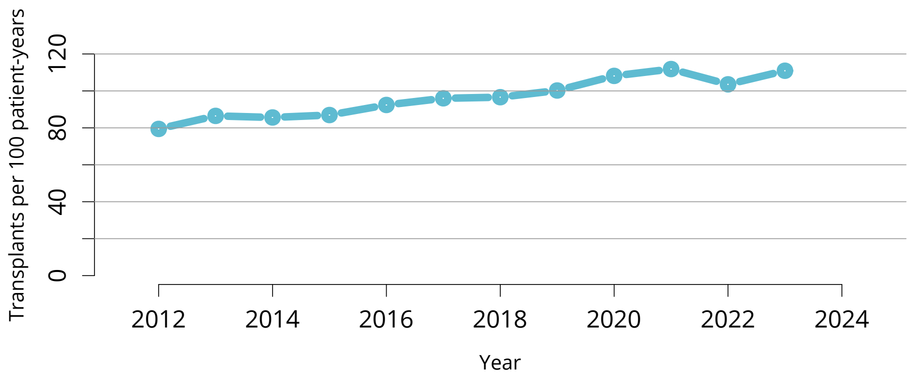

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Figure LI 108: Overall deceased donor liver transplant rates among pediatric waitlist candidates. Transplant rates are computed as the number of deceased donor transplants per 100 patient-years of waiting time in a given year. Individual listings are counted separately.

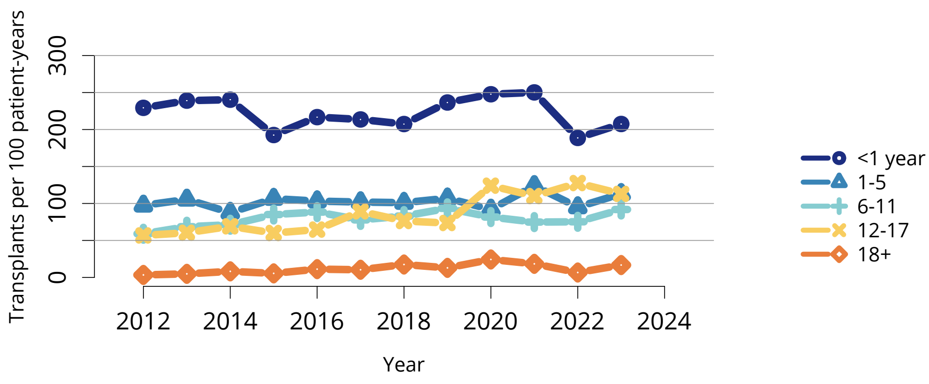

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Figure LI 109: Deceased donor liver transplant rates among pediatric waitlist candidates by age. Transplant rates are computed as the number of deceased donor transplants per 100 patient-years of waiting time in a given year. Individual listings are counted separately. Age is determined at the later of listing date or January 1 of the given year. The 18+ category is for candidates who turned age 18 while waiting.

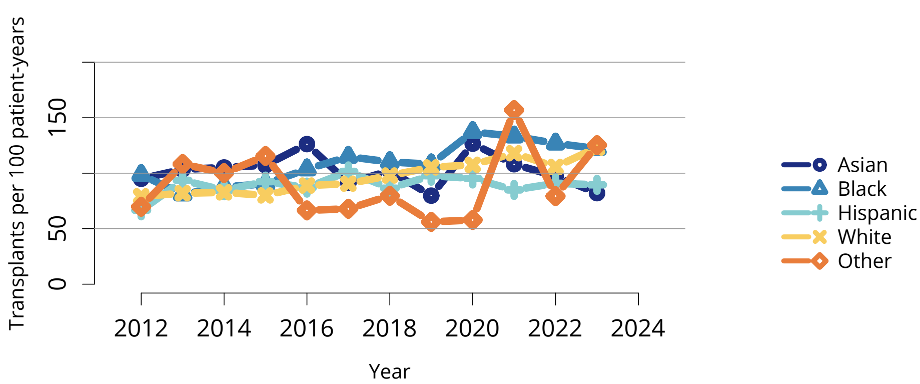

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Figure LI 110: Deceased donor liver transplant rates among pediatric waitlist candidates by race and ethnicity. Transplant rates are computed as the number of deceased donor transplants per 100 patient-years of waiting time in a given year. Individual listings are counted separately. The Other race category is composed of Native American and Multiracial categories.

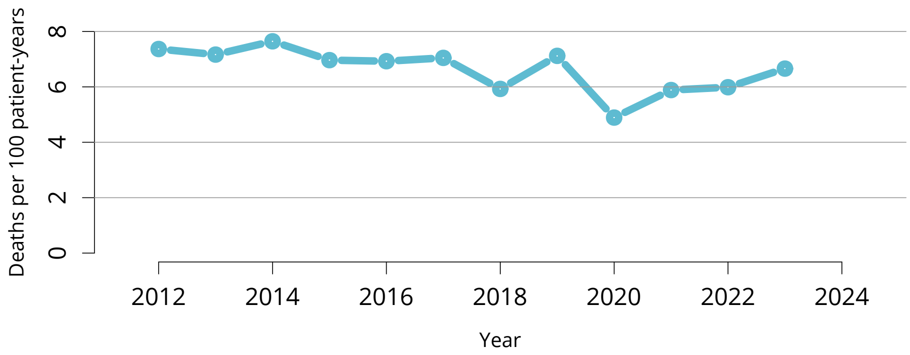

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Figure LI 111: Overall pretransplant mortality rates among pediatric candidates waitlisted for liver. Mortality rates are computed as the number of deaths per 100 patient-years of waiting time in the given year. Waiting time is censored at transplant, death, transfer to another program, removal because of improved condition, or end of cohort. Individual listings are counted separately.

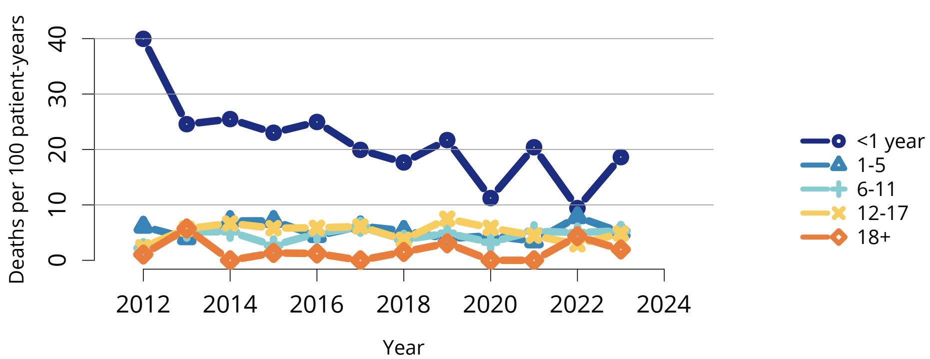

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Figure LI 112: Pretransplant mortality rates among pediatric candidates waitlisted for liver transplant by age. Mortality rates are computed as the number of deaths per 100 patient-years of waiting time in the given year. Waiting time is censored at transplant, death, transfer to another program, removal because of improved condition, or end of cohort. Individual listings are counted separately. Age is determined at the later of listing date or January 1 of the given year. The 18+ category is for candidates who turned age 18 while waiting.

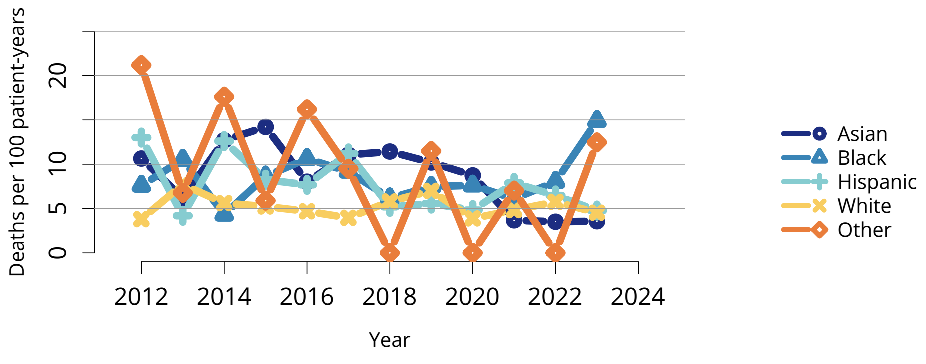

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Figure LI 113: Pretransplant mortality rates among pediatric candidates waitlisted for liver transplant by race and ethnicity. Mortality rates are computed as the number of deaths per 100 patient-years of waiting time in the given year. Waiting time is censored at transplant, death, transfer to another program, removal because of improved condition, or end of cohort. Individual listings are counted separately. The Other race category is composed of Native American and Multiracial categories.

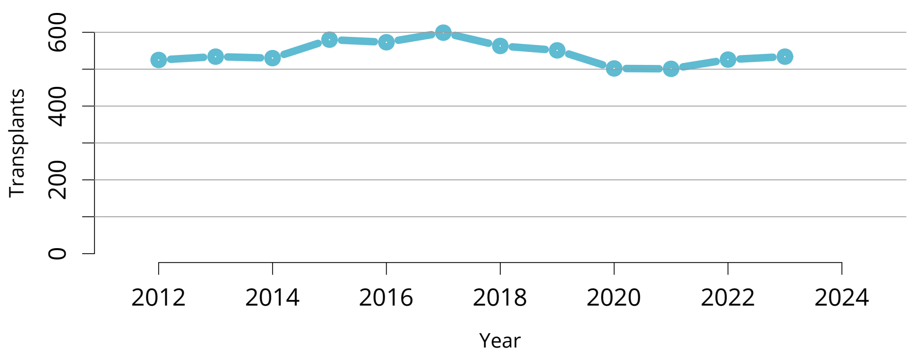

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Figure LI 114: Overall pediatric liver transplants. All pediatric liver transplant recipients, including retransplant and multiorgan recipients.

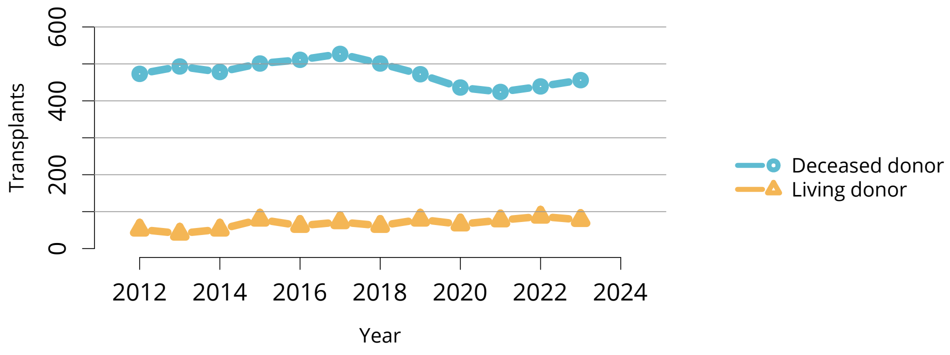

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Figure LI 115: Pediatric liver transplants by donor type. All pediatric liver transplant recipients, including retransplant and multiorgan recipients.

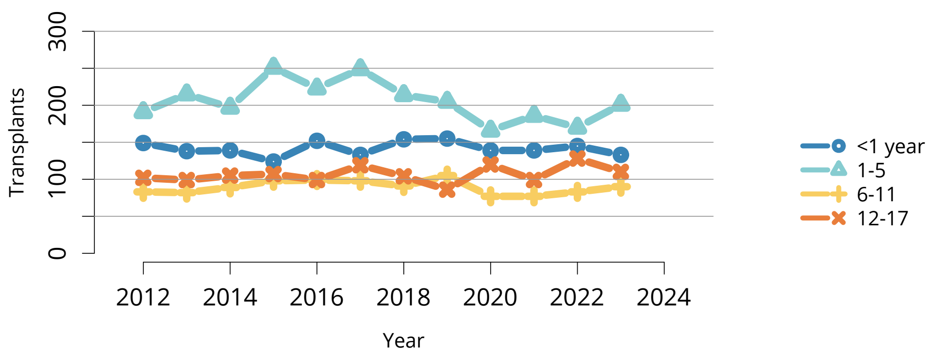

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Figure LI 116: Pediatric liver transplants by recipient age. All pediatric liver transplant recipients, including retransplant and multiorgan recipients.

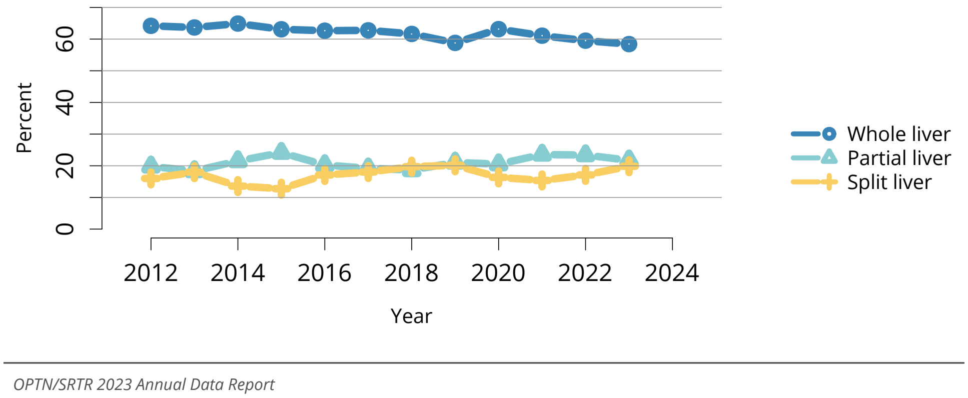

Figure LI 117: Split or partial liver transplants in children. Percent of transplants from a split liver.

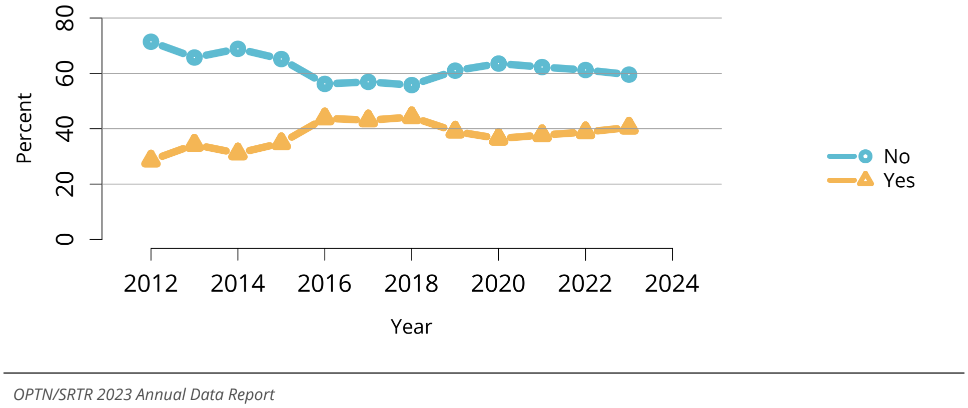

Figure LI 118: Induction agent use in pediatric liver transplant recipients. Immunosuppression at transplant reported to the OPTN.

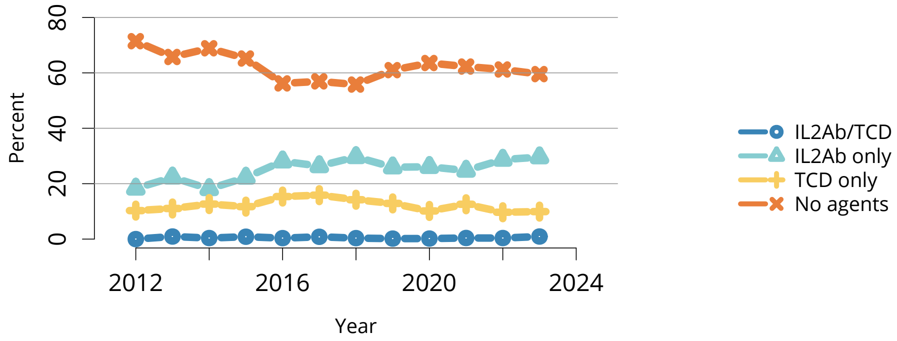

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Figure LI 119: Type of induction agent use in pediatric liver transplant recipients. Immunosuppression at transplant reported to the OPTN. IL2Ab, interleukin-2 receptor antibody; TCD, T-cell depleting.

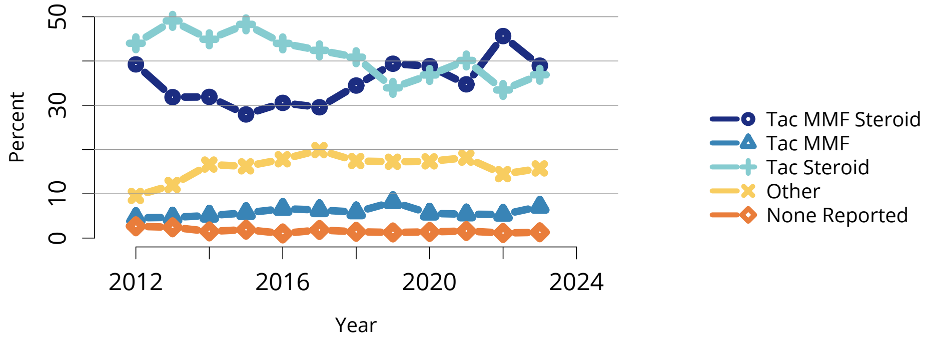

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Figure LI 120: Immunosuppression regimen use in pediatric liver transplant recipients. Immunosuppression regimen at transplant reported to the OPTN. MMF, all mycophenolate agents; Tac, tacrolimus.

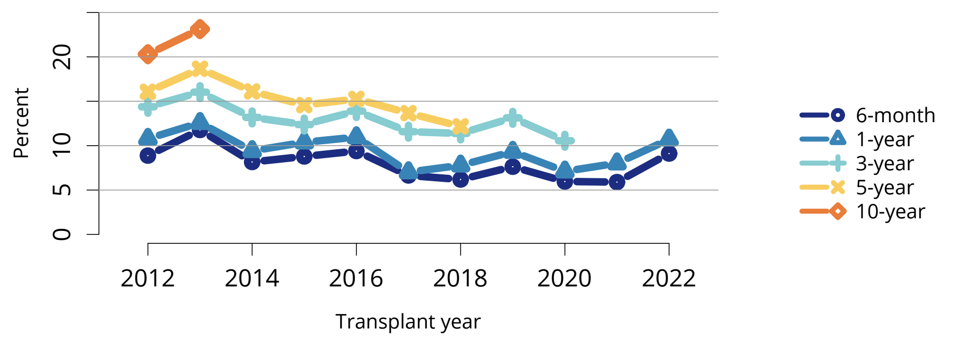

Figure LI 121: Graft failure among pediatric deceased donor liver transplant recipients. All pediatric recipients of deceased donor livers, including multiorgan transplant recipients. Estimates are unadjusted, computed using Kaplan-Meier methods. Recipients are followed to the earliest of retransplant; death; or 6 months, 1, 3, 5, or 10 years posttransplant. All-cause graft failure is defined as any of the prior outcomes prior to 6 months, 1, 3, 5, or 10 years, respectively.

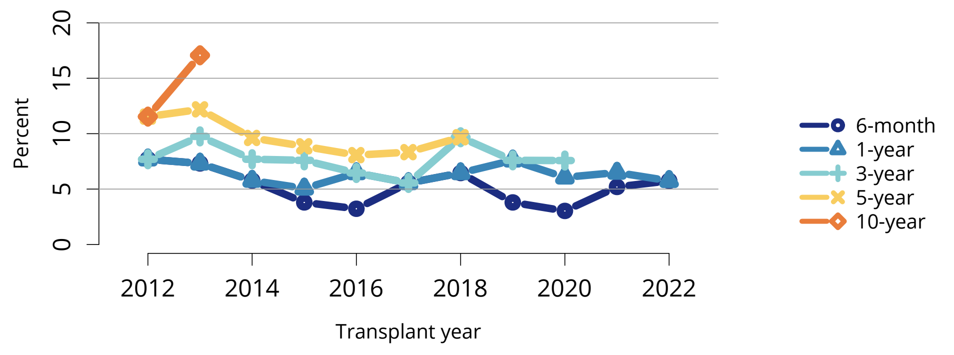

Figure LI 122: Graft failure among pediatric living donor liver transplant recipients. All pediatric recipients of living donor livers, including multiorgan transplant recipients. Estimates are unadjusted, computed using Kaplan-Meier methods. Recipients are followed to the earliest of retransplant; death; or 6 months, 1, 3, 5, or 10 years posttransplant. All-cause graft failure is defined as any of the prior outcomes prior to 6 months, 1, 3, 5, or 10 years, respectively.

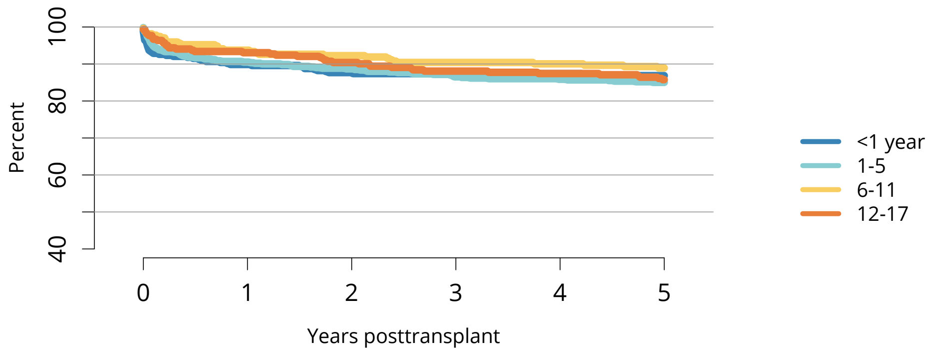

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Figure LI 123: Graft survival among pediatric deceased donor liver transplant recipients, 2016-2018, by age. Recipient survival estimated using unadjusted Kaplan-Meier methods.

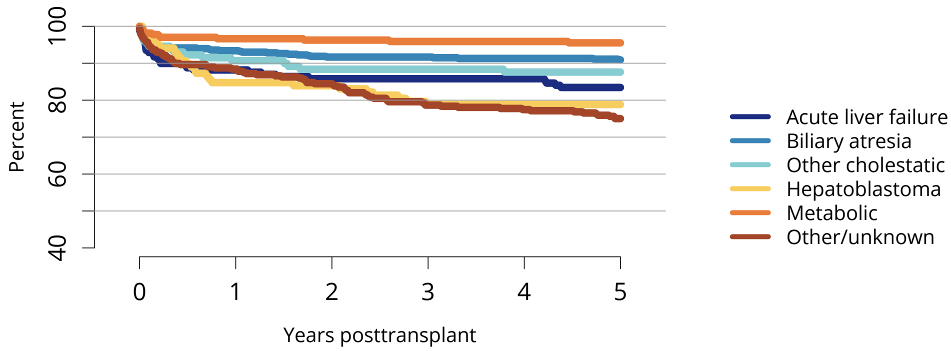

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Figure LI 124: Graft survival among pediatric deceased donor liver transplant recipients, 2016-2018, by diagnosis. Graft survival estimated using unadjusted Kaplan-Meier methods.

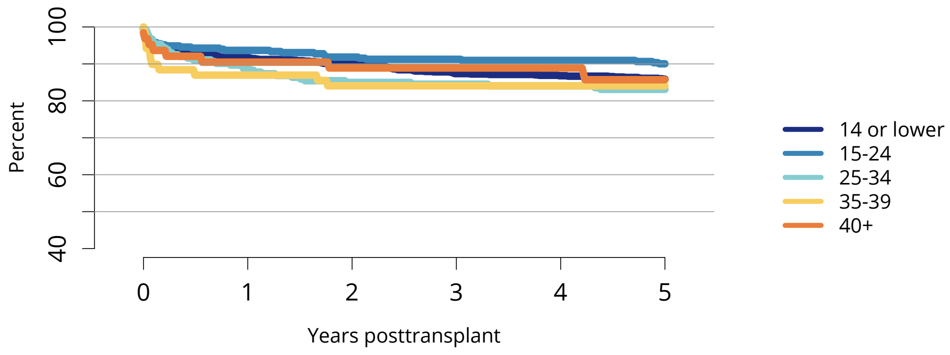

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Figure LI 125: Graft survival among pediatric deceased donor liver transplant recipients, 2016-2018, by laboratory MELD or PELD score. Graft survival estimated using unadjusted Kaplan-Meier methods. Pediatric candidates aged 12-17 years can be assigned MELD or PELD scores. MELD, model for end-stage liver disease; PELD, pediatric end-stage liver disease.

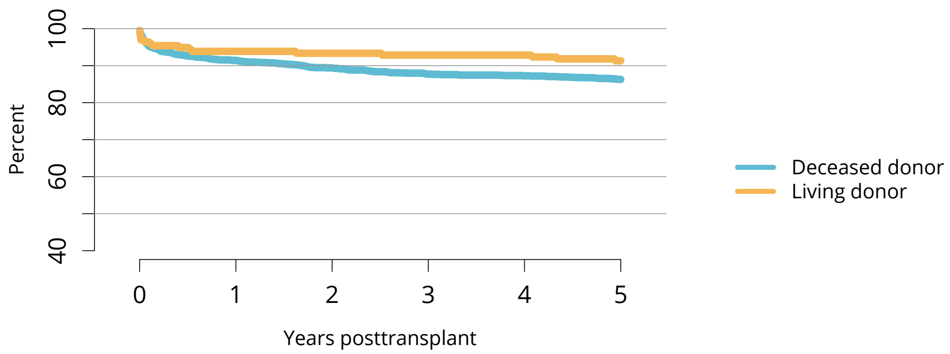

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Figure LI 126: Graft survival among pediatric liver transplant recipients, 2016-2018, by donor type. Recipient survival estimated using unadjusted Kaplan-Meier methods.

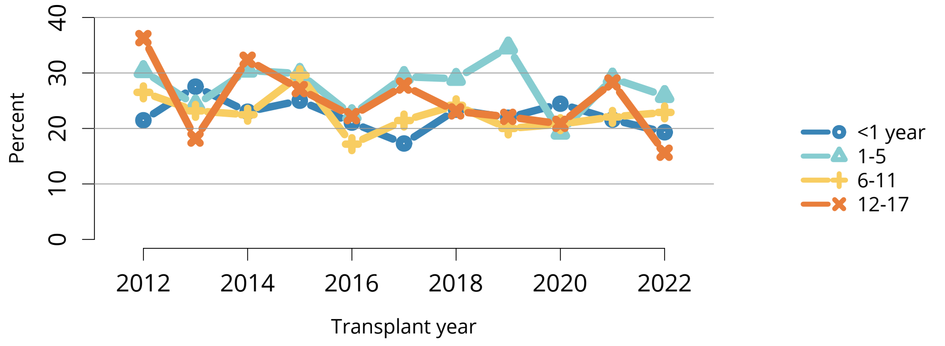

Figure LI 127: Incidence of acute rejection by 1 year posttransplant among pediatric liver transplant recipients by age. Only the first reported rejection event is counted. Cumulative incidence is estimated using the Kaplan-Meier method.

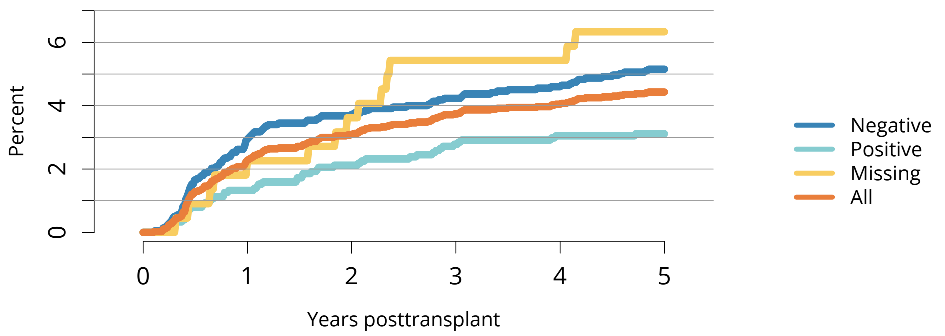

Figure LI 128: Incidence of PTLD among pediatric liver transplant recipients by recipient EBV status at transplant, 2012-2018. Cumulative incidence is estimated using the Kaplan-Meier method. PTLD is identified as a reported complication or cause of death on the OPTN Transplant Recipient Follow-up Form or on the Posttransplant Malignancy Form as polymorphic PTLD, monomorphic PTLD, or Hodgkin's disease. Only the earliest date of PTLD diagnosis is considered. EBV, Epstein-Barr virus; PTLD, posttransplant lymphoproliferative disorder.

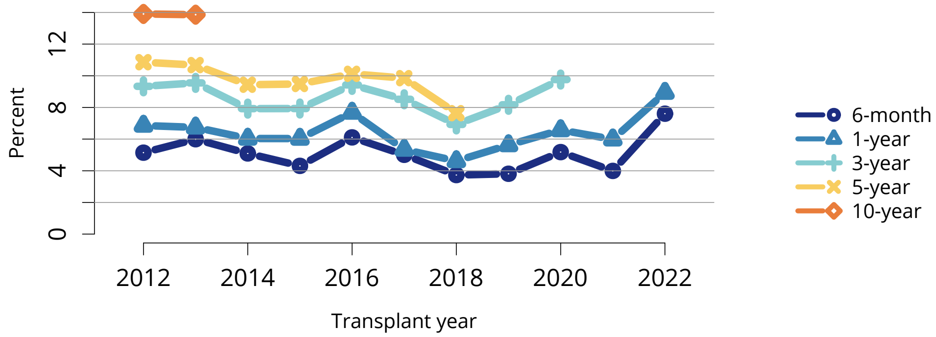

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Figure LI 129: Patient death among pediatric liver transplant recipients. All pediatric recipients of deceased donor livers, including multiorgan transplant recipients. Estimates are unadjusted, computed using Kaplan-Meier methods.

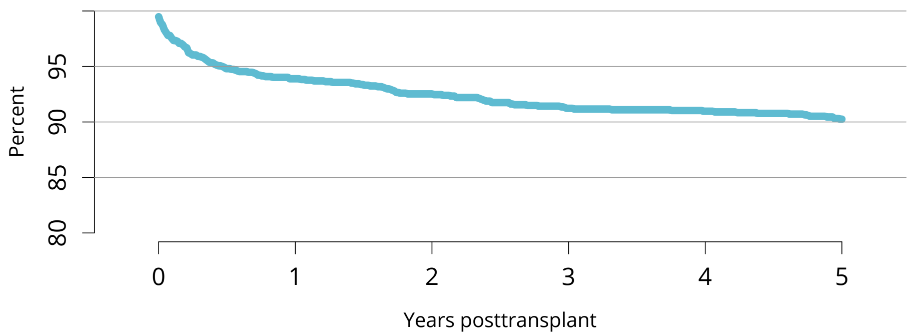

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Figure LI 130: Overall patient survival among pediatric deceased donor liver transplant recipients, 2016-2018. Recipient survival estimated using unadjusted Kaplan-Meier methods.

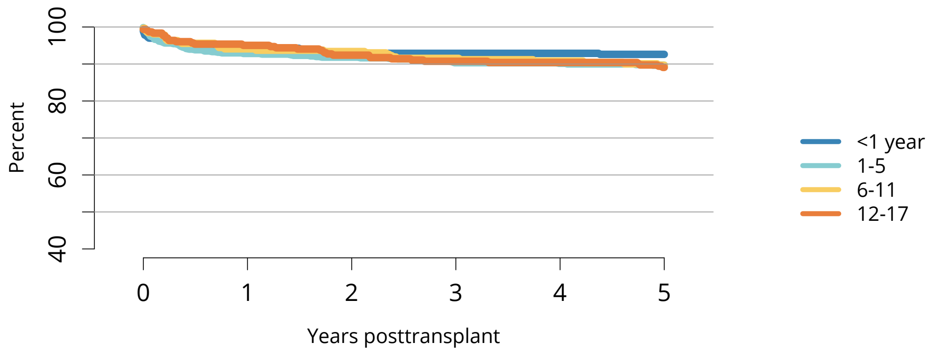

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Figure LI 131: Patient survival among pediatric deceased donor liver transplant recipients, 2016-2018, by recipient age. Recipient survival estimated using unadjusted Kaplan-Meier methods.

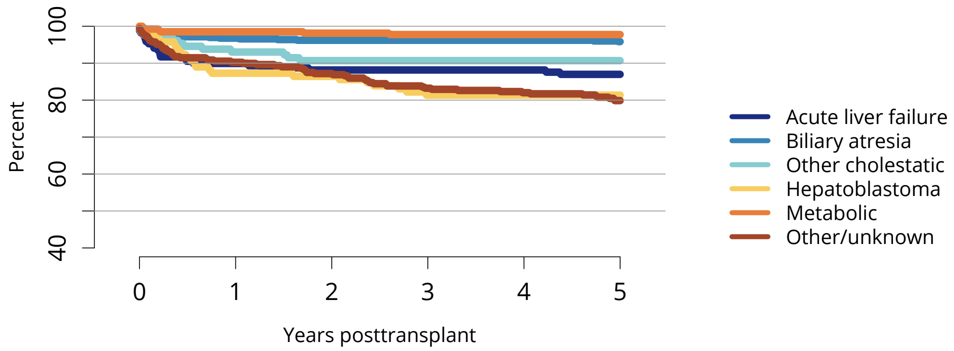

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Figure LI 132: Patient survival among pediatric deceased donor liver transplant recipients, 2016-2018, by diagnosis. Recipient survival estimated using unadjusted Kaplan-Meier methods.

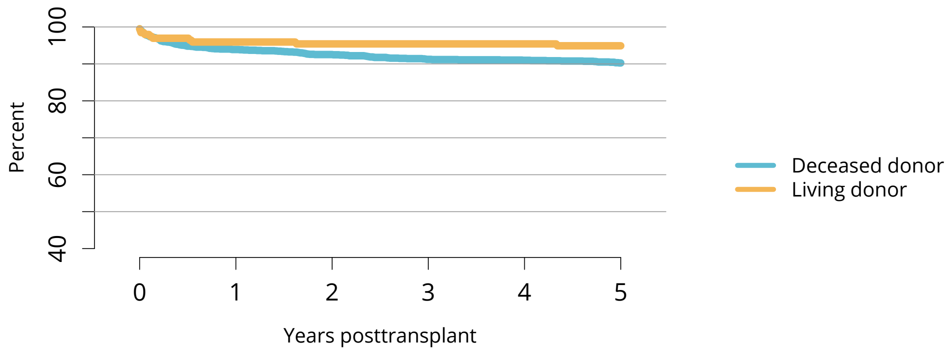

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Figure LI 133: Patient survival among pediatric liver transplant recipients, 2016-2018, by donor type. Recipient survival estimated using unadjusted Kaplan-Meier methods.

Table LI 1: Demographic characteristics of adults on the liver transplant waiting list on December 31, 2013, and December 31, 2023. Candidates waiting for transplant on December 31 of the given year, regardless of first listing date. Distance is computed from candidate's home zip code to the transplant center. Age is determined on December 31 of the year.

	2013		2023	
Characteristic	**N**	**Percent**	**N**	**Percent**
Age (years)				
18-34 years	681	4.3	542	5.6
35-49	2286	14.5	1763	18.1
50-64	9821	62.3	4649	47.7
65+	2973	18.9	2791	28.6
Sex				
Female	5885	37.3	3878	39.8
Male	9876	62.7	5867	60.2
Race and ethnicity				
Asian	797	5.1	482	4.9
Black	1114	7.1	670	6.9
Hispanic	2619	16.6	2004	20.6
Multiracial	44	0.3	56	0.6
Native American	111	0.7	117	1.2
White	11076	70.3	6381	65.5
Unreported	0	0	35	0.4
Geography				
Metropolitan	13253	84.1	8163	83.8
Nonmetropolitan	2349	14.9	1459	15
Missing	159	1	123	1.3
Miles between candidate and center				
<50 miles	9133	57.9	5480	56.2
50-<100	2564	16.3	1821	18.7
100-<150	1358	8.6	903	9.3
150-<250	1309	8.3	757	7.8
250+	1272	8.1	693	7.1
Missing	125	0.8	91	0.9
All candidates				
All candidates	15761	100	9745	100

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Table LI 2: Clinical characteristics of adults on the liver transplant waiting list on December 31, 2013, and December 31, 2023. Candidates waiting for transplant on December 31 of the given year, regardless of first listing date. HCC, hepatocellular carcinoma; HCV, hepatitis C virus; MASH, metabolic dysfunction–associated steatohepatitis.

	2013		2023	
Characteristic	**N**	**Percent**	**N**	**Percent**
Diagnosis				
Acute liver failure	273	1.7	105	1.1
HCV	4668	29.6	746	7.7
Alcohol-associated cirrhosis	3725	23.6	3290	33.8
Alcohol-associated hepatitis	19	0.1	201	2.1
Cholestatic disease	1360	8.6	785	8.1
HCC	1059	6.7	1068	11
MASH	1489	9.4	2067	21.2
Other/unknown	3168	20.1	1483	15.2
Blood type				
A	6106	38.7	3855	39.6
AB	403	2.6	158	1.6
B	1698	10.8	872	8.9
O	7554	47.9	4860	49.9
Urgency status				
14 or lower	7259	46.1	3961	40.6
15-24	4305	27.3	3228	33.1
25-34	1378	8.7	232	2.4
35-39	37	0.2	21	0.2
40+	14	0.1	7	0.1
Status 1A	2	0	2	0
Inactive	2759	17.5	2294	23.5
Missing	7	0	0	0
HCC status for liver candidates				
No HCC exception	14165	89.9	8193	84.1
HCC exception	1589	10.1	1552	15.9
Missing	7	0	0	0
All candidates				
All candidates	15761	100	9745	100

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Table LI 3: Listing characteristics of adults on the liver transplant waiting list on December 31, 2013, and December 31, 2023. Candidates waiting for transplant on December 31 of the given year, regardless of first listing date.

	2013		2023	
Characteristic	**N**	**Percent**	**N**	**Percent**
Previous transplant				
No prior transplant	15348	97.4	9476	97.2
Prior transplant	413	2.6	269	2.8
Waiting time				
<90 days	2100	13.3	2077	21.3
3-<6 months	1717	10.9	1428	14.7
6-<12 months	2331	14.8	1719	17.6
1-<2 years	2810	17.8	1727	17.7
2+ years	6803	43.2	2794	28.7
All candidates				
All candidates	15761	100	9745	100

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Table LI 4: Liver transplant waitlist activity among adults. Candidates listed at more than one center are counted once per listing. Candidates who are listed, undergo transplant, and are relisted are counted more than once. Candidates are not considered to be on the list on the day they are removed; counts on January 1 may differ from counts on December 31 of the prior year. Candidates listed for multiorgan transplants are included.

Waiting list state	**2021**	**2022**	**2023**
Waiting list state			
Patients at start of year	11768	11322	10538
Patients added during year	13165	12862	13954
Patients removed during year	13611	13645	14747
Patients at end of year	11322	10539	9745

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Table LI 5: Removal reason among adult liver transplant candidates. Removal reason as reported to the OPTN. Candidates with death dates that precede removal dates are assumed to have died waiting.

Removal reason	2021	2022	2023
Removal reason			
Deceased donor transplant	8213	8462	9516
Living donor transplant	492	515	579
Transplant outside US	1	2	2
Patient died	1154	1040	943
Patient refused transplant	135	118	146
Improved, transplant not needed	1051	983	1101
Too sick for transplant	1177	1091	973
Other	1388	1434	1487

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Table LI 6: Demographic characteristics of adult liver transplant recipients, 2013 and 2023. Liver transplant recipients, including retransplant recipients. Distance is computed from recipient's home zip code to the transplant center.

	2013		2023	
Characteristic	**N**	**Percent**	**N**	**Percent**
Recipient age (years)				
18-34 years	335	5.7	758	7.5
35-49	976	16.5	2358	23.3
50-64	3645	61.6	4741	46.8
65+	965	16.3	2268	22.4
Sex				
Female	2021	34.1	3925	38.8
Male	3900	65.9	6200	61.2
Race and ethnicity				
Asian	266	4.5	418	4.1
Black	605	10.2	667	6.6
Hispanic	812	13.7	1769	17.5
Multiracial	25	0.4	65	0.6
Native American	29	0.5	101	1
White	4184	70.7	7068	69.8
Unreported	0	0	37	0.4
Body mass index				
<18.5	124	2.1	165	1.6
18.5-<25	1685	28.5	2633	26
25-<30	2053	34.7	3336	32.9
30-<35	1273	21.5	2266	22.4
35+	768	13	1571	15.5
Missing	18	0.3	154	1.5
Insurance				
Private	3201	54.1	5151	50.9
Medicare	1686	28.5	2629	26
Medicaid	774	13.1	1924	19
Other/unknown	260	4.4	421	4.2
Geography				
Metropolitan	4874	82.3	8380	82.8
Nonmetropolitan	927	15.7	1621	16
Missing	120	2	124	1.2
Miles between recipient and center				
<50 miles	3311	55.9	5736	56.7
50-<100	984	16.6	1779	17.6
100-<150	539	9.1	927	9.2
150-<250	468	7.9	793	7.8
250+	551	9.3	804	7.9
Missing	68	1.1	86	0.8
All recipients				
All recipients	5921	100	10125	100

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Table LI 7: Clinical characteristics of adult liver transplant recipients, 2013 and 2023. Liver transplant recipients, including retransplant recipients. HCC, hepatocellular carcinoma; HCV, hepatitis C virus; MASH, metabolic dysfunction–associated steatohepatitis.

	2013		2023	
Characteristic	**N**	**Percent**	**N**	**Percent**
Diagnosis				
Acute liver failure	229	3.9	198	2
HCV	1540	26	423	4.2
Alcohol-associated cirrhosis	1054	17.8	3505	34.6
Alcohol-associated hepatitis	7	0.1	658	6.5
Cholestatic disease	499	8.4	747	7.4
HCC	1041	17.6	1055	10.4
MASH	597	10.1	2055	20.3
Other/unknown	954	16.1	1484	14.7
Blood type				
A	2140	36.1	3661	36.2
AB	287	4.8	478	4.7
B	801	13.5	1267	12.5
O	2693	45.5	4719	46.6
Urgency status at transplant				
14 or lower	168	2.8	1970	19.5
15-24	1753	29.6	3131	30.9
25-34	2444	41.3	2714	26.8
35-39	662	11.2	1094	10.8
40+	695	11.7	985	9.7
Status 1B	1	0	4	0
Status 1A	195	3.3	215	2.1
Inactive	1	0	12	0.1
Missing	2	0	0	0
HCC status for liver recipients				
No HCC exception	4357	73.6	8647	85.4
HCC exception	1562	26.4	1478	14.6
Missing	2	0	0	0
All recipients				
All recipients	5921	100	10125	100

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Table LI 8: Transplant characteristics of adult liver transplant recipients, 2013 and 2023. Liver transplant recipients, including retransplant recipients. DBD, donation after brain death; DCD, donation after circulatory death.

	2013		2023	
Characteristic	**N**	**Percent**	**N**	**Percent**
Waiting time				
<1 day	12	0.2	12	0.1
1-<90 days	2734	46.2	6450	63.7
3-<6 months	992	16.8	1097	10.8
6-<12 months	925	15.6	1382	13.6
1-<2 years	773	13.1	728	7.2
2+ years	483	8.2	456	4.5
Missing	2	0	0	0
Donor type				
Deceased donor	5710	96.4	9545	94.3
Living donor	211	3.6	580	5.7
Split versus whole liver transplant				
Whole liver	5645	95.3	9458	93.4
Partial liver	204	3.4	576	5.7
Split liver	72	1.2	91	0.9
Donation after circulatory death				
DBD	5404	91.3	7850	77.5
DCD	306	5.2	1695	16.7
Living donor	211	3.6	580	5.7
Previous transplant for recipients				
No prior transplant	5626	95	9780	96.6
Prior transplant	295	5	345	3.4
All recipients				
All recipients	5921	100	10125	100

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Table LI 9: Demographic characteristics of pediatric candidates on the liver transplant waiting list on December 31, 2013, and December 31, 2023. Candidates younger than 18 years at listing waiting for transplant on December 31 of the given year, regardless of first listing date. Age is determined on December 31 of the year. The 18+ category is for candidates who turned age 18 while waiting. Distance is computed from candidate's home zip code to the transplant center.

	2013		2023	
Characteristic	**N**	**Percent**	**N**	**Percent**
Age (years)				
<1 year	38	6.6	43	10.6
1-5	178	31	139	34.4
6-11	120	20.9	84	20.8
12-17	148	25.8	89	22
18+	90	15.7	49	12.1
Sex				
Female	291	50.7	200	49.5
Male	283	49.3	204	50.5
Race and ethnicity				
Asian	23	4	29	7.2
Black	85	14.8	54	13.4
Hispanic	127	22.1	124	30.7
Multiracial	11	1.9	18	4.5
Native American	7	1.2	3	0.7
White	321	55.9	172	42.6
Unreported	0	0	4	1
Geography				
Metropolitan	481	83.8	353	87.4
Nonmetropolitan	77	13.4	44	10.9
Missing	16	2.8	7	1.7
Miles between candidate and center				
<50 miles	292	50.9	203	50.2
50-<100	80	13.9	73	18.1
100-<150	55	9.6	28	6.9
150-<250	58	10.1	39	9.7
250+	77	13.4	55	13.6
Missing	12	2.1	6	1.5
All candidates				
All candidates	574	100	404	100

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Table LI 10: Clinical characteristics of pediatric candidates on the liver transplant waiting list on December 31, 2013, and December 31, 2023. Candidates younger than 18 years at listing waiting for transplant on December 31 of the given year, regardless of first listing date. For urgency status, pediatric candidates aged 12-17 years can be assigned model for end-stage liver disease (MELD) or pediatric end-stage liver disease (PELD) scores. HCC, hepatocellular carcinoma.

	2013		2023	
Characteristic	**N**	**Percent**	**N**	**Percent**
Pediatric diagnosis				
Acute liver failure	29	5.1	13	3.2
Biliary atresia	177	30.8	146	36.1
Other cholestatic	52	9.1	35	8.7
Hepatoblastoma	7	1.2	6	1.5
Metabolic	76	13.2	43	10.6
Other/unknown	233	40.6	161	39.9
Blood type				
A	167	29.1	113	28
AB	17	3	6	1.5
B	78	13.6	52	12.9
O	312	54.4	233	57.7
Urgency status				
14 or lower	161	28	125	30.9
15-24	81	14.1	55	13.6
25-34	76	13.2	33	8.2
35-39	13	2.3	5	1.2
40+	32	5.6	4	1
Status 1B	12	2.1	26	6.4
Status 1A	1	0.2	2	0.5
Inactive	196	34.1	154	38.1
Missing	2	0.3	0	0
HCC status for liver candidates				
No HCC exception	569	99.1	403	99.8
HCC exception	3	0.5	1	0.2
Missing	2	0.3	0	0
All candidates				
All candidates	574	100	404	100

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Table LI 11: Listing characteristics of pediatric candidates on the liver transplant waiting list on December 31, 2013, and December 31, 2023. Candidates younger than 18 years at listing waiting for transplant on December 31 of the given year, regardless of first listing date.

	2013		2023	
Characteristic	**N**	**Percent**	**N**	**Percent**
Previous transplant				
No prior transplant	526	91.6	378	93.6
Prior transplant	48	8.4	26	6.4
Waiting time				
<90 days	100	17.4	99	24.5
3-<6 months	79	13.8	54	13.4
6-<12 months	79	13.8	56	13.9
1-<2 years	77	13.4	73	18.1
2+ years	239	41.6	122	30.2
All candidates				
All candidates	574	100	404	100

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Table LI 12: Liver transplant waitlist activity among pediatric candidates. Candidates who are listed, undergo transplant, and are relisted are counted more than once. Candidates are not considered to be on the list on the day they are removed; counts on January 1 may differ from counts on December 31 of the prior year. Candidates listed for multiorgan transplants are included.

Waiting list state	2021	2022	2023
Waiting list state			
Patients at start of year	402	401	438
Patients added during year	666	741	704
Patients removed during year	667	704	738
Patients at end of year	401	438	404

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Table LI 13: Removal reason among pediatric liver transplant candidates. Removal reason as reported to the OPTN. Candidates with death dates that precede removal dates are assumed to have died waiting.

Removal reason	2021	2022	2023
Removal reason			
Deceased donor transplant	435	450	470
Living donor transplant	77	88	79
Transplant outside US	1	0	1
Patient died	20	20	27
Patient refused transplant	5	2	3
Improved, transplant not needed	85	79	96
Too sick for transplant	16	30	13
Other	28	35	49

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Table LI 14: Demographic characteristics of pediatric liver transplant recipients, 2013 and 2023. Pediatric liver transplant recipients, including retransplant recipients. Distance is computed from recipient's home zip code to the transplant center.

	2013		2023	
Characteristic	**N**	**Percent**	**N**	**Percent**
Recipient age (years)				
<1 year	138	25.8	133	24.9
1-5	215	40.3	201	37.6
6-11	82	15.4	90	16.9
12-17	99	18.5	110	20.6
Sex				
Female	256	47.9	275	51.5
Male	278	52.1	259	48.5
Race and ethnicity				
Asian	35	6.6	30	5.6
Black	71	13.3	105	19.7
Hispanic	142	26.6	114	21.3
Multiracial	10	1.9	19	3.6
Native American	6	1.1	7	1.3
White	270	50.6	256	47.9
Unreported	0	0	3	0.6
Insurance				
Private	219	41	202	37.8
Medicare	6	1.1	4	0.7
Medicaid	234	43.8	271	50.7
Other/unknown	75	14	57	10.7
Geography				
Metropolitan	441	82.6	441	82.6
Nonmetropolitan	75	14	83	15.5
Missing	18	3.4	10	1.9
Miles between recipient and center				
<50 miles	238	44.6	241	45.1
50-<100	77	14.4	89	16.7
100-<150	55	10.3	65	12.2
150-<250	70	13.1	61	11.4
250+	80	15	69	12.9
Missing	14	2.6	9	1.7
All recipients				
All recipients	534	100	534	100

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Table LI 15: Clinical characteristics of pediatric liver transplant recipients, 2013 and 2023. Pediatric liver transplant recipients, including retransplant recipients. For urgency status, pediatric candidates aged 12-17 years can be assigned model for end-stage liver disease (MELD) or pediatric end-stage liver disease (PELD) scores. HCC, hepatocellular carcinoma.

	2013		2023	
Characteristic	**N**	**Percent**	**N**	**Percent**
Diagnosis				
Acute liver failure	74	13.9	43	8.1
Biliary atresia	178	33.3	200	37.5
Other cholestatic	54	10.1	39	7.3
Hepatoblastoma	42	7.9	43	8.1
Metabolic	88	16.5	79	14.8
Other/unknown	98	18.4	130	24.3
Blood type				
A	191	35.8	189	35.4
AB	22	4.1	27	5.1
B	72	13.5	74	13.9
O	249	46.6	244	45.7
Urgency status at transplant				
14 or lower	53	9.9	220	41.2
15-24	75	14	90	16.9
25-34	107	20	36	6.7
35-39	32	6	6	1.1
40+	56	10.5	9	1.7
Status 1B	119	22.3	119	22.3
Status 1A	92	17.2	52	9.7
Inactive	0	0	2	0.4
HCC status for liver recipients				
No HCC exception	528	98.9	532	99.6
HCC exception	6	1.1	2	0.4
All recipients				
All recipients	534	100	534	100

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Table LI 16: Transplant characteristics of pediatric liver transplant recipients, 2013 and 2023. Pediatric liver transplant recipients, including retransplant recipients. DBD, donation after brain death; DCD, donation after circulatory death.

	2013		2023	
Characteristic	**N**	**Percent**	**N**	**Percent**
Waiting time				
<1 day	2	0.4	0	0
1-<90 days	334	62.5	349	65.4
3-<6 months	94	17.6	89	16.7
6-<12 months	58	10.9	56	10.5
1-<2 years	27	5.1	27	5.1
2+ years	19	3.6	13	2.4
ABO-Incompatible transplant				
Compatible/Identical	515	96.4	511	95.7
Incompatible	19	3.6	23	4.3
Donor type				
Deceased donor	493	92.3	456	85.4
Living donor	41	7.7	78	14.6
Split versus whole liver transplant				
Whole liver	340	63.7	312	58.4
Partial liver	98	18.4	116	21.7
Split liver	96	18	106	19.9
Donation after circulatory death				
DBD	490	91.8	456	85.4
DCD	3	0.6	0	0
Living donor	41	7.7	78	14.6
Previous transplant for recipients				
No prior transplant	487	91.2	510	95.5
Prior transplant	47	8.8	24	4.5
All recipients				
All recipients	534	100	534	100

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